



V93000 Specifications

V93000 Wave Scale RF Card

Version 8.6.0

Published on 2023-02-24

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RFIM-6 attached to Wave Scale RF card

Following hardware components are part of the specification sections:

- *WSRF 6 GHz (N2600A-OP) with RFIM-6 (N2604A) or RFIM-8 (N2604B)*

Note In case WSPS is included, it must be WSPS 6 GHz (N2601A-OP).

Note Mixing of WSRF 6 GHz (N2600A-OP) and WSRF 8 GHz (N2600B-OP) cards in one system configuration is not supported.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

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Wave Scale RF-6 with RFIM-6 or RFIM-8 specifications

Scope of specifications

The Advantest V93000 specifications are valid at the device with an adequate DUT board and valid calibration, i.e. system calibration (see section "Calibration and Operation", below) and fixture delay measurement (TDR). Since design and manufacturing of the DUT board are beyond Advantest's control, Advantest specifies and verifies the specifications up to the pogo level. All specifications assume a 50 Ω resistive source or load.

Specifications describe warranted product performance.

Characteristics, included as typical values, provide additional useful information, but do not represent warranted performance.

These specifications are valid for recommended system configurations.

These specifications are applicable for SmarTest 8.6.0.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

Calibration and operation

Warm up time	60 minutes
Calibration period	3 months
RF interface module activation lifetime	12 months after first activation
RF performance check	Optional: You can run it after system installation, after a repair or if any issue is suspected.

The RFIM activation lifetime is the elapsed time between the first activation and the need to have this RFIM exchanged by another one that has been factory calibrated.

RFIM calibration can only be done at Advantest factory. It cannot be done at customer site.

The external calibration and external adjustment period for a newly shipped system begins on the date of activation at your site, and not on the factory calibration and adjustment date disclosed on the Certificate of Calibration (CoC). As long as the shelf life (one year) is not expired at date of activation, the factory calibration and adjustment is still valid for the subsequent activation lifetime.

Valid at ambient temperature within ± 2.5 K of calibration temperature. For temperature requirements during calibration, please refer to the "Maintenance Guide".

WSRF calibration is required to meet the specifications outlined in this document.

WS system calibration with SmarTest 8.0.4 (or higher)

WSRF system calibration interval	Stim	Measure
3 months	0.2dB (typ) off vs calibration	0.15dB (typ) off vs calibration

Environmental

Operating temperature	15°C to 30°C (59°F to 86°F)
Specification warranted temperature	20°C to 30°C (68°F to 86°F)
Maximum humidity at 30°C	<80% R.H., non condensing
Storage (without water)	-40°C to +70°C

Compatibility

Wave Scale RF can be installed in a slot powered by universal DC/DC power units (E8003AA). It is compatible with Smart Scale generation or later digital cards and power supplies for example, PS1600, PS9G, DPS64/128) but can also be used without any other type of card.

System software

- Wave Scale RF requires SmarTest release 8.0.2 or higher
- Linux Red Hat Enterprise 7 or later releases supported by SmarTest

Wave Scale RF-6 with RFIM-6 or RFIM-8 stimulus**RFIM-6 stimulus maximum output power**

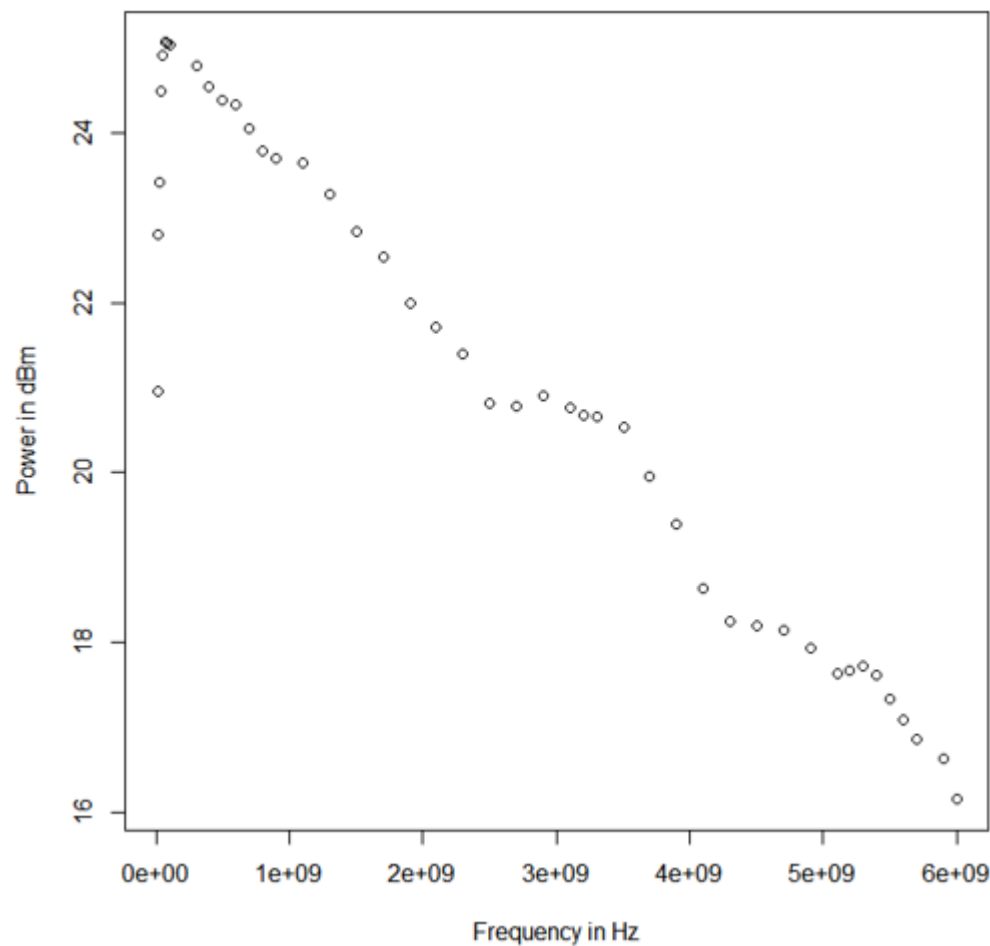
Shows absolute maximum output power, measured as saturation power.

It is not recommended to use the Wave Scale Precision Source for power levels less than -80 dBm.

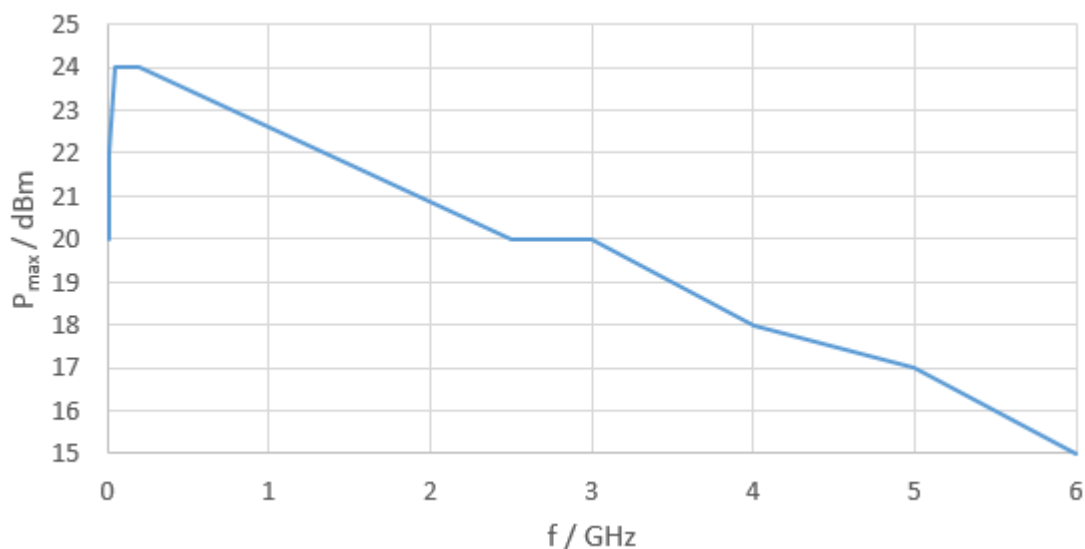
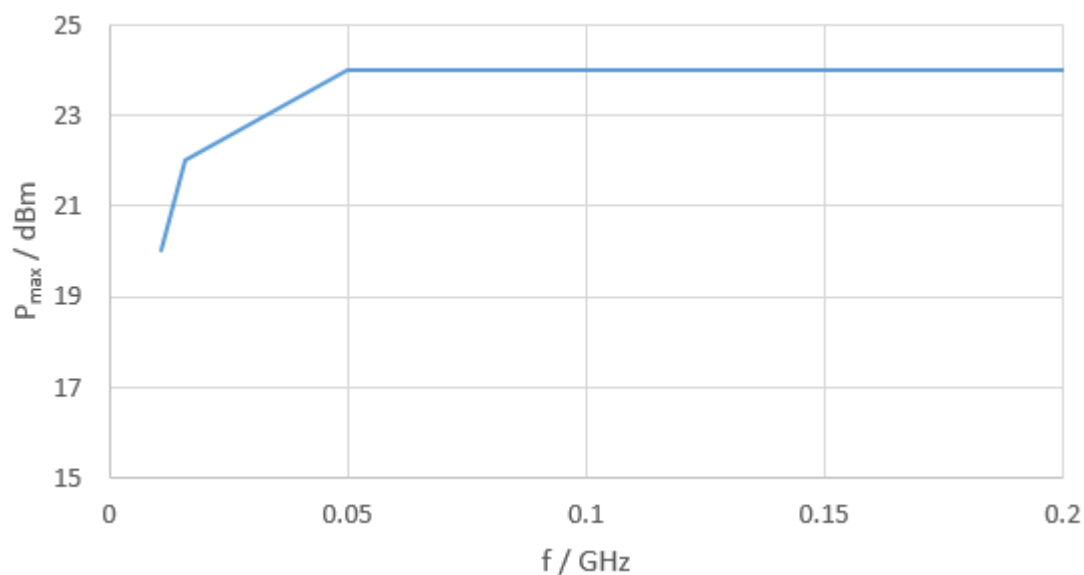
Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
0.011	17	20.5
0.016	18	22.5
0.031	21	24
0.051	21.5	24.5
0.1	21.5	24.5
1	21	23
2	18	21.5
2.5	18	20.8
3	17	20.5
4	14	18.5

Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
5	13	17.5
6	11.5	15.5

Wave Scale RF stimulus typical maximum power



Wave Scale RF stimulus typical maximum power

Wave Scale RF stimulus typical maximum output power at 0-6 GHz**Wave Scale RF stimulus typical saturated power at 0-6 GHz****Wave Scale RF stimulus typical maximum output power at 0-0.2 GHz****Wave Scale RF stimulus typical saturated power at 0-0.2 GHz****RFIM-6 stimulus modulation flatness**

The measured performance for Wave Scale RF stimulus modulation flatness is shown in the table below.

Carrier frequency range (GHz)	Maximum modulation bandwidth (MHz)	10 MHz	20 MHz	40 MHz	90 MHz	100 MHz	200 MHz
0.06 - 0.1	10	±0.15					
0.1 - 0.96	90	±0.15	±0.3	±0.5	±1.2		
0.96 - 3	200	±0.1	±0.2	±0.3		±0.8	±1.3
3 - 6	200	±0.1	±0.2	±0.4		±1.2	±1.7

RFIM-6 stimulus accuracy

Stimulus power accuracy conditions:

- Standard deviation index $k=3$ (coverage factor of expanded uncertainty) unless specified differently.
- $\pm 0.5K$ relative water temperature stability after calibration.
- All power levels are CW signals in a 50Ω system unless noted.

Stimulus power accuracy

Power range (dBm)	-110...	-100...	-90...	-80...	-70...	-60...	-50...	-40...	-30...	-20...	-10...	0...5
Frequency range (GHz)	0.05...	± 0.81	± 0.70	± 0.54	± 0.48	± 0.48	± 0.48	± 0.48	± 0.48	± 0.48	± 0.48	± 0.51
1...2	± 0.71	± 0.63	± 0.55	± 0.49	± 0.49	± 0.49	± 0.49	± 0.49	± 0.49	± 0.49	± 0.49	± 0.49
2...3	± 0.72	± 0.72	± 0.64	± 0.57	± 0.57	± 0.57	± 0.52	± 0.52	± 0.52	± 0.52	± 0.52	± 0.52
3...4	± 1.8 [1]	± 1.18	± 0.82	± 0.66	± 0.66	± 0.59	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53
4...5	± 1.8 [1]	± 1.18 [1]	± 0.82	± 0.66	± 0.66	± 0.59	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53	± 0.53
5...6	± 1.8 [1]	± 1.19 [1]	± 0.84	± 0.68	± 0.68	± 0.68	± 0.57	± 0.57	± 0.57	± 0.57	± 0.57	± 0.57

RFIM-6 stimulus tone-combiner TOI (third-order intercept)

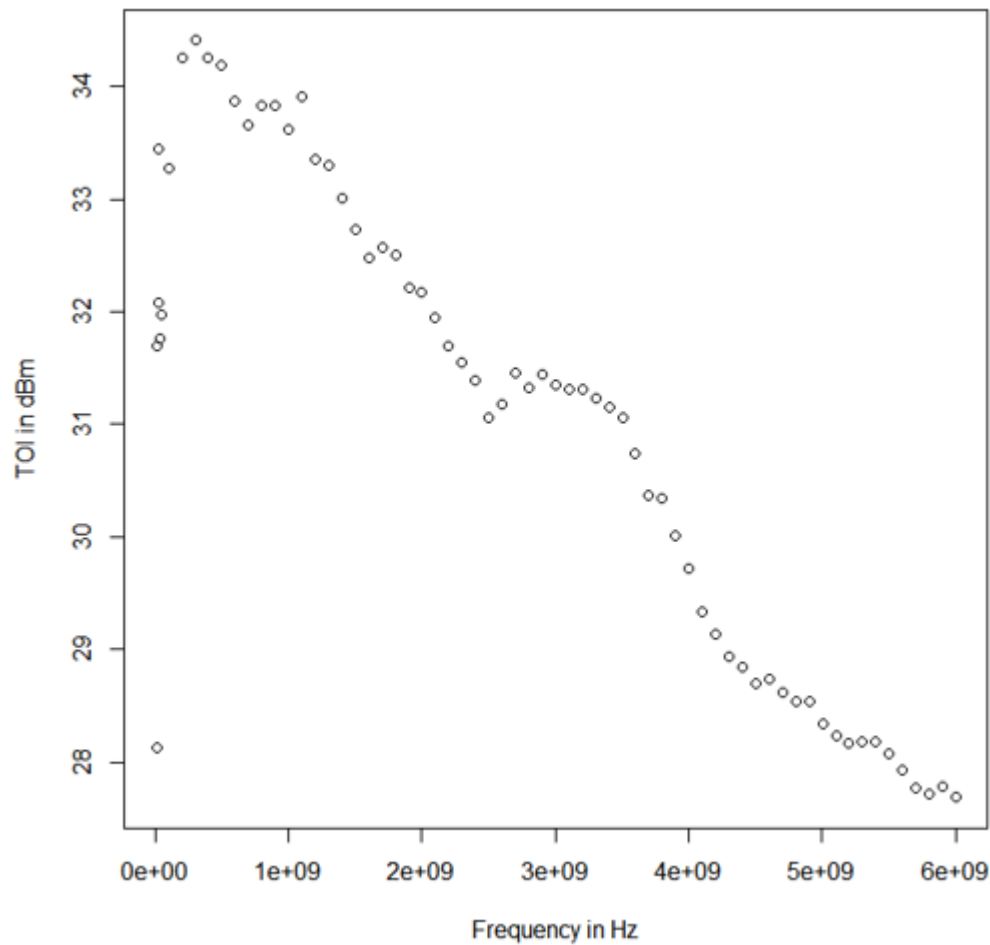
Conditions ^[2]: Tone-combiner mode, 100kHz tone spacing, optimizer balanced, 10 kHz Noise-Bandwidth.

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.1	0	28	32.5
0.2	0	29	34
0.5	0	29	34
1	0	29	34
2	0	28	32
3	0	27	31.5
4	0	25	30
5	0	24	28
6	0	23	27.5

^[1] Standard deviation index $k=2$ (Typical)

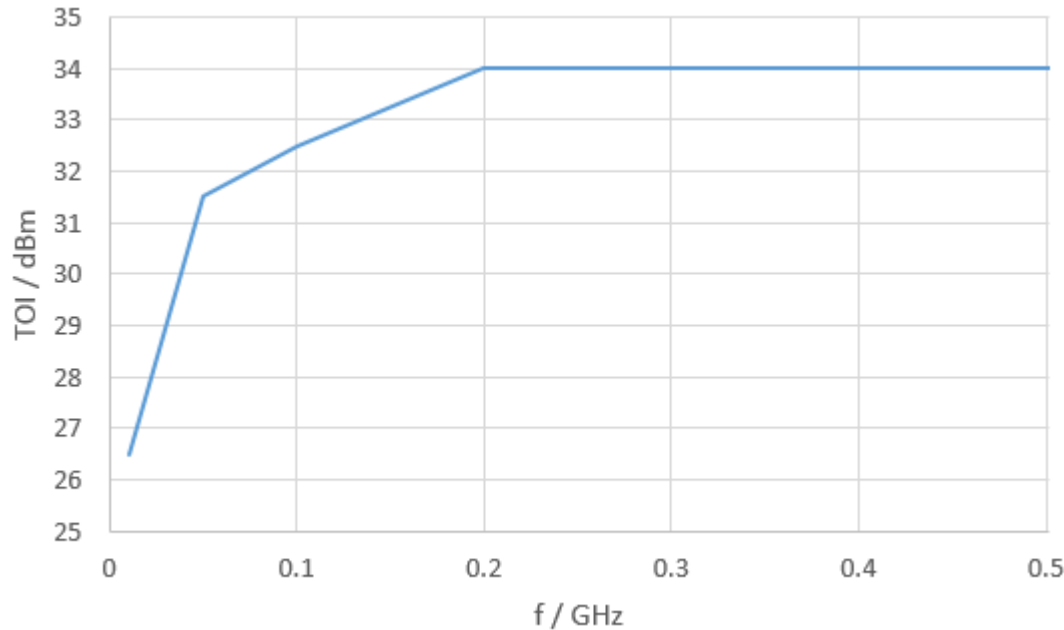
^[2] The maximum power delta between the two tones that the WSRF-6 card can guarantee is 60 dB for the above conditions.

Wave Scale RF stimulus typical TOI tone-combiner mode



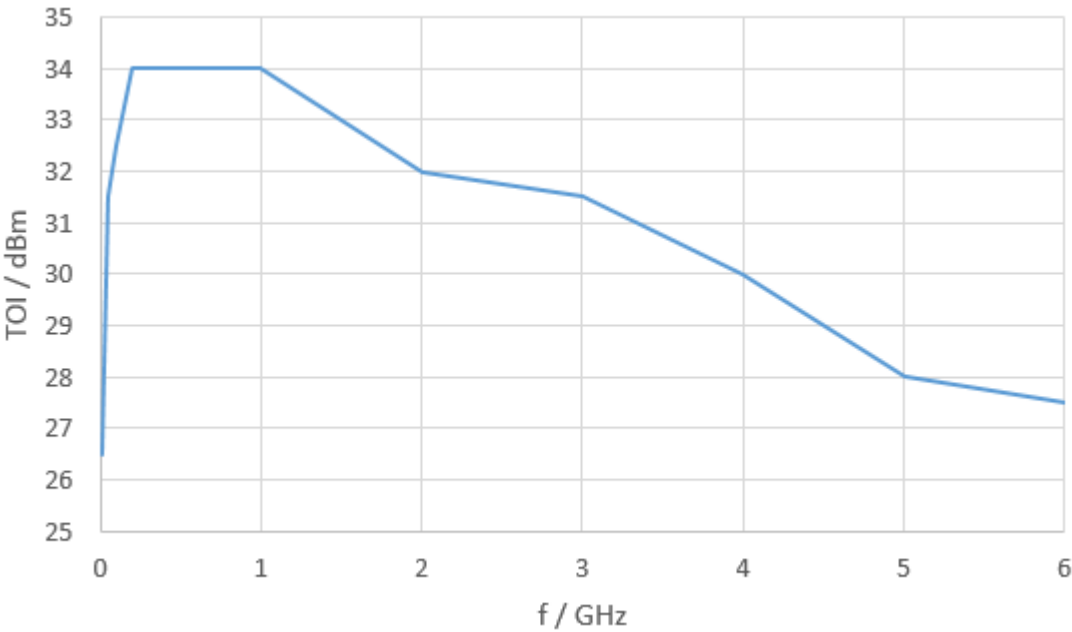
Wave Scale RF stimulus typical TOI tone-combiner mode

Wave Scale RF stimulus TOI tone-combiner typical performance 0-0.5 GHz



Wave Scale RF stimulus TOI tone-combiner typical performance 0-0.5 GHz

Wave Scale RF stimulus TOI tone-combiner typical performance 0-6 GHz



Wave Scale RF stimulus TOI tone-combiner typical performance 0-6 GHz

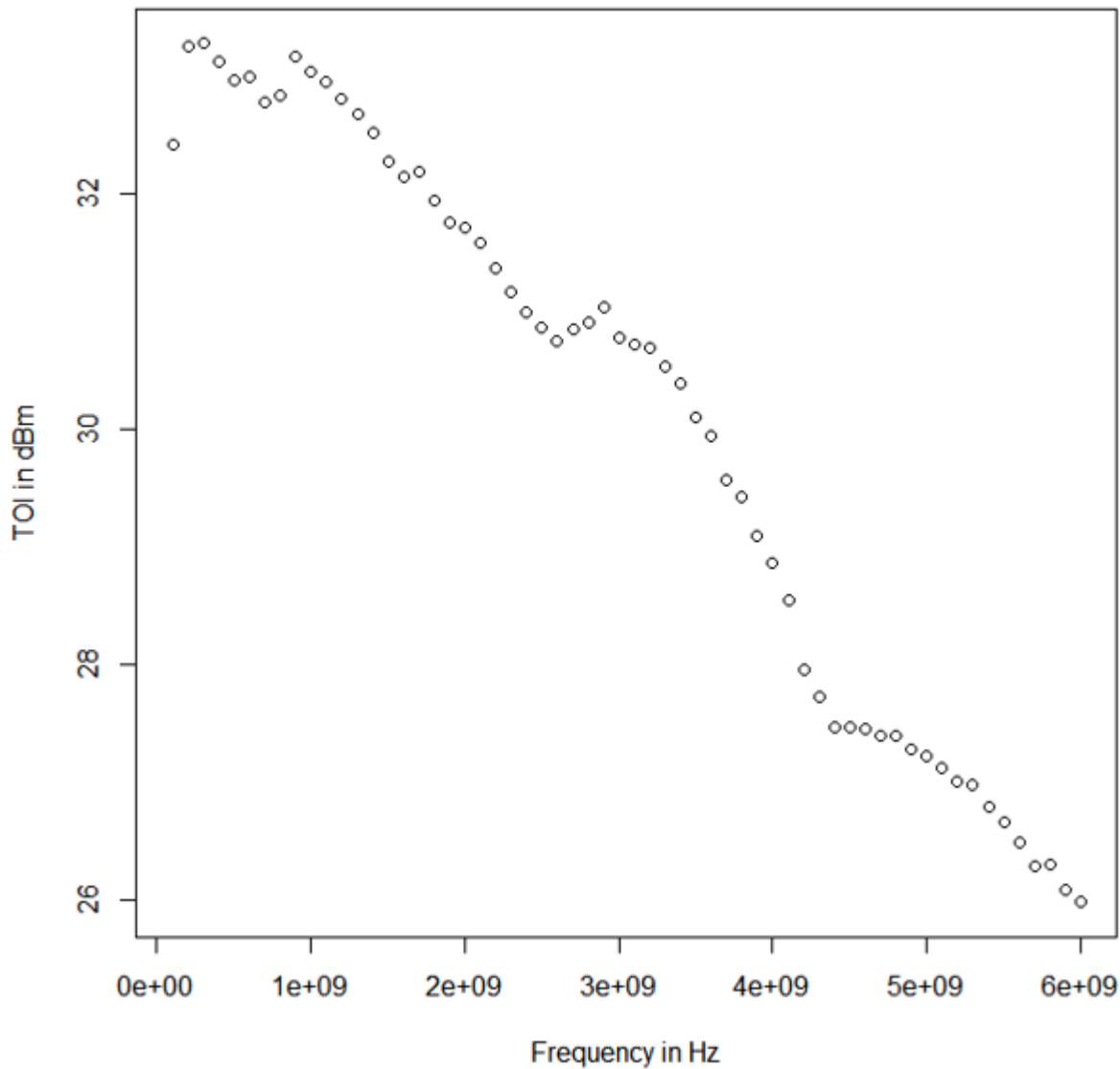
RFIM-6 stimulus modulated TOI (third-order intercept)

IQ-modulated 2-tone signal, tone-spacing 100kHz ^[3]

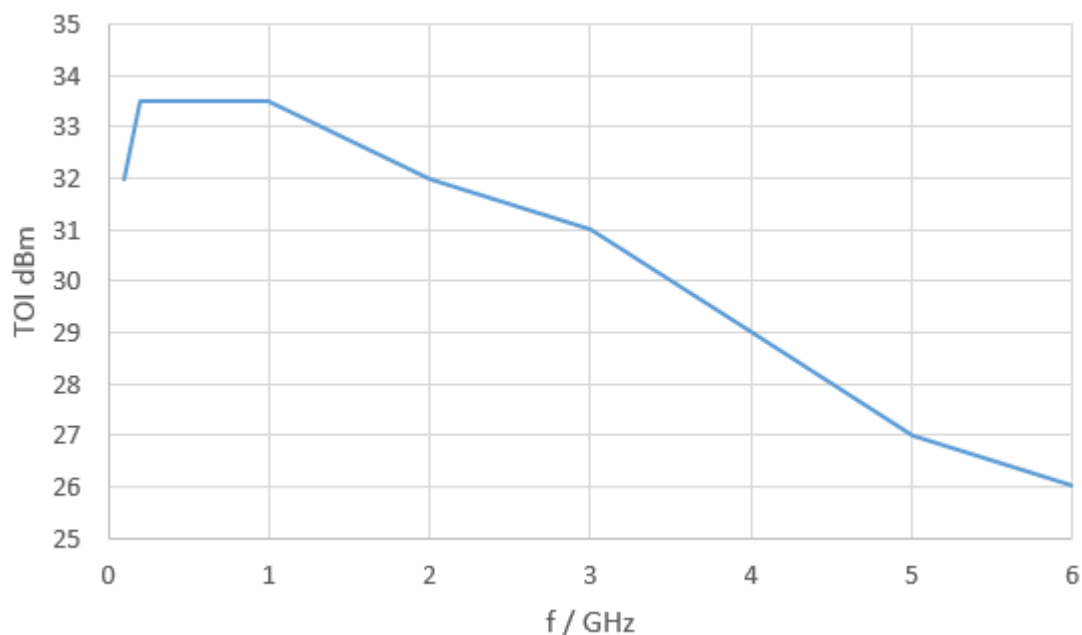
Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.5	0	28.5	33.5
1	0	28.5	33.5
2	0	27.5	32
3	0	27	31
4	0	24	29
5	0	22.5	27
6	0	21.5	26

^[3] 2-tone IQ Signal, 250Msps, (tones at -10MHz, -10.1MHz, 0dbm each tone power), total power +3dBm, optimizer balanced, 10kHz Noise-Bandwidth

Wave Scale RF stimulus typical TOI modulated



Wave Scale RF stimulus typical TOI modulated

Wave Scale RF stimulus modulated TOI (third-order intercept) at 0-6 GHz

Wave Scale RF stimulus modulated TOI (third-order intercept) at 0-6 GHz

RFIM-6 stimulus harmonics

Carrier frequency range (GHz)	Output power (dBm)	Harmonics, 2nd and above (dBc)	Subharmonics, 2nd (dBc)	Subharmonics, 3rd (dBc)
0.01 - 6	≤ 0	< -30	< -30	< -30

RFIM-6 stimulus phase noise

Specified CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power.

WSRF Stimulus Phase Noise (dBc/Hz)

Frequency [MHz]	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
50	-90	-104	-114	-116	-132
320	-90	-104	-114	-116	-132
545	-103	-117	-126	-130	-143
600	-103	-117	-126	-130	-143
885	-99	-113	-124	-127	-142
1900	-92	-107	-117	-119	-138
2400	-90	-106	-116	-119	-136
5800	-82	-98	-108	-111	-129

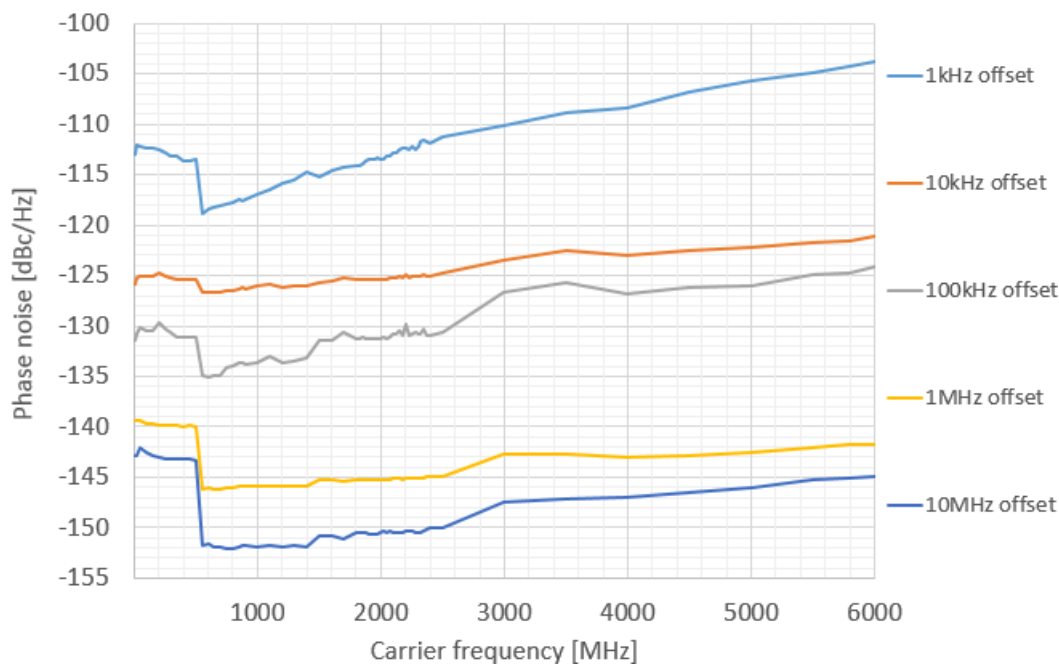
WSRF + WSPS stimulus phase noise max [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-112	-121	-129	-142	-148
600	-112	-121	-129	-142	-148
885	-111	-120	-128	-142	-148
1900	-108	-119	-125	-141	-147
2400	-105	-119	-125	-141	-147
5800	-95	-115	-117	-138	-141

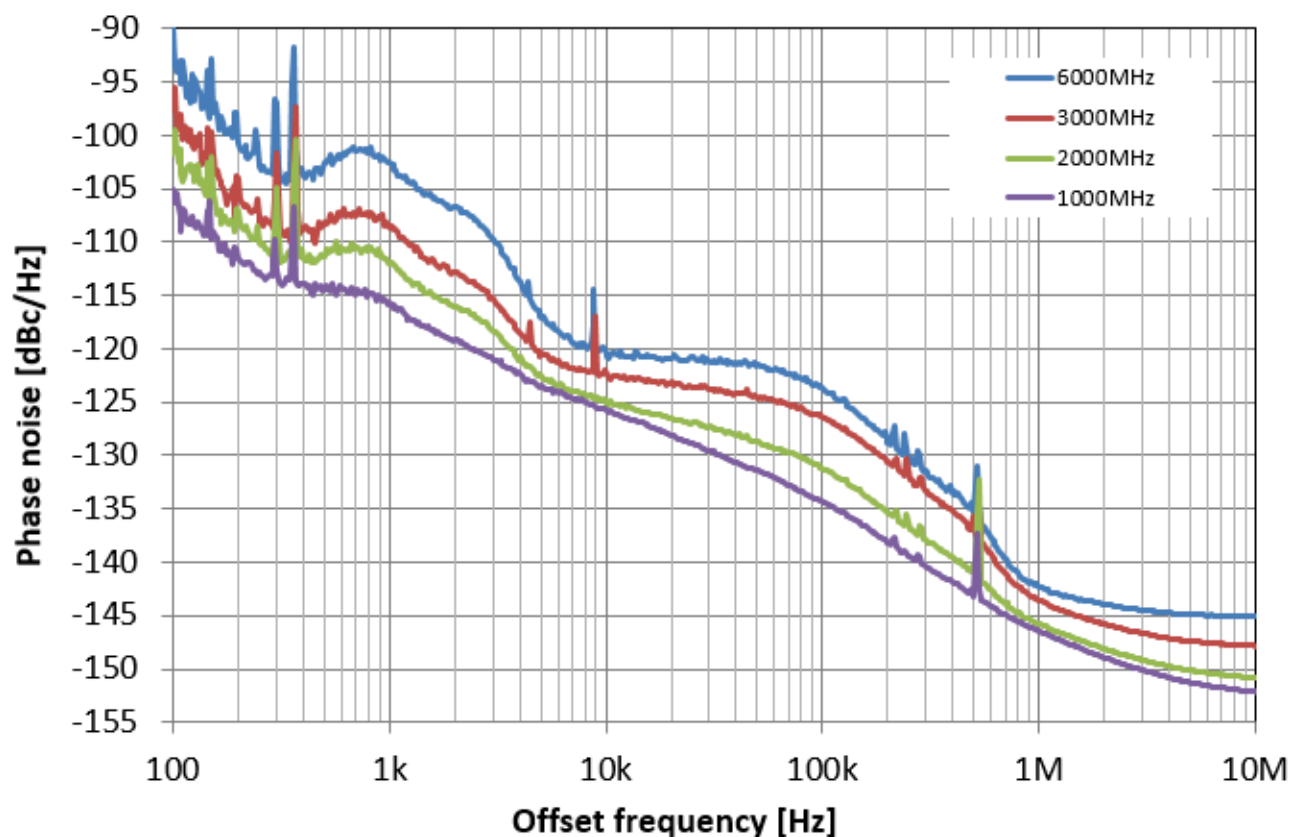
WSRF + WSPS stimulus phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-111	-124	-130	-139	-141
320	-111	-125	-130	-139	-143
545	-118	-126	-134	-145	-151
600	-117	-126	-134	-145	-151
885	-116	-125	-133	-145	-151
1900	-112	-125	-131	-144	-150
2400	-110	-124	-130	-144	-149
5800	-103	-121	-124	-141	-144

The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus carrier frequency for several offset frequencies.



The following graph for WSRF+WSPS shows the typical measured stimulus phase noise versus offset frequency for several carrier frequencies.



RFIM-6 stimulus random jitter

Measured at 0 dBm stimulus power.

WSRF Stimulus random jitter

Frequency (MHz)	RMS jitter bandwidth	Random jitter (fs_{rms}) Typ	Random jitter (fs_{rms}) Max
600-6000	10 kHz to 10 MHz	110	155

Integration bandwidth : 10kHz to 10MHz

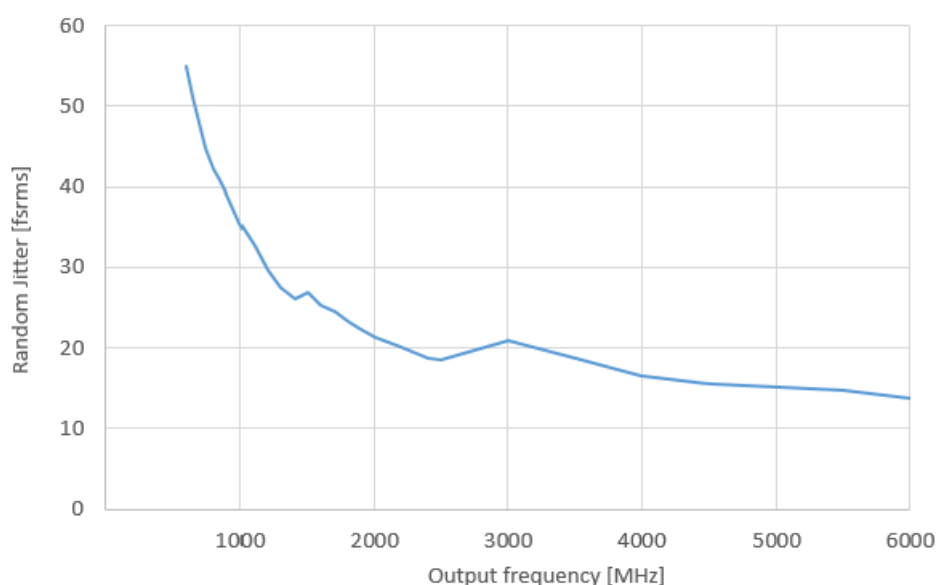
WSRF + WSPS stimulus random jitter max

Frequency [MHz]	Random jitter max [fs_{rms}]
600 $\leq f < 800$	75
800 $\leq f < 1000$	59
1000 $\leq f < 1400$	46
1400 $\leq f < 1600$	37
1600 $\leq f < 2000$	36
2000 $\leq f < 4000$	32
4000 $\leq f < 5000$	26
5000 $\leq f \leq 6000$	22

WSRF + WSPS stimulus random jitter typ

Frequency [MHz]	Random jitter typ [fs_{rms}]
600	56
885	40
1900	23
2400	19
5800	15

The following figure shows the typical measured stimulus random jitter as a function of output carrier frequency.



RFIM-6 stimulus EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB, using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.80%	0.5%
5.6 (-15 dBm to -2 dBm)	0.80%	0.55%

Optimizer: balanced, NoiseBandwidth=80 MHz

802.11ac 160MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.90%	0.65%
5.6 (-15 dBm to -2 dBm)	0.90%	0.7%

Optimizer: balanced, NoiseBandwidth=40 MHz

WSRF + WSPS stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.45%
5.6 (-15 dBm to -2 dBm)	0.55%	0.47%

Optimizer: balanced, NoiseBandwidth=80 MHz

802.11ac 160MHz

Frequency [GHz]	EVM
5.2 (-15 dBm to -2 dBm)	0.7% (max)
5.2 (-11 dBm to -2 dBm)	0.54% (typ)
5.2 (-15 dBm to -11 dBm)	0.62% (typ)
5.6 (-15 dBm to -2 dBm)	0.75% (max)
5.6 (-11 dBm to -2 dBm)	0.58% (typ)
5.6 (-15 dBm to -11 dBm)	0.66% (typ)

Optimizer: balanced, NoiseBandwidth=40 MHz

RFIM-6 stimulus ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	ACLR [dBc]	ACLR2 [dBc]
2 (-30 dBm to +4 dBm)	-65 (typ)	-72 (typ)
2 (-25 dBm to +1 dBm)	-63 (max) ^[4]	-70 (max) ^[4]

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 6dB

WSRF + WSPS stimulus ACLR

^[4] Some baseband frequency offset values might result in dominant spur yielding higher ACLR than spec. To avoid this, change the basebandFrequencyOffset value.

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	ACLR [dBc]	ACLR2 [dBc]
2 (-30 dBm to +4 dBm)	-68 (typ)	-72.5 (typ)
2 (-25 dBm to +1 dBm)	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 6dB

RFIM-6 stimulus noise floor

Stimulus noise floor

CW 0 dBm, optimizer mode setting: noiseFloor, measured at +40 MHz offset relative to carrier frequency

Carrier frequency range (GHz)	Noise-floor (dBm/Hz)
0.05 - 0.545	-138
0.545 - 3	-142
3 - 5	-140
5 - 6	-138

RFIM-6 stimulus spurs

Measured at 0 dBm stimulus power.

Note

Frequencies defined in *RFIM-6 stimulus harmonics* on page 14 are excluded from this specification.

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF-6 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-57
0.55	1.2	-65
1.2	1.7	-62
1.7	2	-60
2	2.05	-59
2.05	2.55	-60
2.55	3.1	-57
3.1	3.15	-53
3.15	3.4	-57
3.4	4	-54
4	4.1	-50

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
4.1	5.4	-53.5
5.4	5.45	-52
5.45	5.65	-54
5.65	5.9	-51
5.9	6	-54

Valid for spur frequency offset above +/- 40MHz

WSRF stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	2.5	-63
2.5	3.5	-60
3.5	4.5	-57
4.5	6	-50

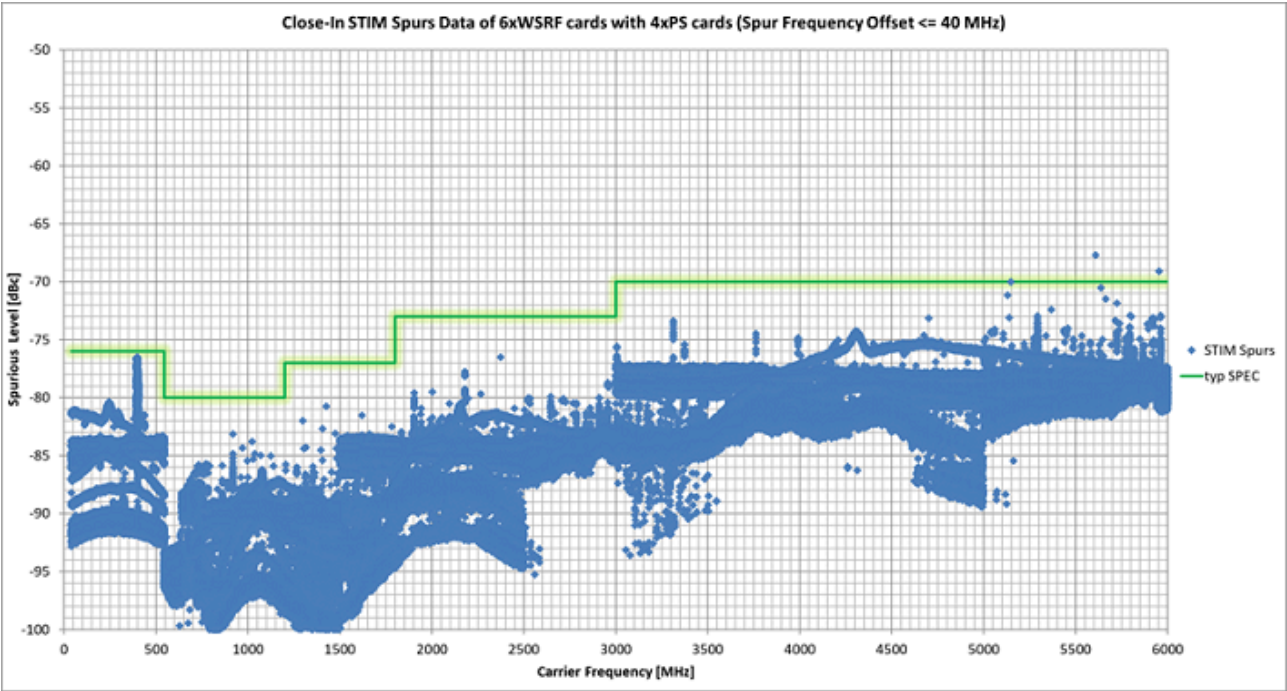
Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF + WSPS stimulus spurs max

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.01	6	-63

WSRF + WSPS stimulus spurs typ

Frequency from [GHz]	Frequency to [GHz]	Typ spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77
1.8	3	-73
3	6	-70



WSPS stimulus nominal spurious performance

Valid for spur frequency offset above +/- 40MHz

WSRF + WSPS stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63

Wave Scale RF-6 with RFIM-6 or RFIM-8 measure

RFIM-6 measure power

Applicable for high resolution ADC only.

Applicable for manual path selection and attenuation settings.

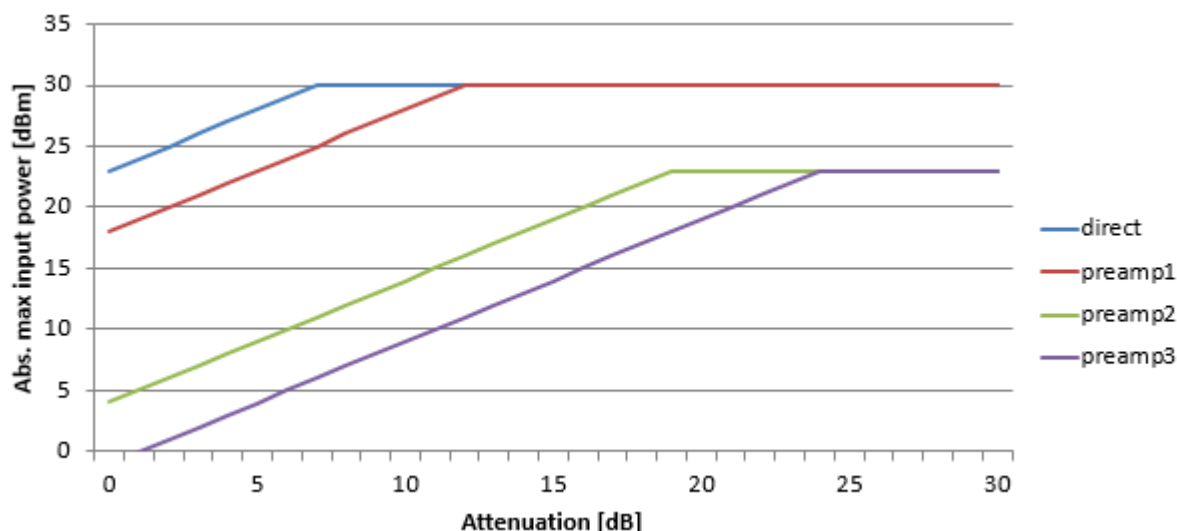
Note

The specified maximum RF input power values below are only valid for the splitch in switch mode.

Maximum RF-6 input power to all ports

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
direct	0.01 to < 0.05	-1	3.0	29
	0.05 to < 1	3	6.9	30
	1 to < 2	5	8.8	30
	2 to < 3	7	10.2	30
	3 to < 4	7	10.7	30

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
	4 to < 5	7	11.2	30
	5 to 6	9	12.3	30
preamp1	0.01 to < 0.05	-10	-6.1	20
	0.05 to < 1	-6	-1.8	24
	1 to < 2	-3	0.3	27
	2 to < 3	-2	1.5	28
	3 to < 4	-1	2.6	29
	4 to < 5	-1	3.0	29
	5 to 6	0	3.5	30
preamp2	0.01 to < 0.1	-18	-14.7	12
	0.1 to < 0.2	-16	-12.1	14
	0.2 to < 1	-15	-12.0	15
	1 to < 2	-13	-9.9	17
	2 to < 3	-12	-8.5	18
	3 to < 4	-11	-7.7	19
	4 to < 5	-11	-7.2	19
preamp3	5 to 6	-11	-6.8	19
	0.01 to < 0.1	-28	-23.7	2
	0.1 to < 0.2	-24	-20.7	6
	0.2 to < 1	-24	-20.5	6
	1 to < 2	-22	-18.4	8
	2 to < 3	-21	-17.0	9
	3 to < 4	-20	-16.1	10
	4 to < 5	-19	-15.5	11
	5 to 6	-19	-15.2	11

Absolute maximum power damage level (dBm)

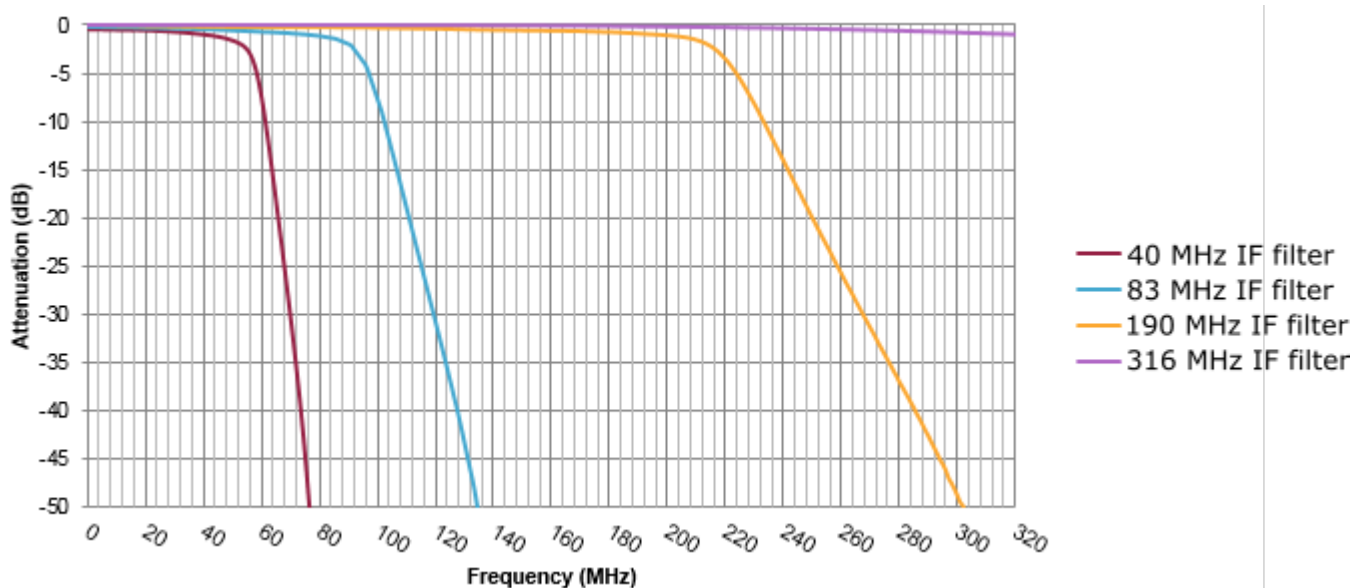
If the expected power value is larger than the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)", SmartTest will issue a warning.

If the expected power value is larger than the "Absolute maximum power damage level(dBm)", SmartTest will issue an error.

Note

When setting *rfmeas actions*, the following aspects should be taken into account:

- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Absolute maximum power damage level (dBm)}$ on page 24 (see figure above).
- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Maximum measurable power with 0dB att, spec(dBm)} + \text{Attenuation(dB)}$ (see table above)
- If the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)" is higher than "Absolute maximum power damage level(dBm)", the latter has priority.

Typical IF filter response within the RF path

RFIM-6 measure accuracy up to 6 GHz

Measure power accuracy conditions:

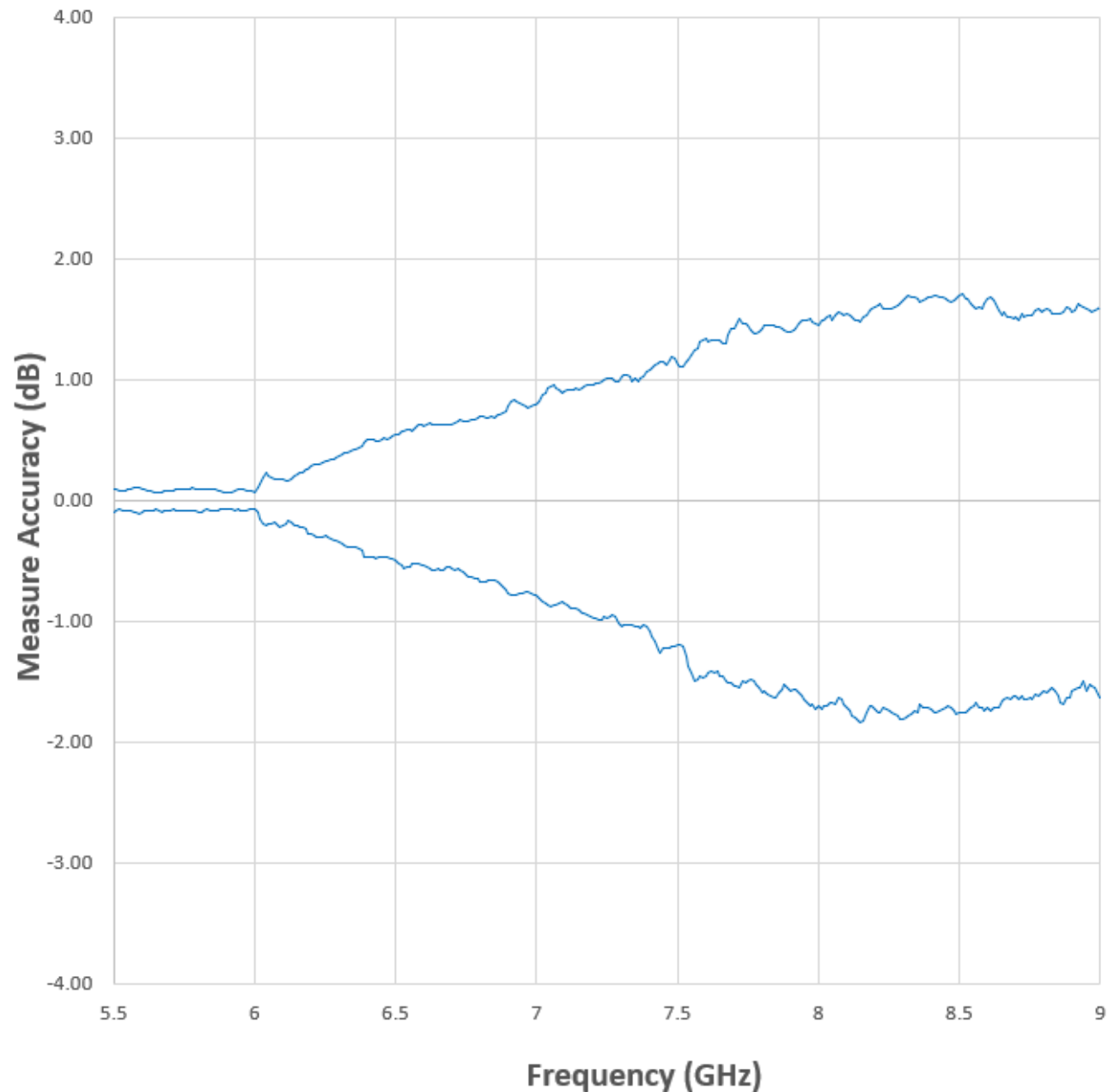
- All power levels are CW signals in a 50 ohm system unless noted.
- Standard deviation index $k=3$ (coverage factor of expanded uncertainty) unless specified differently.
- $\pm 0.5K$ relative water temperature stability after calibration.
- Level accuracy is valid for High Resolution ADC, only.
- To achieve specified level accuracy for very low power levels appropriate bandwidth and result averaging must be applied.

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
$0.050 \leq f < 0.545$	$+1 \leq L_v \leq +20$	± 0.38
	$-8 \leq L_v < +1$	± 0.38
	$-18 \leq L_v < -8$	± 0.38
	$-26 \leq L_v < -18$	± 0.38
	$L_v < -26$	± 0.38
$0.545 \leq f < 1.100$	$+3 \leq L_v \leq +20$	± 0.34
	$-6 \leq L_v < +3$	± 0.34
	$-16 \leq L_v < -6$	± 0.32
	$-24 \leq L_v < -16$	± 0.32
	$L_v < -24$	± 0.32
$1.100 \leq f < 1.990$	$+4 \leq L_v \leq +20$	± 0.36
	$-5 \leq L_v < +4$	± 0.34
	$-15 \leq L_v < -5$	± 0.34
	$-23 \leq L_v < -15$	± 0.34
	$L_v < -23$	± 0.34
$1.990 \leq f < 3.880$	$5 \leq L_v \leq +20$	± 0.40
	$-4 \leq L_v < 5$	± 0.40
	$-14 \leq L_v < -4$	± 0.40
	$-22 \leq L_v < -14$	± 0.40
	$L_v < -22$	± 0.40
$3.880 \leq f < 5.400$	$+6 \leq L_v \leq +20$	± 0.48
	$-3 \leq L_v < +6$	± 0.46
	$-13 \leq L_v < -3$	± 0.48
	$-21 \leq L_v < -13$	± 0.50

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
	$L_v < -21$	± 0.50
$5.400 \leq f \leq 6.000$	$+8 \leq L_v \leq +20$	± 0.42
	$-1 \leq L_v < +8$	± 0.40
	$-11 \leq L_v < -1$	± 0.40
	$-19 \leq L_v < -11$	± 0.40
	$L_v < -19$	± 0.40

RFIM-6 measure accuracy between 6 to 9 GHz

The measure accuracy for frequencies between 6 GHz and 9 GHz is shown below. This "measured" data represents 80% of the distribution of all measured curves.



WSRF measure accuracy between 6 to 9 GHz

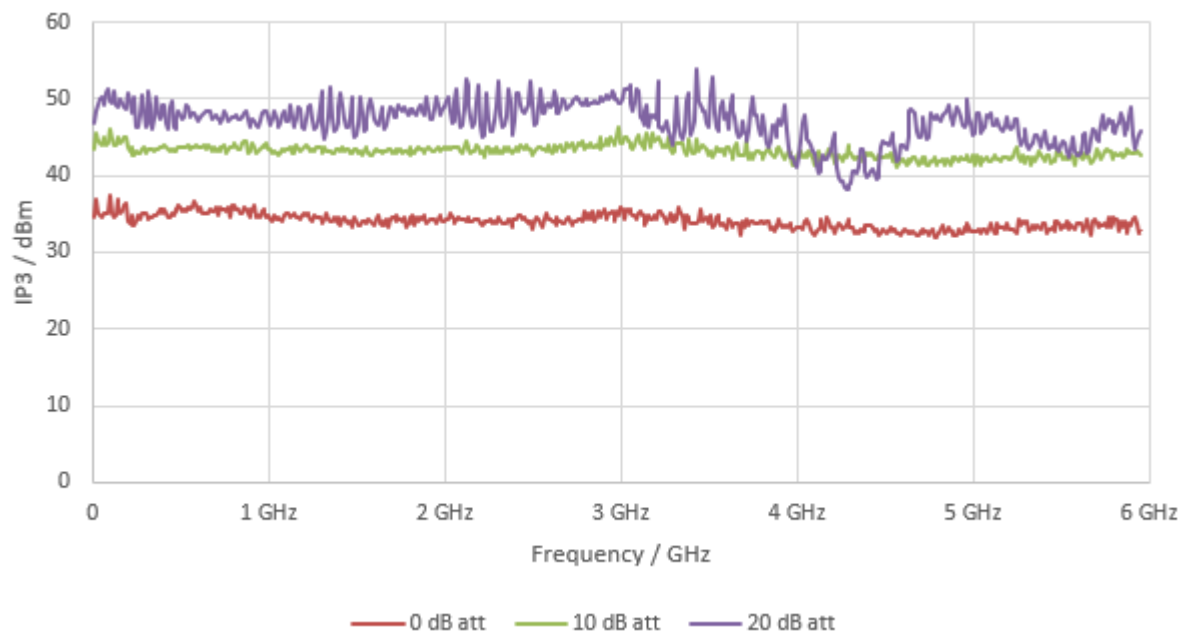
RFIM-6 measure two-tone intermodulation (IP3)

Conditions: HR ADC, 40 MHz filter, $f_{IF} = 0.533$ MHz, $\Delta f = 100$ kHz, -5 dBm tone power, with 0 dB receiver attenuation. IP3 increases with receiver attenuation.

Frequency: 50 MHz - 6 GHz

Measure Path	TOI (dBm)	
	Spec	Typ
direct	24	29
preamp1	15	19
preamp2	5	10

Measure Path	TOI (dBm)	
	Spec	Typ
preamp3	-4	1

Measured data

RFIM-6 measure two-tone intermodulation IP3 measured data

RFIM-6 measure EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM
2 (-40 dBm to +2 dBm)	0.80% spec
2 (-40 dBm to +14 dBm)	0.60% typ

802.11ac 80MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to +0 dBm)	0.80% spec
5.2 (-26 dBm to +10 dBm)	0.65% typ
5.6 (-26 dBm to +0 dBm)	0.90% spec
5.6 (-26 dBm to +10 dBm)	0.71% typ

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to +0 dBm)	1.00% spec

Frequency [GHz]	EVM
5.2 (-26 dBm to +10 dBm)	0.80% typ
5.6 (-26 dBm to +0 dBm)	1.10% spec
5.6 (-26 dBm to +10 dBm)	0.93% typ

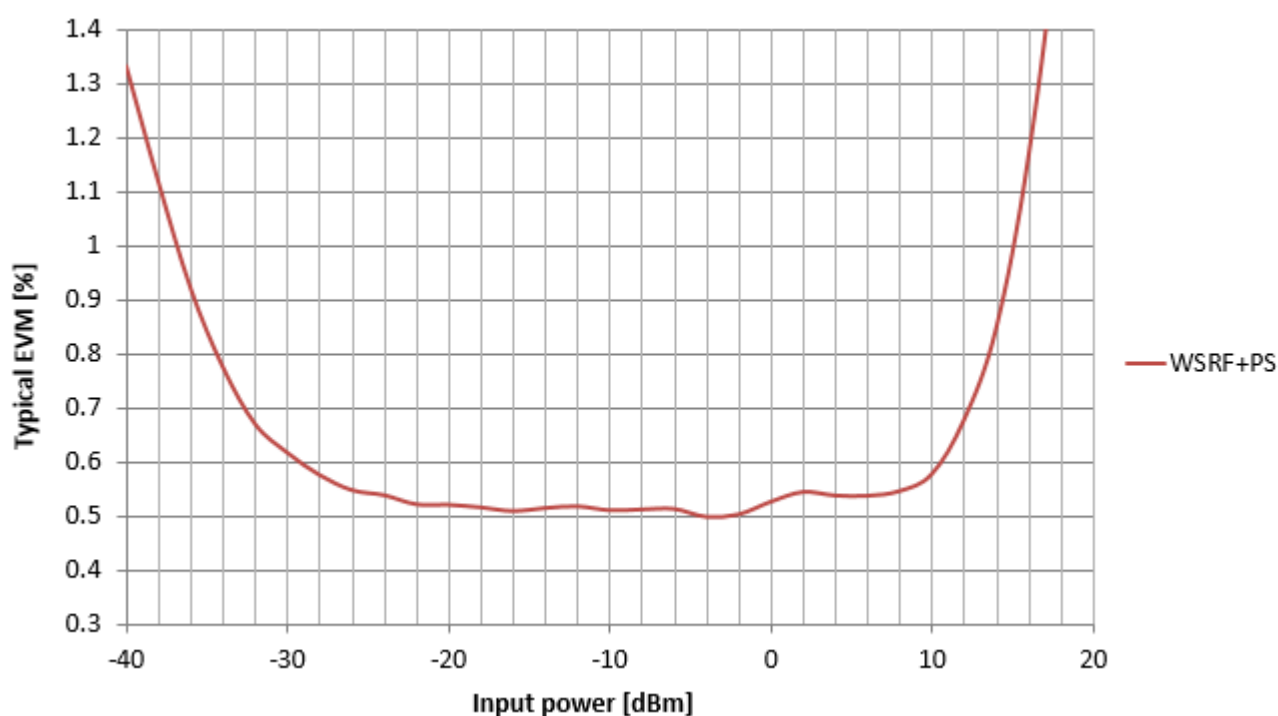
WSRF + WSPS measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM
2 (-40 dBm to +2 dBm)	0.8% (max)
2 (-40 dBm to +14 dBm)	0.54% (typ)

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM
5.2 (-26 dBm to + 2 dBm)	1.0% (max)
5.2 (-26 dBm to +8 dBm)	0.54% (typ)



Wave Scale PS measure typ EVM 802.11ac 160MHz

RFIM-6 measure ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +6 dBm)	-63.0 (max)

Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +14 dBm) ^[5]	-65.7 (typ) ^[5]

WSRF + WSPS measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	ACLR [dBc]
2 (-16 dBm to +6 dBm)	-67 (max)
2 (-16 dBm to +14 dBm)	-69 (typ)

RFIM-6 measure noise floor

Applicable for high resolution ADC only.

Measured at 0 dB receiver attenuation.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
direct	0.01 to < 0.05	-130	-136
	0.05 to < 1	-138	-143
	1 to < 2	-136	-141
	2 to < 3	-135	-141
	3 to < 4	-135	-140
	4 to < 5	-133	-139
	5 to 6	-132	-137
preamp1	0.01 to < 0.05	-139	-144
	0.05 to < 1	-145	-150
	1 to < 2	-144	-149
	2 to < 3	-143	-148
	3 to < 4	-142	-147
	4 to < 5	-141	-146
	5 to 6	-139	-144
preamp2	0.01 to < 0.1	-140	-145
	0.1 to < 0.2	-152	-157

^[5] Some IF value might result in dominant spur which yields to higher ACLR than spec. To avoid this, change IF value.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
	0.2 to < 1	-152	-157
	1 to < 2	-152	-157
	2 to < 3	-152	-156
	3 to < 4	-151	-156
	4 to < 5	-151	-155
	5 to 6	-150	-154
preamp3	0.01 to < 0.1	-142	-146
	0.1 to < 0.2	-153	-157
	0.2 to < 1	-155	-159
	1 to < 2	-155	-159
	2 to < 3	-155	-159
	3 to < 4	-155	-159
	4 to < 5	-154	-158
	5 to 6	-154	-157

RFIM-6 measure dynamic range

Applicable for high resolution ADC only.

Note

Dynamic range is the difference between noise floor (in dBm/Hz) and the maximum measurable power. It applies for 1 Hz measurement bandwidth and the dynamic range decreases with increasing bandwidth as $10\log(\text{bandwidth})$.

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
direct	0.01 to < 0.05	129	139
	0.05 to < 1	141	149.8
	1 to < 2	141	149.8
	2 to < 3	142	151.1
	3 to < 4	142	150.7
	4 to < 5	140	150.2
	5 to 6	141	149.3
preamp1	0.01 to < 0.05	129	137.9
	0.05 to < 1	139	148.2

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	1 to < 2	141	149.2
	2 to < 3	141	149.5
	3 to < 4	141	149.6
	4 to < 5	140	149
	5 to 6	139	147.4
preamp2	0.01 to < 0.1	122	130.3
	0.1 to < 0.2	136	144.9
	0.2 to < 1	137	144.9
	1 to < 2	139	147.1
	2 to < 3	140	147.4
	3 to < 4	140	148.2
	4 to < 5	140	147.7
	5 to 6	139	147.2
preamp3	0.01 to < 0.1	114	122.2
	0.1 to < 0.2	129	136.2
	0.2 to < 1	131	138.5
	1 to < 2	133	140.6
	2 to < 3	134	141.9
	3 to < 4	135	142.9
	4 to < 5	135	142.5
	5 to 6	135	141.7

RFIM-6 measure spurs

Measured at 0dBm stimulus power

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

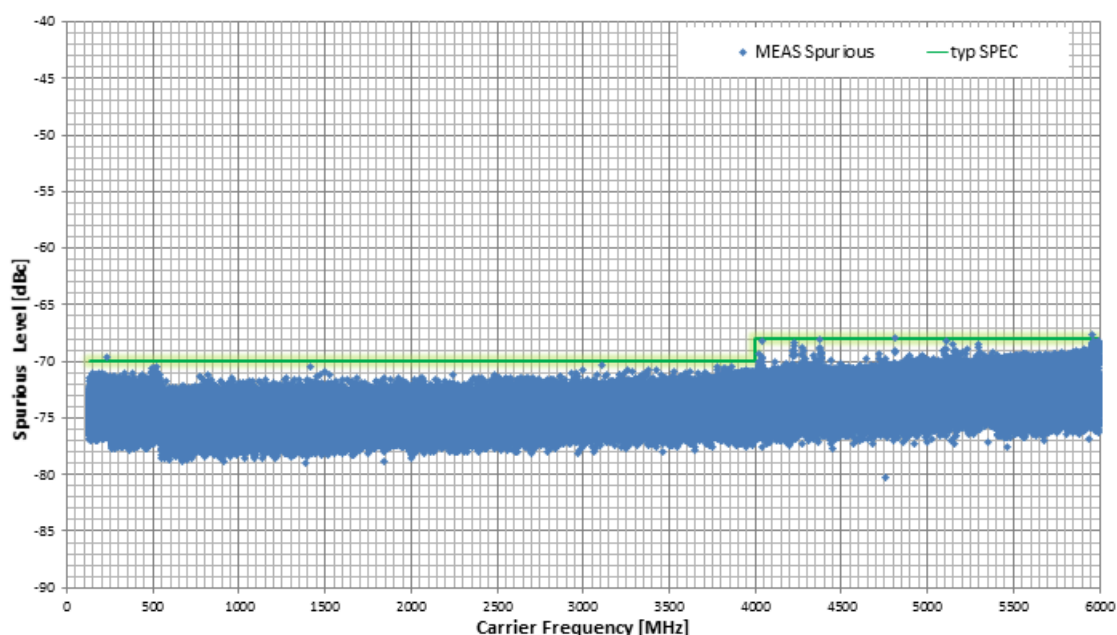
WSRF-6 measure spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
0.13	1	-67
1	1.2	-64
1.2	1.6	-61.5

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
1.6	1.7	-54
1.7	3.4	-56.5
3.4	4	-53
4	4.1	-51
4.1	5.15	-53
5.15	5.25	-52.5
5.25	5.9	-50.5
5.9	6	-53

WSRF + WSPS measure spurs

Input frequency from [GHz]	Input frequency to [GHz]	Typ spurs level (<0.01%) [dBc]
0.13	4	-70
4	6	-68



WS PS measure typical spurious performance

Wave Scale RF-6 with RFIM-6 or RFIM-8 terminate

RF ports that are turned-off, have an internal 50-Ohm termination to GND. In order to avoid thermal damage to the termination resistors, the following abs-max restrictions have to be observed per logical board (half of RFIM).

Number of ports in termination mode	Maximum power applied per port (dBm)
1 or 2	30

Number of ports in termination mode	Maximum power applied per port (dBm)
3 to 9	27
10 or more	26

RFIM-8 attached to Wave Scale RF card

Following hardware components are part of the specification sections:

- *WSRF 8 GHz (N2600B-OP) with RFIM-8 (N2604B)*

Note In case WSPS is included, it must be WSPS 8 GHz (N2601A-OP).

Note Mixing of WSRF 6 GHz (N2600A-OP) and WSRF 8 GHz (N2600B-OP) cards in one system configuration is not supported.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

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▲ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

▲ Warning A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

Wave Scale RF-8 with RFIM-8 specifications

The RF test solution with an RFIM-8 module attached to a WSRF-8 card provides the capabilities to generate and process RF signals with a frequency coverage from 10 MHz to 8 GHz.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

WS system calibration

WSRF-8 system calibration interval	Stim	Measure
3 months	0.2dB (typ) off vs calibration lot additional 0.15dB off in the first month for new produced cards	0.15dB (typ) off vs calibration lot additional 0.3dB off in the first month for new produced cards

Wave Scale RF-8 with RFIM-8 stimulus

RFIM-8 stimulus maximum output power

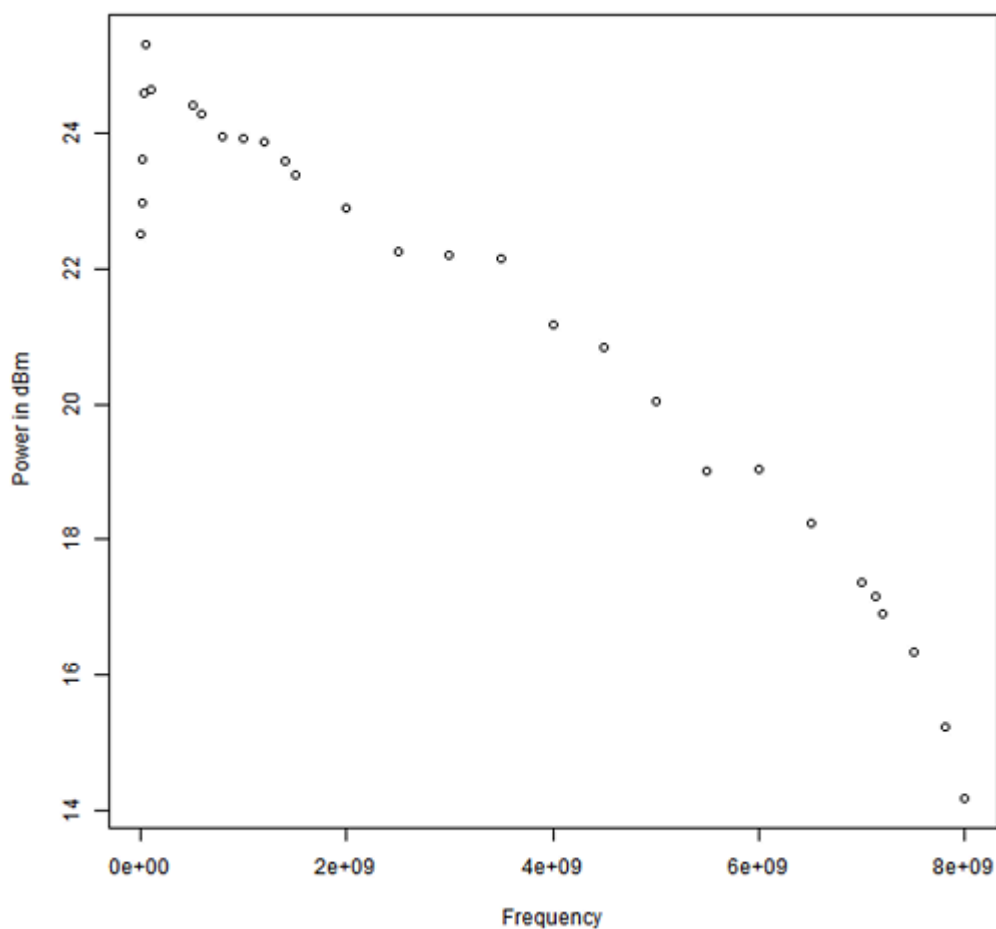
Shows absolute maximum output power, measured as saturated power.

It is not recommended to use the Wave Scale Precision Source for power levels less than -80 dBm.

Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
0.011	18.4	22.8
0.016	19.6	23.3
0.031	20.6	24.3
0.051	21.2	24.9
0.1	20.7	24.3
1	20	23.8
2	18.9	22.7
2.5	18.4	22.2
3	17.9	21.8
4	16.7	20.5

Carrier frequency (GHz)	Maximum output power (dBm) Spec	Maximum output power (dBm) Typ
5	16.4	20.3
6	15.1	19.3
7	13.3	16.8
7.125	12.3	16.7
7.5	10.7	15.8
8	9.7	14.2

Wave Scale RF-8 stimulus typical maximum power (average)



RFIM-8 stimulus modulation flatness

The measured performance for Wave Scale RF-8 stimulus modulation flatness is shown in the table below.

Carrier frequency range (GHz)	Maximum modulation bandwidth (MHz)	10 MHz	20 MHz	40 MHz	90 MHz	100 MHz	200 MHz
0.06 - 0.1	10	±0.15					
0.1 - 0.96	90	±0.15	±0.3	±0.5	±1.2		

Carrier frequency range (GHz)	Maximum modulation bandwidth (MHz)	10 MHz	20 MHz	40 MHz	90 MHz	100 MHz	200 MHz
0.96 - 3	200	±0.1	±0.2	±0.3		±0.8	±1.3
3 - 8	200	±0.1	±0.2	±0.4		±1.2	±1.7

RFIM-8 stimulus accuracy

Stimulus power accuracy

Power range (dBm)	-110...	-100...	-90...	-80...	-70...	-60...	-50...	-40...	-30...	-20...	-10...	0...5
Frequency range (GHz)	0.05...	0.1...	0.2...	0.3...	0.4...	0.5...	0.6...	0.7...	0.8...	0.9...	1.0...	1.1...
0.05...	±0.73	±0.64	±0.56	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49	±0.49
1...2	±0.74	±0.66	±0.58	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52	±0.52
2...3	±0.76	±0.76	±0.67	±0.60	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53	±0.53
3...4	±1.05 [6]	±1.14	±0.86	±0.62	±0.62	±0.62	±0.56	±0.56	±0.56	±0.56	±0.56	±0.56
4...5	±1.21 [6]	±0.64 [6]	±0.87	±0.62	±0.62	±0.62	±0.56	±0.56	±0.56	±0.56	±0.56	±0.56
5...6	±1.89 [6]	±0.79 [6]	±0.89	±0.65	±0.65	±0.65	±0.58	±0.58	±0.58	±0.58	±0.58	±0.58
6...7	±1.77 [6]	±0.82 [6]	±1.11	±0.76	±0.76	±0.76	±0.62	±0.62	±0.62	±0.62	±0.62	±0.62
7...7.5		±1.73 [6]	±1.21	±0.85	±0.78	±0.78	±0.65	±0.65	±0.65	±0.65	±0.65	±0.65
7.5...8			±1.44 [6]	±1.12	±0.99	±0.91	±0.76	±0.76	±0.76	±0.76	±0.71	±0.76

RFIM-8 stimulus tone-combiner TOI (third-order intercept)

Conditions ^[7]: Tone-combiner mode, 100kHz tone spacing, optimizer balanced, 10 kHz Noise-Bandwidth.

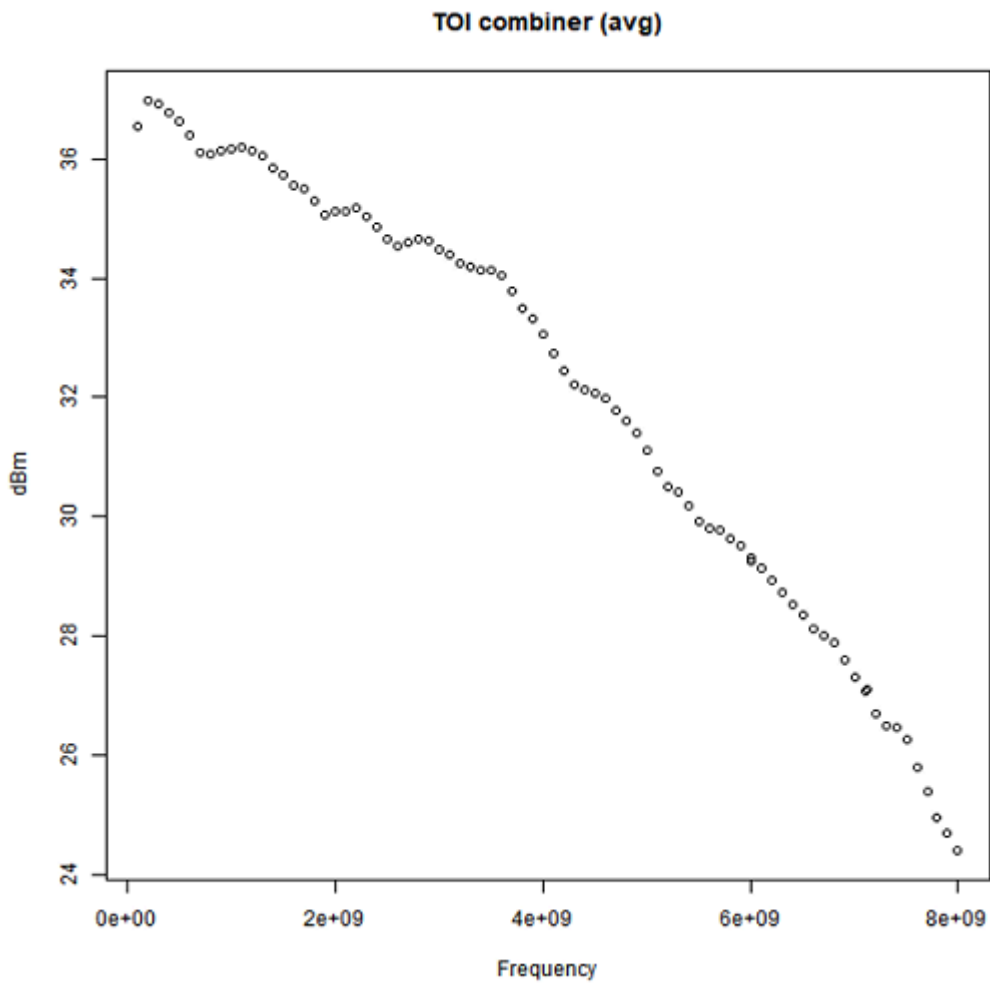
Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.1	0	30	36.2
0.2	0	30	36.8
0.5	0	30	36.4
1	0	29	36.0
2	0	29	34.9
3	0	28	34.2

^[6] Typical

^[7] The maximum power delta between the two tones that the WSRF-8 card can guarantee is 60 dB for the above conditions.

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
4	0	27	32.6
5	0	25	30.8
6	0	23	29
7	0	21.6	27
7.5	0	19.6	25.7
8	0	17.6	23.9

Wave Scale RF-8 stimulus typical TOI tone-combiner mode (average)



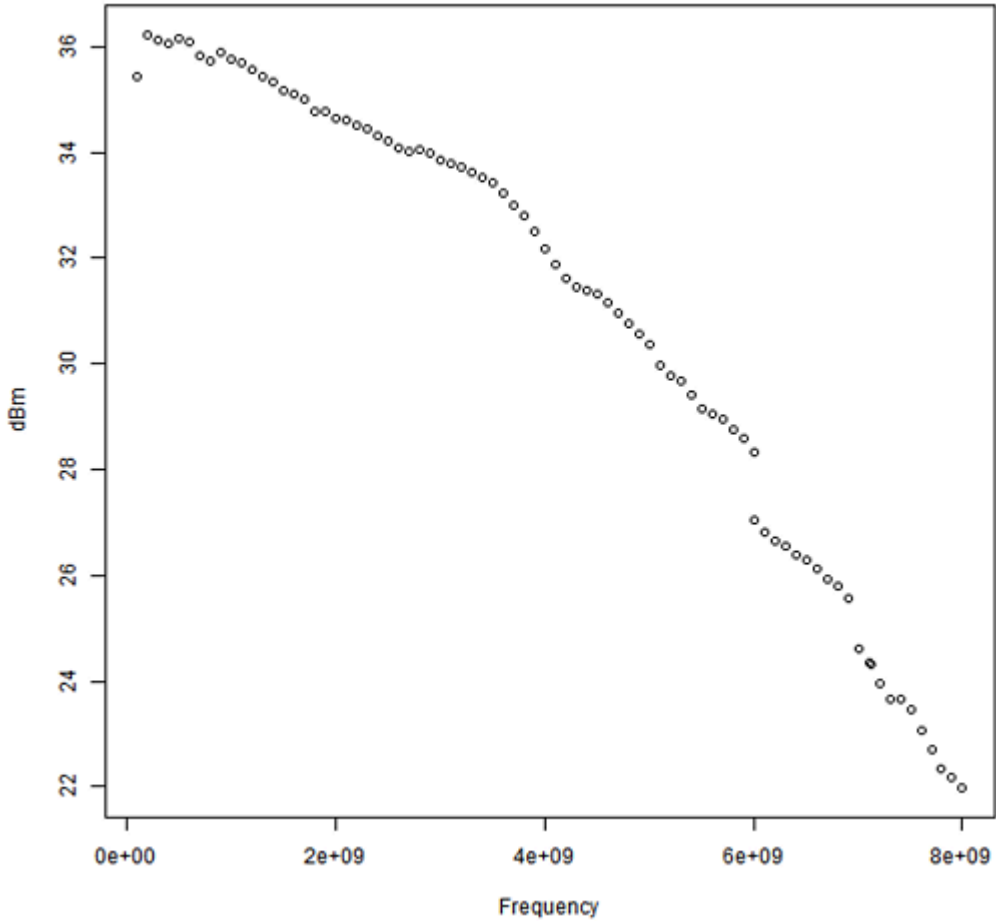
RFIM-8 stimulus modulated TOI (third-order intercept)

IQ-modulated 2-tone signal, tone-spacing 100kHz ^[8]

^[8] 2-tone IQ Signal, 250Msps, (tones at -10MHz, -10.1MHz, 0dBm or -3dBm each tone power), total power +3dBm or 0dBm, optimizer balanced, 10kHz Noise-Bandwidth

Carrier frequency (GHz)	Each tone power output at test port (dBm)	TOI (dBm) Spec	TOI (dBm) Typ
0.5	0	29.3	35.4
1	0	29.3	35.1
2	0	27.8	34.2
3	0	27.8	33.5
4	0	25.8	31.7
5	0	23.8	30.0
6	0	21.6	26.6
7	-3	18.6	24.2
7.5	-3	16.6	23.0
8	-3	14.6	21.5

Wave Scale RF-8 stimulus typical TOI modulated (average)



RFIM-8 stimulus harmonics

CW Pout≤0dBm, Optimizer mode IMD

Carrier frequency range (GHz)	Output power (dBm)	Harmonics, 2nd and above (dBc)	Subharmonics, 2nd (dBc)	Subharmonics, 3rd (dBc)
0.01-8	≤0	< -30	< -30	< -30

RFIM-8 stimulus phase noise

Specified CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power.

WSRF-8 stimulus phase noise [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-90	-104	-114	-116	-132
320	-90	-104	-114	-116	-132
545	-103	-117	-126	-130	-143
600	-103	-117	-126	-130	-143
885	-99	-113	-124	-127	-142
1900	-92	-107	-117	-119	-138
2400	-90	-106	-116	-119	-136
5800	-82	-98	-108	-111	-129
7000	-80	-96	-106	-109	-127

WSRF-8 + WSPS-8 stimulus phase noise [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-104	-119	-124	-132	-134
320	-104	-119	-124	-132	-134
545	-111	-120	-129	-142	-148
600	-111	-120	-129	-142	-148
885	-110	-120	-128	-142	-148
1900	-107	-119	-125	-141	-147
2400	-105	-118	-124	-141	-147
5800	-94	-115	-117	-138	-141
7000	-94	-112	-116	-135	-139

WSRF-8 + WSPS-8 stimulus phase noise typ [dBc/Hz]

Frequency [MHz]	Offset frequency				
	1kHz	10kHz	100kHz	1MHz	10MHz
50	-113.5	-124.9	-131.3	-139.7	-142.4
320	-113.5	-124.9	-131.1	-140.1	-143.3
545	-116.9	-126.2	-136.7	-146.9	-152.6
600	-116.9	-126.3	-136.8	-146.9	-152.5
885	-116.1	-125.5	-135	-146.3	-152
1900	-113.9	-124.9	-132	-145.7	-150.9
2400	-112.4	-124.5	-131.5	-145.5	-150.4
5800	-106.3	-119.4	-124.6	-141.7	-145.2
7000	-104.5	-117.7	-121.2	-139.5	-143.2

RFIM-8 stimulus random jitter

Measured at 0 dBm stimulus power.

WSRF-8 Stimulus random jitter

Frequency (MHz)	RMS jitter bandwidth	Random jitter (fs_{rms}) Typ	Random jitter (fs_{rms}) Max
600-7000	10 kHz to 10 MHz	107	173

Integration bandwidth : 10kHz to 10MHz

WSRF-8 + WSPS-8 stimulus random jitter max

Frequency [MHz]	Random jitter max [fs_{rms}]
600 <= f < 800	83
800 <= f < 1000	64
1000 <= f < 1400	53
1400 <= f < 1600	40
1600 <= f < 2000	36
2000 <= f < 2500	33
2500 <= f < 4000	31
4000 <= f < 5000	24
5000 <= f < 6000	21
6000 < f <= 7000	33

WSRF-8 + WSPS-8 stimulus random jitter typ

Frequency [MHz]	Random jitter typ [$f_{s_{rms}}$]
600	54
885	41
1900	23
2400	19
5800	15
7000	18

RFIM-8 stimulus EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.5%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB, using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.8%	0.5%
5.6 (-15 dBm to -2 dBm)	0.8%	0.55%

Optimizer: balanced, NoiseBandwidth=20 MHz

802.11ac 160MHz

Frequency [GHz]	EVM	EVM typ
5.2 (-15 dBm to -2 dBm)	0.9%	0.61%
5.6 (-15 dBm to -2 dBm)	0.9%	0.67%

Optimizer: balanced, NoiseBandwidth=10 MHz

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -6	0.77%	-42.27
	-5 < to 1	0.84%	-41.51

Optimizer: balanced, NoiseBandwidth=9 MHz

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -5	0.92%	-40.72
	-5 < to 1	0.95%	-40.45

Optimizer: balanced, NoiseBandwidth=20 MHz

WSRF-8 + WSPS-8 stimulus EVM

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM max	EVM typ
2 (-25 dBm to +10 dBm)	0.8%	0.52%

Optimizer: balanced, NoiseBandwidth= 80 MHz, baseband attenuation 0dB , using a baseband frequency offset of 20 MHz.

Loopback configuration: measurement performed with the measure path of the Wave Scale RF card using the Wave Scale Precision Source as a receiver LO.

802.11ac 80MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-15 dBm to -2 dBm)	0.55%	0.4%
5.6 (-15 dBm to -2 dBm)	0.55%	0.42%

Optimizer: balanced, NoiseBandwidth=20 MHz

802.11ac 160MHz

Frequency [GHz]	EVM max	EVM typ
5.2 (-11 dBm to -2 dBm)	0.7%	0.53%
5.2 (-15 dBm to -11 dBm)	0.7%	0.54%
5.6 (-11 dBm to -2 dBm)	0.75%	0.56%
5.6 (-15 dBm to -11 dBm)	0.75%	0.57%

Optimizer: balanced, NoiseBandwidth=10 MHz

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83
	-30 to < -25	0.59%	-44.58
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

Optimizer: balanced, NoiseBandwidth=9 MHz

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

Optimizer: balanced, NoiseBandwidth=20 MHz

RFIM-8 stimulus ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, Center frequency: 2 GHz

Power level	ACLR [dBc]	ACLR2 [dBc]
-30 dBm to +4 dBm	-65.1 (typ)	-72 (typ)
-25 dBm to +1 dBm	-63 (max) ^[9]	-70 (max) ^[9]

Optimizer: balanced, NoiseBandwidth= 60 MHz, baseband attenuation 6 dB, baseband frequency offset 0 Hz

WSRF-8 + WSPS-8 stimulus ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, Center frequency: 2 GHz

Power level	ACLR [dBc]	ACLR2 [dBc]
-30 dBm to +4 dBm	-67.6 (typ)	-72.7 (typ)
-25 dBm to +1 dBm	-66.5 (max)	-71.5 (max)

Optimizer: balanced, NoiseBandwidth= 60 MHz, baseband attenuation 6 dB

^[9] Some baseband frequency offset values might result in dominant spur yielding higher ACLR than spec. To avoid this, change the basebandFrequencyOffset value.

RFIM-8 stimulus noise floor

CW 0 dBm, optimizer mode setting: noiseFloor, measured at +40 MHz offset relative to carrier frequency

Carrier frequency range (GHz)	Noise-floor (dBm/Hz)
0.05 - 0.545	-138
0.545 - 3.0	-142
3.0 - 5.0	-140
5.0 - 6.0	-138
6.0 - 8.0	-135

RFIM-8 stimulus spurs

Measured at 0 dBm stimulus power

Note

Frequencies defined in *RFIM-8 stimulus harmonics* on page 42 are excluded from this specification.

Valid for spur frequency offset ranging from +/-10kHz to +/-100MHz

WSRF-8 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-57
0.55	1.2	-65
1.2	1.7	-62
1.7	2	-60
2	2.05	-59
2.05	2.55	-60
2.55	3.1	-57
3.1	3.15	-53
3.15	3.4	-57
3.4	4	-54
4	4.1	-50
4.1	5.4	-53.5
5.4	5.45	-52
5.45	5.65	-54
5.65	6.9	-51
6.9	8	-50

Valid for spur frequency offset above +/-40MHz

WSRF-8 stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	2.5	-63
2.5	3.5	-60
3.5	4.5	-57
4.5	6	-50
6	7	-46
7	7.4	-44
7.4	7.8	-40
7.8	7.9	-38
7.9	8	-30

Valid for spur frequency offset ranging from +/-10kHz to +/-100MHz

WSRF-8 + WSPS-8 stimulus spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.04	0.55	-76
0.55	1.2	-80
1.2	1.8	-77
1.8	3	-73
3	6	-70
6	8	-68

Valid for spur frequency offset above +/-40MHz

WSRF-8 + WSPS-8 stimulus spurs (spec)

Frequency from [GHz]	Frequency to [GHz]	Max spur [dBc]
0.1	6	-63
6	7	-63
7	7.4	-60
7.4	7.8	-54
7.8	8	-48

Wave Scale RF-8 with RFIM-8 measure

RFIM-8 measure power

Applicable for high resolution ADC only.

Applicable for manual path selection and attenuation settings.

Note

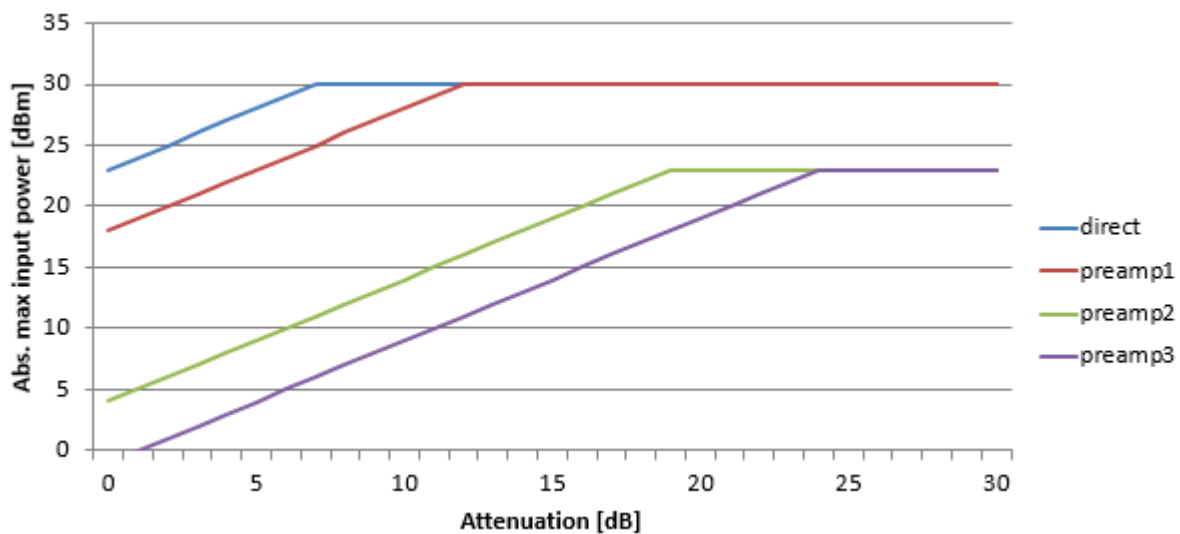
The specified maximum RF input power values below are only valid for the splitch in switch mode.

Maximum RF-8 input power to all ports

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
direct	0.01 to < 0.05	-4	-0.3	26
	0.05 to < 1	3	6.4	30
	1 to < 2	5	8.7	30
	2 to < 3	7	10	30
	3 to < 4	7	10.5	30
	4 to < 5	7	11.2	30
	5 to < 6	9	12.1	30
	6 to 8	9	14.6	30
preamp1	0.01 to < 0.05	-13	-8.9	17
	0.05 to < 1	-6	-1.8	24
	1 to < 2	-4	0	26
	2 to < 3	-2	1.5	28
	3 to < 4	-1	2.2	29
	4 to < 5	-1	3	29
	5 to < 6	0	3.4	30
	6 to 8	0	7.4	30
preamp2	0.01 to < 0.05	-21	-17.1	9
	0.05 to < 1	-16	-12.1	14
	1 to < 2	-14	-10.2	16
	2 to < 3	-12	-8.6	18
	3 to < 4	-12	-7.9	18
	4 to < 5	-11	-7	19

Measure Path	Receiver frequency range (CW) (GHz)	Maximum measurable power (dBm) with		
		0 dB att spec	0 dB att typ	30 dB att spec
preamp3	5 to <6	-11	-6.3	19
	6 to 8	-11	-3.3	19
	0.01 to < 0.05	-30	-25.9	0
	0.05 to < 1	-24	-20.8	6
	0.2 to < 1	-22	-18.9	8
	2 to < 3	-21	-17.2	9
	3 to < 4	-20	-16.3	10
	4 to < 5	-19	-15.3	11
	5 to < 6	-18	-15	12
	6 to 8	-18	-10.9	12

Absolute maximum power damage level (dBm)



If the expected power value is larger than the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)", SmartTest will issue a warning.

If the expected power value is larger than the "Absolute maximum power damage level(dBm)", SmartTest will issue an error.

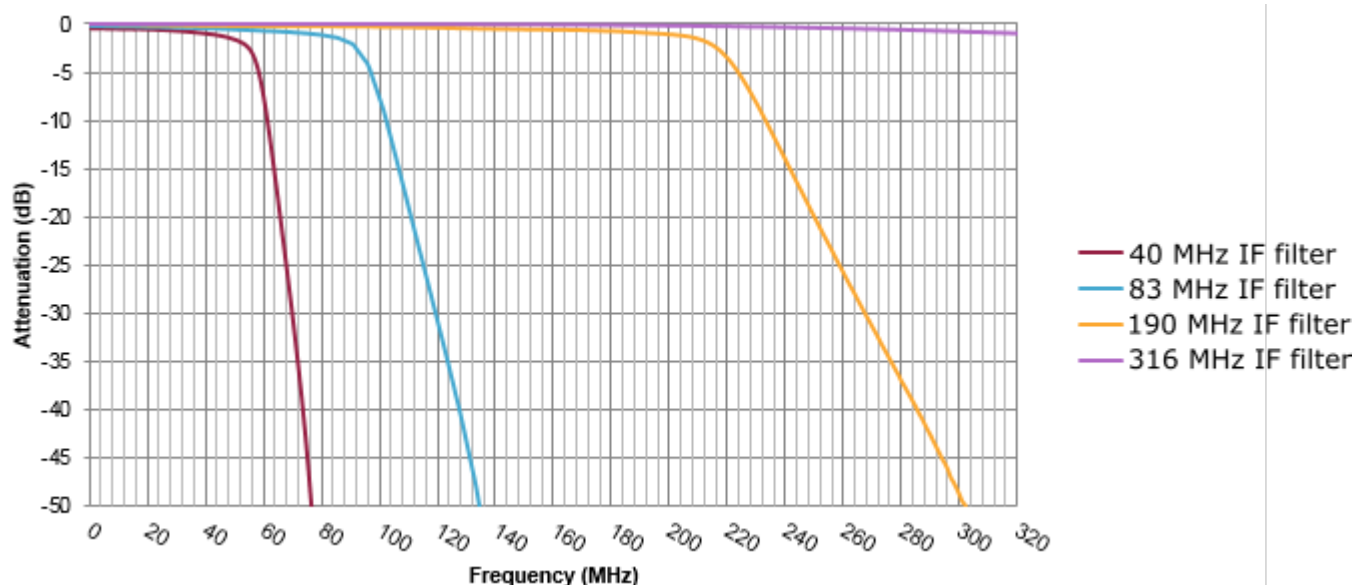
Note

When setting *rfmeas actions*, the following aspects should be taken into account:

- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Absolute maximum power damage level (dBm)}$ on page 50 (see figure above).
- $\text{expectedMaxPower} + \text{peakToAvgPowerRatio} < \text{Maximum measurable power with 0dB att, spec(dBm)} + \text{Attenuation(dB)}$ (see table above)

- If the sum of "Maximum measurable power with 0dB att, spec(dBm)" and "Attenuation(dB)" is higher than "Absolute maximum power damage level(dBm)", the latter has priority.

Typical IF filter response within the RF path



RFIM-8 measure accuracy up to 8 GHz

Measure power accuracy conditions:

- Level accuracy is valid for CW signals in a 50 ohm system.
- Level accuracy is valid for High Resolution ADC, only.
- To achieve specified level accuracy for very low power levels appropriate bandwidth and result averaging must be applied.
- Specifications apply after a 30-minute warm-up and executions of all recommended WSRF calibrations.

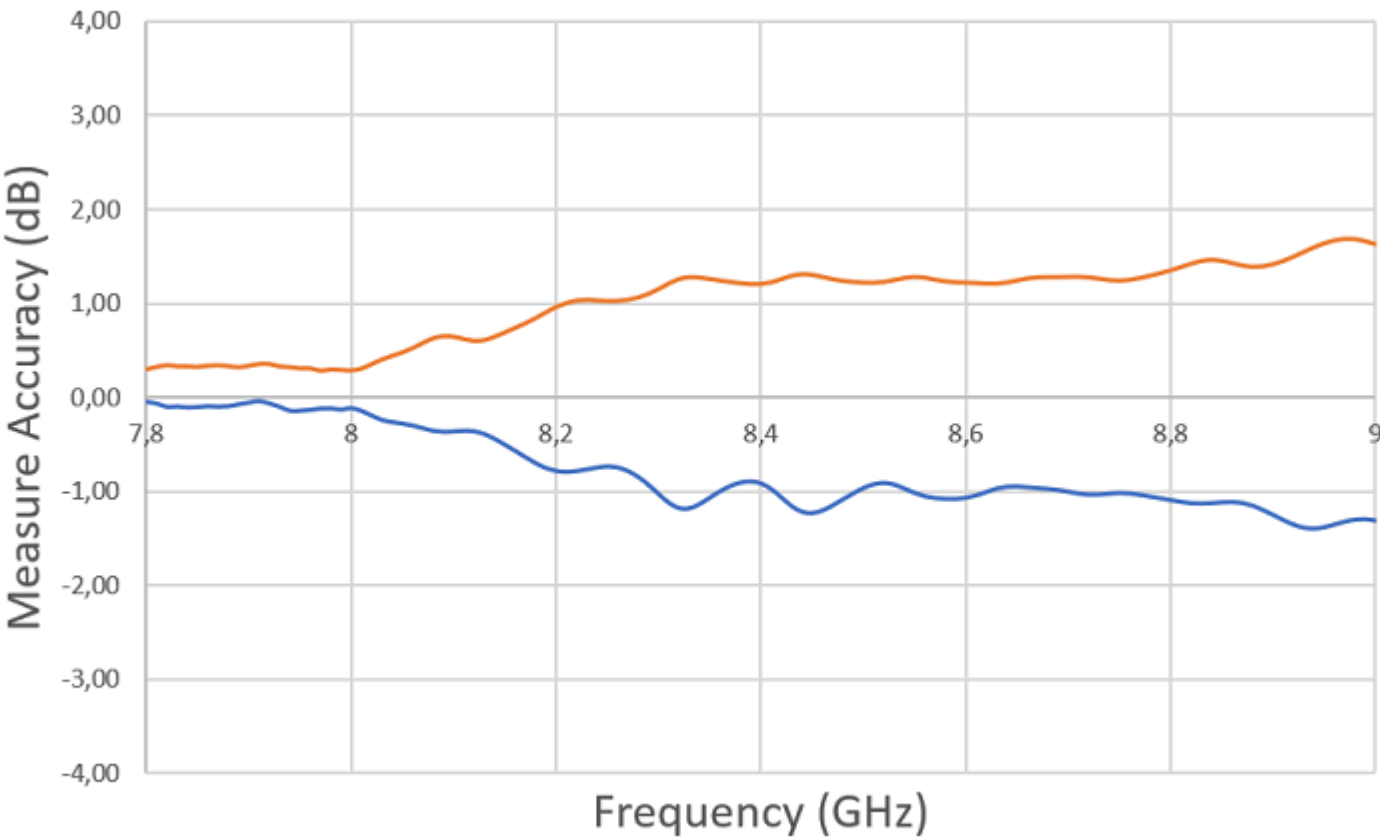
Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
$0.050 \leq f < 0.545$	$-1 \leq L_v \leq +20$	± 0.37
	$-10 \leq L_v < -1$	± 0.37
	$-18 \leq L_v < -10$	± 0.36
	$-28 \leq L_v < -18$	± 0.37
	$L_v < -28$	± 0.37
$0.545 \leq f < 1.100$	$+3 \leq L_v \leq +20$	± 0.33
	$-6 \leq L_v < +3$	± 0.33
	$-16 \leq L_v < -6$	± 0.33
	$-24 \leq L_v < -16$	± 0.32
	$L_v < -24$	± 0.32
$1.100 \leq f < 1.990$	$+4 \leq L_v \leq +20$	± 0.34

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
	$-5 \leq L_v < +4$	± 0.34
	$-15 \leq L_v < -5$	± 0.34
	$-23 \leq L_v < -15$	± 0.34
	$L_v < -23$	± 0.34
$1.990 \leq f < 3.880$	$5 \leq L_v \leq +20$	± 0.40
	$-4 \leq L_v < 5$	± 0.40
	$-14 \leq L_v < -4$	± 0.40
	$-22 \leq L_v < -14$	± 0.40
	$L_v < -22$	± 0.40
$3.880 \leq f < 5.400$	$+6 \leq L_v \leq +20$	± 0.44
	$-3 \leq L_v < +6$	± 0.43
	$-13 \leq L_v < -3$	± 0.43
	$-21 \leq L_v < -13$	± 0.43
	$L_v < -21$	± 0.43
$5.400 \leq f \leq 6.000$	$+8 \leq L_v \leq +20$	± 0.42
	$-1 \leq L_v < +8$	± 0.41
	$-11 \leq L_v < -1$	± 0.41
	$-19 \leq L_v < -11$	± 0.41
	$L_v < -19$	± 0.41
$6.000 < f \leq 7.000$	$+10 \leq L_v \leq +20$	± 0.45
	$+2 \leq L_v < +10$	± 0.45
	$-8 \leq L_v < +2$	± 0.45
	$-15 \leq L_v < -8$	± 0.45
	$L_v < -15$	± 0.45
$7.000 < f \leq 7.200$	$+10 \leq L_v \leq +20$	± 0.46
	$+2 \leq L_v < +10$	± 0.46
	$-17 \leq L_v < +2$	± 0.46
	$-16 \leq L_v < -7$	± 0.46
	$L_v < -16$	± 0.46

Input frequency (GHz)	Input level (dBm)	Measure accuracy (dB)
7.200 < f ≤ 7.500	+10 ≤ Lv ≤ +20	±0.50
	+2 ≤ Lv < +10	±0.50
	-17 ≤ Lv < +2	±0.50
	-16 ≤ Lv < -7	±0.50
	Lv < -16	±0.50
7.500 < f ≤ 8.000	+10 ≤ Lv ≤ +20	±0.57
	+2 ≤ Lv < +10	±0.58
	-17 ≤ Lv < +2	±0.57
	-16 ≤ Lv < -7	±0.58
	Lv < -16	±0.58

RFIM-8 measure accuracy between 8 to 9 GHz

The measure accuracy for frequencies between 8 GHz and 9 GHz is shown below. This "measured" data represents 80% of the distribution of all measured curves.



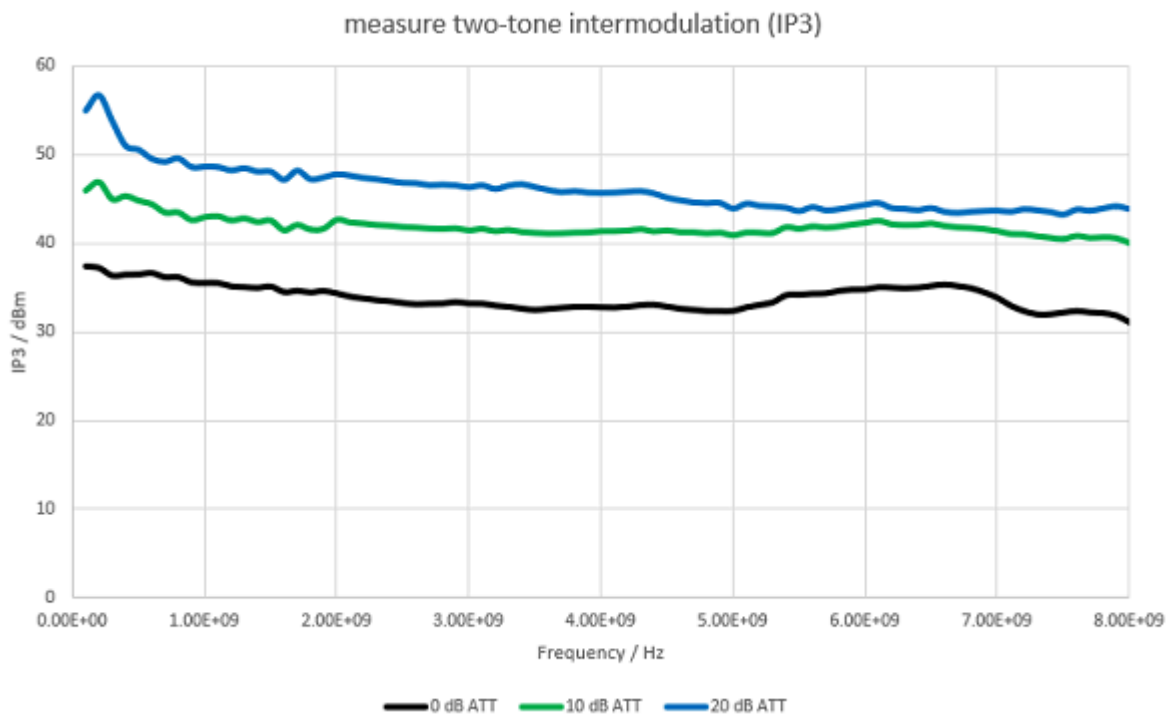
RFIM-8 measure two-tone intermodulation (IP3)

Conditions: HR ADC, 40 MHz filter, $f_{IF} = 0.533\text{ MHz}$, $\Delta f = 100\text{ kHz}$, -5 dBm tone power, with 0 dB receiver attenuation. IP3 increases with receiver attenuation.

Frequency: 10 MHz - 7.2 GHz

Measure Path	TOI (dBm)	
	Spec	Typ
direct	24	29.5
preamp1	15	20.9
preamp2	5	12.4
preamp3	-4	2.4

Measured data



RFIM-8 measure EVM (error vector magnitude)

WCDMA, Uplink, 1 Carrier, DPCCH

Frequency [GHz]	EVM spec	EVM typ
2 (-40 dBm to +2 dBm) Data obtained with loopback configuration	0.8%	
2 (-40 dBm to +10 dBm)		0.5%

802.11ac 80MHz

Frequency [GHz]	Power range [dBm]	EVM spec	EVM typ
5.2	-26 to -6	0.8%	0.54%

Frequency [GHz]	Power range [dBm]	EVM spec	EVM typ
	-6 < to 0	0.8%	0.6%
5.6	-26 to -6	0.9%	0.58%
	-6 < to 0	0.9%	0.68%

802.11ac 160MHz

Frequency [GHz]	Power range [dBm]	EVM spec	EVM typ
5.2	-26 to -6	1%	0.67%
	-6 < to 0	1%	0.68%
5.6	-26 to -6	1.1%	0.7%
	-6 < to 0	1.1%	0.78%

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-5 to -6	0.77%	-42.27
	-5 < to 1	0.84%	-41.51

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-25 to -5	0.92%	-40.72
	-5 < to 1	0.95%	-40.45

WSRF-8 + WSPS-8 measure EVM

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC

Frequency [GHz]	EVM (max)	EVM (typ)
2 (-40 dBm to +10 dBm)	0.8%	0.47%

802.11ac 160MHz bandwidth, HS ADC

Frequency [GHz]	EVM (max)	EVM (typ)
5.2 (-16 dBm to + 2 dBm)	1.0%	0.52%
5.2 (-26 dBm to -16 dBm)	1.0%	0.58%

802.11ax 80MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-33 to < -30	0.81%	-41.83

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
	-30 to < -25	0.59%	-44.58
	-25 to < -16	0.46%	-46.74
	-16 to < -7	0.40%	-47.96
	-7 to < -2	0.45%	-46.94
	-2 to 0	0.54%	-45.35
	0 < to 3	0.72%	-42.85

802.11ax 160MHz

Frequency [GHz]	Power range [dBm]	EVM typ	typ [dB]
7.125	-30 to < -25	0.83%	-41.62
	-25 to < -18	0.66%	-43.61
	-18 to -5	0.61%	-44.29
	-5 < to < -2	0.65%	-43.74
	-2 to 0	0.68%	-43.35
	0 < to 3	0.86%	-41.31

RFIM-8 measure ACLR (adjacent channel leakage ratio)

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC, Center frequency: 2 GHz

Power level	ACLR [dBc]
-16 dBm to +6 dBm	-63.0 (max)
-16 dBm to +6 dBm	-65.3 (typ)

Some IF value might result in dominant spur which yields to higher ACLR than spec. To avoid this, change IF value.

WSRF-8 + WSPS-8 measure ACLR

WCDMA, Uplink, 1 Carrier, DPCCH, HR ADC, Center frequency: 2 GHz

Power level	ACLR [dBc]
-16 dBm to +6 dBm	-67.0 (max)
-16 dBm to +6 dBm	-69.0 (typ)

RFIM-8 measure noise floor

Applicable for high resolution ADC only.

Measured at 0 dB receiver attenuation.

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
direct	0.01 to < 0.05	-130	-142
	0.05 to < 1	-138	-145
	1 to < 2	-136	-142
	2 to < 3	-135	-141
	3 to < 4	-135	-141
	4 to < 5	-133	-139
	5 to 6	-132	-137
	6 < to < 7	-132	-135
	7 to 7.2	-132	-136
	7.2 < to 8	-131	-135
preamp1	0.01 to < 0.05	-139	-150
	0.05 to < 1	-145	-152
	1 to < 2	-144	-150
	2 to < 3	-143	-148
	3 to < 4	-142	-148
	4 to < 5	-141	-146
	5 to 6	-139	-145
	6 < to < 7	-139	-143
	7 to 7.2	-140	-143
	7.2 < to 8	-138	-142
preamp2	0.01 to < 0.1	-140	-152
	0.1 to < 0.2	-152	-157
	0.2 to < 1	-152	-158
	1 to < 2	-152	-157
	2 to < 3	-152	-157
	3 to < 4	-151	-156
	4 to < 5	-151	-155
	5 to 6	-150	-154

Measure Path	Receiver frequency range CW (GHz)	Avrg. noise floor (dBm/Hz) Spec	Avrg. noise floor (dBm/Hz) Typ
	6 < to < 7	-149	-152
	7 to 7.2	-149	-152
	7.2 < to 8	-147	-151
preamp3	0.01 to < 0.1	-142	-152
	0.1 to < 0.2	-153	-158
	0.2 to < 1	-155	-159
	1 to < 2	-155	-159
	2 to < 3	-155	-159
	3 to < 4	-155	-159
	4 to < 5	-154	-158
	5 to 6	-154	-158
	6 < to < 7	-154	-156
	7 to 7.2	-153	-156
	7.2 < to 8	-152	-155

RFIM-8 measure dynamic range

Applicable for high resolution ADC only.

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
direct	0.01 to < 0.05	126	141.7
	0.05 to < 1	141	151.4
	1 to < 2	141	150.7
	2 to < 3	142	151
	3 to < 4	142	150.5
	4 to < 5	140	150.2
	5 to < 6	141	149.1
	6 to < 7	141	149.6
	7 to < 7.2	141	150.6
	7.2 to 8	140	149.6
preamp1	0.01 to < 0.05	126	141.1

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	0.05 to < 1	139	150.2
	1 to < 2	140	150
	2 to < 3	141	149.5
	3 to < 4	141	150.2
	4 to < 5	140	149
	5 to < 6	139	148.4
	6 to < 7	139	150.4
	7 to < 7.2	140	150.4
	7.2 to 8	138	149.4
preamp2	0.01 to < 0.1	119	134.9
	0.1 to < 0.2	136	144.9
	0.2 to < 1	136	145.9
	1 to < 2	138	146.8
	2 to < 3	140	148.4
	3 to < 4	139	148.1
	4 to < 5	140	148
	5 to < 6	139	147.7
	6 to < 7	138	148.7
	7 to < 7.2	138	148.7
	7.2 to 8	136	147.7
preamp3	0.01 to < 0.1	112	126.1
	0.1 to < 0.2	129	137.2
	0.2 to < 1	131	138.2
	1 to < 2	133	140.1
	2 to < 3	134	141.8
	3 to < 4	135	142.7
	4 to < 5	135	142.7
	5 to < 6	136	143
	6 to < 7	136	145.1

Measure Path	Receiver frequency range CW (GHz)	Dynamic range Spec (dB)	Dynamic range Typ (dB)
	7 to < 7.2	135	145.1
	7.2 to 8	134	144.1

RFIM-8 measure spurs

Measured at 0 dBm stimulus power

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF-8 measure spurs (measured values)

Frequency from [GHz]	Frequency to [GHz]	Max Spur [dBc]
0.13	1	-67
1	1.2	-64
1.2	1.6	-61.5
1.6	1.7	-54
1.7	3.4	-56.5
3.4	4	-53
4	4.1	-51
4.1	5.15	-53
5.15	5.25	-52.5
5.25	5.9	-50.5
5.9	6.9	-51
6.9	8	-50

Valid for spur frequency offset ranging from +/-10kHz to +/-40MHz

WSRF + WSPS measure spurs (measured values)

Input frequency from [GHz]	Input frequency to [GHz]	Max spurs [dBc]
0.13	4	-70
4	6	-68
6	8	-67

Wave Scale RF-8 with RFIM-8 terminate

RF ports that are turned-off, have an internal 50-Ohm termination to GND. In order to avoid thermal damage to the termination resistors, the following abs-max restrictions have to be observed per logical board (half of RFIM).

Number of ports in termination mode	Maximum power applied per port (dBm)
1 or 2	30
3 to 9	27
10 or more	26

RFIM-18 attached to Wave Scale RF card (WSRF N2660)

Following hardware components are part of the specification sections:

- *WSRF 6 GHz or 8 GHz (N2600A-OP or N2600B-OP) with RFIM-18*

Note

In case WSPS is included, it must be WSPS 6 GHz (N2601A-OP) for WSRF 6 GHz card (N2600A-OP), or WSPS 8 GHz (N2601B-OP) for WSRF 8 GHz card (N2600B-OP).

Note

Mixing of WSRF 6 GHz (N2600A-OP) and WSRF 8 GHz (N2600B-OP) cards in one system configuration is not supported.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

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▲ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

▲ Warning A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

Wave Scale RF with RFIM-18 specifications

In this section, Wave Scale RF is either WSRF-6 or WSRF-8 card.

Note The specifications provided here are preliminary and are subject to change without notice.

Note The temperature of the water circulating in the secondary cooling loop must be set to +30 # to guarantee the specified performance. For more details, see *Water specification secondary cooling loop of STH*.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

Wave Scale RF with RFIM-18 stimulus

RFIM-18 stimulus maximum output power

Shows absolute maximum output power, measured as saturated power.

Conditions: LO source WSRF. Measured in loopback. Receiver and mix receiver path direct. Programmed output power 25 dBm.

Carrier frequency (GHz)	Maximum output power (dBm) Spec
5.85 <= f < 13.75	10
13.75 <= f < 15.75	6
15.75 <= f < 17.75	10
17.75 <= f <= 18	9

RFIM-18 stimulus power accuracy

Conditions:

- Output power: -10 dBm
- Optimization: noisefloor
- Valid only when driving the stimulus into a 50 Ohm matched power sensor directly attached to the pogo. Any further cabling and/or mismatch will induce additional error.
- Option: powerAccuracyOptimization

Characteristic stimulus power accuracy (dB)

Power range	Frequency range (GHz)												
	5.85 ≤ f < 6	6 ≤ f < 7	7 ≤ f < 8	8 ≤ f < 9	9 ≤ f < 10	10 ≤ f < 11	11 ≤ f < 12	12 ≤ f < 13	13 ≤ f < 14	14 ≤ f < 15	15 ≤ f < 16	16 ≤ f < 17	17 ≤ f < 18
-80 ≤ Lv < -70	±1.2	±1.3	±1.4	±1.6	±1.6	±1.3	±1.0	±1.2	±1.3	±1.4	±1.4	±1.5	±1.4
-70 ≤ Lv < -60	±0.7	±1.0	±1.1	±1.3	±1.3	±1.1	±1.0	±1.0	±1.1	±1.1	±1.1	±1.4	±1.2
-60 ≤ Lv < -50	±0.7	±0.8	±0.9	±1.1	±1.1	±1.0	±0.9	±1.0	±1.0	±1.1	±1.1	±1.3	±1.2
-50 ≤ Lv < -40	±0.6	±0.8	±0.9	±1.1	±1.0	±1.0	±0.9	±0.9	±1.0	±1.0	±1.1	±1.2	±1.1
-40 ≤ Lv < -30	±0.6	±0.8	±0.9	±1.0	±1.0	±1.0	±0.9	±0.9	±1.0	±1.0	±1.1	±1.2	±1.1
-30 ≤ Lv < -20	±0.6	±0.8	±0.9	±0.9	±0.9	±0.9	±0.8	±0.9	±1.0	±1.0	±1.0	±1.1	±1.1
-20 ≤ Lv < -10	±0.6	±0.7	±0.8	±0.8	±0.7	±0.9	±0.7	±0.7	±0.8	±0.9	±0.9	±0.9	±0.9
-10 ≤ Lv < -8	±0.7	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9
-8 ≤ Lv < -6	±0.7	±0.7	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9
-6 ≤ Lv < -4	±0.7	±0.7	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.7	±0.8	±0.7	±0.9	±0.9
-4 ≤ Lv < -2	±0.7	±0.8	±0.8	±0.8	±0.8	±0.7	±0.7	±0.6	±0.8	±0.9	±0.8	±0.9	±0.9
-2 ≤ Lv < 0	±0.7	±0.8	±0.9	±0.8	±0.8	±0.7	±0.8	±0.7	±0.9	±1.0	±1.0	±0.9	±1.0

Power range	Frequency range (GHz)												
	5.85 ≤ f < 6	6 ≤ f < 7	7 ≤ f < 8	8 ≤ f < 9	9 ≤ f < 10	10 ≤ f < 11	11 ≤ f < 12	12 ≤ f < 13	13 ≤ f < 14	14 ≤ f < 15	15 ≤ f < 16	16 ≤ f < 17	17 ≤ f < 18
0 ≤ Lv < 2	±0.7	±0.8	±0.9	±0.9	±0.8	±0.7	±0.9	±0.7	±1.0	±1.1	±1.0	±1.0	±1.1
2 ≤ Lv < 4	±0.7	±0.8	±0.9	±0.9	±0.9	±0.8	±1.0	±0.8	±1.1	±1.4	±1.3	±1.1	±1.2
4 ≤ Lv < 6	±0.7	±0.8	±0.9	±0.9	±0.9	±0.9	±1.3	±1.0	±1.3				
6 ≤ Lv < 8	±0.7	±0.8	±0.9	±0.9	±1.0	±1.1	±1.3	±1.2					
8 ≤ Lv < 10	±0.8	±0.9	±1.0	±1.0	±1.1	±1.1							

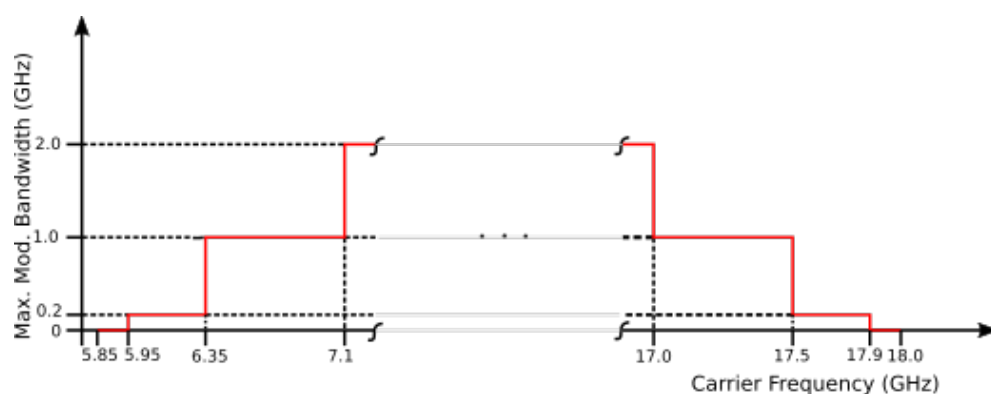
Typical stimulus power accuracy (dB)

Power range	Frequency range (GHz)												
	5.85 ≤ f < 6	6 ≤ f < 7	7 ≤ f < 8	8 ≤ f < 9	9 ≤ f < 10	10 ≤ f < 11	11 ≤ f < 12	12 ≤ f < 13	13 ≤ f < 14	14 ≤ f < 15	15 ≤ f < 16	16 ≤ f < 17	17 ≤ f < 18
-80 ≤ Lv < -70	±0.8	±0.8	±0.9	±1.1	±1.1	±0.9	±0.9	±0.8	±0.9	±0.9	±0.9	±1.0	±0.9
-70 ≤ Lv < -60	±0.6	±0.7	±0.8	±1.0	±1.0	±0.8	±0.7	±0.8	±0.8	±0.8	±0.8	±1.0	±0.8
-60 ≤ Lv < -50	±0.5	±0.6	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.8	±0.8	±0.8	±0.9	±0.8
-50 ≤ Lv < -40	±0.5	±0.6	±0.7	±0.8	±0.8	±0.7	±0.7	±0.7	±0.8	±0.7	±0.8	±0.9	±0.8
-40 ≤ Lv < -30	±0.5	±0.6	±0.7	±0.8	±0.7	±0.7	±0.7	±0.7	±0.7	±0.7	±0.8	±0.9	±0.8
-30 ≤ Lv < -20	±0.5	±0.6	±0.7	±0.7	±0.6	±0.7	±0.7	±0.6	±0.7	±0.7	±0.7	±0.8	±0.7
-20 ≤ Lv < -10	±0.5	±0.5	±0.6	±0.6	±0.6	±0.6	±0.6	±0.5	±0.6	±0.7	±0.6	±0.6	±0.6

Power range	Frequency range (GHz)												
	5.85 ≤ f < 6	6 ≤ f < 7	7 ≤ f < 8	8 ≤ f < 9	9 ≤ f < 10	10 ≤ f < 11	11 ≤ f < 12	12 ≤ f < 13	13 ≤ f < 14	14 ≤ f < 15	15 ≤ f < 16	16 ≤ f < 17	17 ≤ f < 18
-10 ≤ Lv < -8	±0.5	±0.5	±0.6	±0.6	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.6	±0.6
-8 ≤ Lv < -6	±0.5	±0.5	±0.6	±0.6	±0.6	±0.5	±0.5	±0.5	±0.5	±0.5	±0.5	±0.6	±0.6
-6 ≤ Lv < -4	±0.5	±0.5	±0.6	±0.7	±0.6	±0.5	±0.5	±0.5	±0.6	±0.6	±0.5	±0.6	±0.6
-4 ≤ Lv < -2	±0.5	±0.6	±0.6	±0.7	±0.6	±0.5	±0.5	±0.5	±0.6	±0.7	±0.6	±0.7	±0.7
-2 ≤ Lv < 0	±0.5	±0.6	±0.7	±0.7	±0.6	±0.5	±0.5	±0.5	±0.7	±0.8	±0.7	±0.7	±0.7
0 ≤ Lv < 2	±0.5	±0.6	±0.7	±0.7	±0.6	±0.5	±0.6	±0.6	±0.8	±0.8	±0.8	±0.8	±0.8
2 ≤ Lv < 4	±0.5	±0.6	±0.7	±0.7	±0.6	±0.6	±0.7	±0.7	±0.9	±1.1	±1.1	±0.8	±0.9
4 ≤ Lv < 6	±0.5	±0.6	±0.7	±0.7	±0.7	±0.7	±0.8	±0.8	±1.0				
6 ≤ Lv < 8	±0.5	±0.6	±0.7	±0.7	±0.8	±0.8	±1.0	±0.9					
8 ≤ Lv < 10	±0.6	±0.7	±0.7	±0.7	±0.9	±0.9							

RFIM-18 stimulus modulation bandwidth

To meet the RFIM-18 specifications, the restrictions on a modulated stimulus are: (Carrier Frequency - Mod. Bandwidth/2) ≥ 5.85 GHz. However, SmarTest only checks carrier frequency against 5.2 GHz. If you use a modulated signal at a carrier frequency of 5.2 GHz, the result will be attenuated by a filter.



Stim Mod. Bandwidth vs. Carrier Frequency

RFIM-18 stimulus phase noise

CW stimulus phase noise (dBc/Hz), measured at 0 dBm stimulus power. Optimization is "noiseFloor".

LO source WSRF stimulus phase noise

Frequency (GHz)	Characteristic stimulus phase noise (dBc/Hz)				
	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-80.8	-94.9	-104.5	-104.5	-129.3
18	-75.6	-89.9	-98.0	-100.8	-129.3

LO source WSPS stimulus phase noise

Frequency (GHz)	Characteristic stimulus phase noise (dBc/Hz)				
	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-83.6	-101.1	-109.7	-112.1	-129.4
18	-85.8	-104.8	-108.8	-115.5	-130.0

RFIM-18 stimulus TOI (third-order intercept)

Conditions: Tone power is 0 dBm, Optimization is "imd", SmT source is "modulated".

WSRF base band 100 kHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	22	26
6	22	25
7.2	20	23
8	21	24
9.2	21	25
10	17	21
11.2	16	19
12	14	18

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
13.2	14	18
14	13	16
15.2	9	12
16	14	19
17.2	15	20
18	11	16

WSRF base band 40 MHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	21	25
6	22	25
7.2	22	25
8	23	26
9.2	20	25
10	17	21
11.2	15	18
12	14	18
13.2	13	16
14	12	15
15.2	8	12
16	14	18
17.2	16	20
17.96	11	16

AUX base band 100 kHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	22	26
6	23	26
7.2	20	25
8	21	25
9.2	18	25
10	17	22
11.2	18	22
12	17	22
13.2	16	19
14	15	18

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
15.2	10	13
16	15	20
17.2	16	21
18	12	17

AUX base band 40 MHz spacing

Lower side carrier frequency (GHz)	TOI (dBm) Characteristic	TOI (dBm) Typ
5.2	20	24
6	21	24
7.2	23	26
8	23	26
9.2	21	26
10	16	21
11.2	16	20
12	15	19
13.2	15	18
14	13	16
15.2	9	12
16	15	19
17.2	17	21
17.96	13	18

RFIM-18 stimulus spurs Second harmonic spurs

Conditions: LO source WSRF or WSPS. Programmed output power 0 dBm.

WSRF path

Carrier frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-34.7	-38.0
5.85	-34.9	-36.8
9		
12		
15		
18		

AUX path

Carrier frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-26.8	-36.2
5.85	-33.1	-36.4
9		
12		
15		
18		

Stimulus LO feedthrough spurs

Conditions: LO source WSRF. Programmed output power 0 dBm.

WSRF path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-2.4	-5.1
5.85	-37.3	-47.9
9		
12		
15		
18		

AUX path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-7.3	-17.8
5.85	-37.9	-48.3
9		
12		
15		
18		

Unknown spurs

Conditions: LO source WSRF. Programmed output power 0 dBm.

From the result of the unknown spurs, the known ones are removed. Known spurs are considered the following:

- $(RF \times m) + (BB \times n)$
- $abs((RF \times m) - (BB \times n))$
- $(BB \times n) + (LO \times p)$

- $\text{abs}((\text{BB} \times n) - (\text{LO} \times p))$
- $(\text{LO} \times p) + (\text{RF} \times m)$ and $\text{abs}((\text{LO} \times p) - (\text{RF} \times m))$

using the following variables:

- m: 0, 1, 2, 3, 4, 5
- n: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- p: 0, 1/4, 1/2, 3/4, 1, 2, 3

WSRF path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-47.1	-50.9
5.85	-43.5	-53.2
9		
12		
15		
18		

AUX path

Frequency (GHz)	Max spurs (dBc) Characteristic	Max spurs (dBc) Typ
5.2	-44.6	-51.5
5.85	-44.6	-54.0
9		
12		
15		
18		

RFIM-18 stimulus ACLR (adjacent channel leakage ratio)

5GNR 1Carrier ACLR, stim power: -15 dBm to 0 dBm

WSRF path

Frequency (GHz)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ	ACLR2 (dBc) Characteristic	ACLR2 (dBc) Typ
10	-33	-37	-45	-49
14	-34	-37	-46	-49
17.5	-34	-37	-43	-48

Optimizer: balanced, NoiseBandwidth = 100 MHz

AUX path

Frequency (GHz)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ	ACLR2 (dBc) Characteristic	ACLR2 (dBc) Typ
10	-34	-37	-48	-52
14	-33	-37	-46	-52
17.5	-34	-38	-45	-50

Optimizer: balanced, NoiseBandwidth = 100 MHz

RFIM-18 stimulus EVM (error vector magnitude)

Conditions:

- Signal format: 5GNR
- Signal bandwidth: 100 MHz

LO source WSRF, Stim path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-5	1.94	1.59
14	-5	2.17	1.75
17.5	-5	2.45	2.02

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO Source WSPS, Stim path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-5	1.62	1.14
14	-5	1.61	1.32
17.5	-5	1.71	1.45

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO source WSRF, Stim path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-5	1.72	1.48
14	-5	1.98	1.73
17.5	-5	2.12	1.84

Optimizer: balanced, NoiseBandwidth= 5 MHz

LO source WSPS, Stim path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-5	1.12	0.97

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
14	-5	1.53	1.3
17.5	-5	1.37	1.22

Optimizer: balanced, NoiseBandwidth= 5 MHz

Wave Scale RF with RFIM-18 measure

RFIM-18 measure power

Conditions:

- WSRF receiver path: direct
- Actual input power setting (dBm): +25 (direct path), +15 (preamp path)
- SmarTest expected input power setting (dBm): +20 (direct path), +20 (preamp path)

Maximum RF input power to all ports

Mix receiver path	Receiver frequency range CW (GHz)	Maximum measurable power with 0 dB att. (dBm) Characteristic
direct	5.85 <= f < 12	17.5
	12 <= f < 13	17.0
	13 <= f < 17	16.5
	17 <= f <= 18	15.5
preamp	5.85 <= f <= 18	0.5

Absolute maximum power damage level (dBm)

- For direct path: 24
- For preamp path: 22

RFIM-18 measure power accuracy

Characteristic measure power accuracy - direct path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -30dBm appropriate bandwidth of interest and result averaging must be applied.

Characteristic power accuracy (dB) direct path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -30	±0.40	±0.46	±0.49
-30 <= 0	±0.40	±0.46	±0.49
0 <= 10	±0.38	±0.48	±0.55
10 <= 14	±0.65	±0.72	±0.89 ^[10]

^[10] 17 GHz <= 18 GHz; 10 dBm <= 12 dBm

Characteristic measure power accuracy - preamp path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -50dBm appropriate bandwidth of interest and result averaging must be applied.

Characteristic power accuracy (dB) preamp path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -50	±0.45	±0.53	±0.54
-50 <= -10	±0.45	±0.53	±0.54
-10 <= -6	±0.42	±0.56	±0.68
-6 <= -3	±0.57	±0.78	±0.81 ^[11]

Typical measure power accuracy - direct path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -30dBm appropriate bandwidth of interest and result averaging must be applied.

Typical power accuracy (dB) direct path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -30	±0.29	±0.33	±0.40
-30 <= 0	±0.29	±0.33	±0.40
0 <= 10	±0.28	±0.36	±0.45
10 <= 14	±0.60	±0.64	±0.81 ^[10]

Typical measure power accuracy - preamp path

Conditions: Bandwidth of interest is set to 1 MHz. To achieve the specified level accuracy for power levels < -50dBm appropriate bandwidth of interest and result averaging must be applied.

Typical power accuracy (dB) preamp path

Power range (dBm)	Frequency range		
	6 GHz <= 12 GHz	12 GHz <= 16 GHz	16 GHz <= 18 GHz
< -50	±0.34	±0.44	±0.44
-50 <= -10	±0.34	±0.44	±0.44
-10 <= -6	±0.33	±0.47	±0.57
-6 <= -3	±0.50	±0.70	±0.70 ^[11]

RFIM-18 measure modulation flatness

The typical and characteristic performance for RFIM-18 measure modulation flatness is shown in the table below.

Conditions: RF power is 0 dBm, expected input power is +5 dBm, mix receiver path is direct, Bandwidth of interest (For WSRF numbers) is 210 MHz.

^[11] 16 GHz <= 17 GHz; -6 dBm <= -4 dBm 17 GHz <= 18 GHz; -6 dBm <= -3 dBm

The flatness is measured using a fixed LO and a swept BB. The BB center frequency is set to default.

Measure path	RF center frequency (GHz)	Maximum modulation bandwidth (GHz)	Modulation flatness (dB) Typ	Modulation flatness (dB) Characteristic
WSRF	6.5	0.2	1.7	3.7
	17.8	0.2		
AUX	6.5	1	4.9	9.1
	17.3	1		
	7.4	2	4.6	8.3
	8	2		
	8.9	2		
	10.5	2		
	12	2		
	14	2		
	16	2		
	16.8	2		

RFIM-18 measure noise floor

Conditions: Measured at 0 dB receiver attenuation using the internal termination. The Stim path is set to 18 GHz and -90 dBm.

Mix receiver path	Receiver frequency range CW (GHz)	Avg. noise floor (dBm/Hz) Spec
direct	5.2 <= f < 8.2	-144
	8.2 <= f < 11.2	-143
	11.2 <= f < 13.2	-142
	13.2 <= f < 15.6	-141
	15.6 <= f <= 18.0	-138
preamp	5.2 <= f < 8.2	-157
	8.2 <= f < 10.2	-156
	10.2 <= f < 13.0	-155
	13.0 <= f < 15.6	-154
	15.6 <= f <= 18.0	-153

RFIM-18 measure phase noise

CW measure phase noise (dBc/Hz), measured at 0 dBm RF power.

LO source WSRF measure phase noise (dBc/Hz)

Input carrier frequency (GHz)	Characteristic measure phase noise (dBc/Hz)				
	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-78.7	-95.5	-103.8	-105.6	-127.6
18	-72.6	-89.3	-96.1	-101.3	-127.1

LO source WSPS measure phase noise (dBc/Hz)

Input carrier frequency (GHz)	Characteristic measure phase noise (dBc/Hz)				
	Offset frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
6	-80.4	-101.2	-109.4	-112.9	-128.0
18	-85.3	-103.0	-106.9	-113.2	-128.9

RFIM-18 measure two-tone intermodulation (IP3)

Conditions: Measured at -5 dBm for direct path and -20 dBm for preamp path. WSRF receiver path is preamp1. Frequency range $5.85 \leq f \leq 18$ GHz.

WSRF path

Mix receiver path	100 kHz tone spacing TOI (dBm) Characteristic	40 MHz tone spacing TOI (dBm) Characteristic
direct	19.5	20.3
preamp	3.6	4.4

AUX path

Mix receiver path	100 kHz tone spacing TOI (dBm) Characteristic	40 MHz tone spacing TOI (dBm) Characteristic
direct	23.2	23.6
preamp	6.4	6.3

RFIM-18 measure spurs

WSRF baseband: valid for spurs frequency offset +/- 100 MHz

Frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-23	-27
5.85	-49	-54
9	-46	-50
12	-54	-57

Frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
15	-42	-46
18	-42	-45

AUX baseband: valid for spurs frequency offset +/- 100 MHz

Frequency (GHz)	Max spur (dBc) Characteristic	Max spur (dBc) Typ
5.2	-41	-51
5.85	-44	-52
9	-41	-48
12	-44	-54
15	-34	-44
18	-36	-43

RFIM-18 measure ACLR (adjacent channel leakage ratio)

Conditions:

- LO source WSRF or WSPS
- Signal format: 5GNR
- SmarTest expected input power: Signal power + PeakToAvgPowerRatio
- SmarTest PeakToAvgPowerRatio(5GNR): 12.0 dB
- SmarTest resultAveraging: 8

WSRF path

Frequency (GHz)	Signal Power (dBm)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ
10	-28 <= P <= 0	-44.1	-45.8
14	-28 <= P <= 0	-44.0	-45.4
17.5	-28 <= P < -4	-40.4	-43.7
	-4 <= P <= 0	-38.3	-42.2

AUX path

Frequency (GHz)	Signal Power (dBm)	ACLR1 (dBc) Characteristic	ACLR1 (dBc) Typ
10	-28 <= P <= 0	-45.7	-48.1
14	-28 <= P <= 0	-42.6	-45.6
17.5	-28 <= P < -4	-44.6	-48.2
	-4 <= P <= 0	-41.9	-44.1

RFIM-18 measure EVM (error vector magnitude)

Conditions:

- Power range: -4 to -16 dBm for direct path and -16 to -28 dBm for preamp path.
- Signal format: 5GNR
- SmarTest Bandwidth of Interest: 100 MHz
- SmarTest expected input power: Signal power + PeakToAvgPowerRatio
- SmarTest PeakToAvgPowerRatio (5GNR): 12.0 dB

LO source WSRF, meas path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.80	1.40
14	-28 <= P <= -4	2.00	1.55
17.5	-28 <= P <= -4	2.45	1.85

LO source WSPS, meas path WSRF

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.00	0.80
14	-28 <= P <= -4	1.00	0.80
17.5	-28 <= P <= -4	1.05	0.90

LO source WSRF, meas path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.80	1.50
14	-28 <= P <= -4	2.20	1.60
17.5	-28 <= P <= -4	2.40	1.85

LO source WSPS, meas path AUX

Frequency (GHz)	Signal Power (dBm)	EVM (%) Characteristic	EVM (%) Typ
10	-28 <= P <= -4	1.15	0.95
14	-28 <= P <= -4	1.20	1.00
17.5	-28 <= P <= -4	1.35	1.00

Wave Scale RF with RFIM-18 loopback

RFIM-18 loopback ACLR (adjacent channel leakage ratio)

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Mix receiver path	Frequency (GHz)	Power range (dBm)	ACLR1 upper/ lower (dBc) Spec	ACLR2 upper/ lower (dBc) Spec
direct	10.0	-12 to -4	-39	-40
	14.0		-39	-40
	17.5		-38	-38
preamp	10.0	-20 to -12	-38	-40
	14.0		-39	-41
	17.5		-38	-39

Conditions: Optimizer: balanced, NoiseBandwidth = 75 MHz, baseband attenuation = 6dB

RFIM-18 loopback EVM (error vector magnitude)**LO source WSRF**

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Frequency (GHz)	Power (dBm)	EVM (%) Spec
10.0	-10	2.2
14.0		2.6
17.5		2.7

Optimizer: balanced, NoiseBandwidth= 5 MHz, baseband attenuation 0dB

LO source WSPS

5GNR, Uplink, 100 MHz bandwidth, HS ADC

Frequency (GHz)	Power (dBm)	EVM (%) Spec
10.0	-10	1.5
14.0		1.4
17.5		1.5

Optimizer: balanced, NoiseBandwidth= 5 MHz, baseband attenuation 0dB

Wave Scale Millimeter solution

Following hardware components are part of the specification sections:

- *WSRF 6 GHz or 8 GHz (N2600A-OP or N2600B-OP) with RFIM-18*

Note

In case WSPS is included, it must either be WSPS 6 GHz (N2601A-OP) for WSRF 6 GHz card (N2600A-OP), or WSPS 8 GHz (N2601B-OP) for WSRF 8 GHz card (N2600B-OP).

Note

Mixing of WSRF 6 GHz (N2600A-OP) and WSRF 8 GHz (N2600B-OP) cards in one system configuration is not supported.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

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Safety Notices

▲ CAUTION A CAUTION notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in damage to the product or loss of important data. Do not proceed beyond a CAUTION notice until the indicated conditions are fully understood and met.

▲ Warning A WARNING notice denotes a hazard. It calls attention to an operating procedure, practice, or the like that, if not correctly performed or adhered to, could result in death or severe injury. Do not proceed beyond a WARNING notice until the indicated conditions are fully understood and met.

Wave Scale Millimeter specifications

The Wave Scale Millimeter (WSMM) solution is an extension of the V93000 DUT interface providing an integrated test solution for 5G and WiGig applications. The solution is based on the Wave Scale RF base card (WSRF-6 or WSRF-8) and extends its capabilities for IF (5.85 to 18 GHz) using RFIM-18 modules, and for millimeter wave frequencies using Remote Mixing Modules (RMMs).

Three versions of RMMs are supported: RMM-44 (operational limit 24 to 30 GHz, 35 to 44 GHz) and RMM-53 (operational limit 24 to 52.8 GHz) for 5G applications, and RMM-72 (operational limit 57 to 70 GHz) for WiGig applications.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

Note All Wave Scale Millimeter (WSMM) specifications are including Advantest's 1.85 mm DUT board connector, providing a standardized coaxial interface.

Note The Wave Scale Millimeter (WSMM) specifications are only valid with LO source WSPS .

WSMM stimulus

WSMM stimulus accuracy

Stimulus accuracy conditions and notes:

- All values are typical according to *Terms and conditions* on page 83, unless noted.
- All power levels are CW signals in a 50 Ω system, unless noted.
- All values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.
- Non-default parameters in SmarTest 8:
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e-100 Hz

Note All values are in dB.

RMM-44 without Switch

Typical

Frequency range (GHz)	Power level range (dBm)	
	$-80 \leq L_v < -50$	$-50 \leq L_v \leq +4$
$24 \leq f \leq 30$	± 1.27	± 0.69

Frequency range (GHz)	Power level range (dBm)	
	$-80 \leq L_v < -50$	$-50 \leq L_v \leq +4$
$35 \leq f \leq 42$	± 1.14	± 0.90
$42 < f \leq 44$	± 1.89	± 1.59 ^[12]

RMM-44 with 1:8 Switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -50$	$-50 \leq L_v \leq -2$	$-2 < L_v \leq +2$
$24 \leq f \leq 30$	± 0.98	± 0.47	± 0.90
$35 \leq f \leq 40$	± 1.09	± 0.63	± 0.81 ^[13]

RMM-53 without switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v \leq +4$
$24 \leq f \leq 27$	± 0.95	± 0.46	± 0.49
$27 < f \leq 35$	± 0.89 ^[14]	± 0.53	± 0.52
$35 < f \leq 50$	± 0.95	± 0.60	± 0.58
$50 < f \leq 52.8$	± 0.43 ^[15]	± 0.44 ^[15]	± 0.67 ^[15]

RMM-53 with 1:8 switch

Typical

Frequency range (GHz)	Power level range (dBm)			
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v < -4$	$-4 \leq L_v \leq +2$
$24 \leq f \leq 27$	± 0.58	± 0.44	± 0.46	± 0.59
$27 < f \leq 31$	± 0.69 ^[14]	± 0.52	± 0.50	± 0.60
$31 < f \leq 42$	± 0.69	± 0.54	± 0.55	± 0.75 ^[16]
$42 < f \leq 50$	± 0.72	± 0.60	± 0.64	± 0.99 ^[17]
$50 < f \leq 52.8$	± 0.45 ^[15]	± 0.43 ^[15]	± 0.74 ^{[15][18]}	± 0.76 ^{[15][17][19]}

RMM-72 without Switch

Typical

^[12] for $-50 \leq L_v \leq +2$ ^[13] for $-2 < L_v \leq 0$ ^[14] for $-70 \leq L_v < -60$ ^[15] Measured^[16] $-4 \leq L_v < -1$ for $f > 40$ GHz^[17] $-4 \leq L_v < -2$ ^[18] $-10 \leq L_v < -7$ for $f \geq 52$ GHz^[19] $50 < f \leq 50.5$ GHz

Frequency range (GHz)	Power level range (dBm)		
	$-65 \leq L_v < -60$	$-60 \leq L_v < -50$	$-50 \leq L_v \leq +6$
$57 \leq f \leq 67$	± 1.07	± 0.73	± 0.58

RMM-72 with 1:8 Switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-65 \leq L_v < -60$	$-60 \leq L_v \leq -1$	$-1 < L_v \leq +1$
$57 \leq f \leq 67$	± 0.81	± 0.68	± 1.18

WSMM stimulus noise floor

Stimulus noise floor conditions and notes:

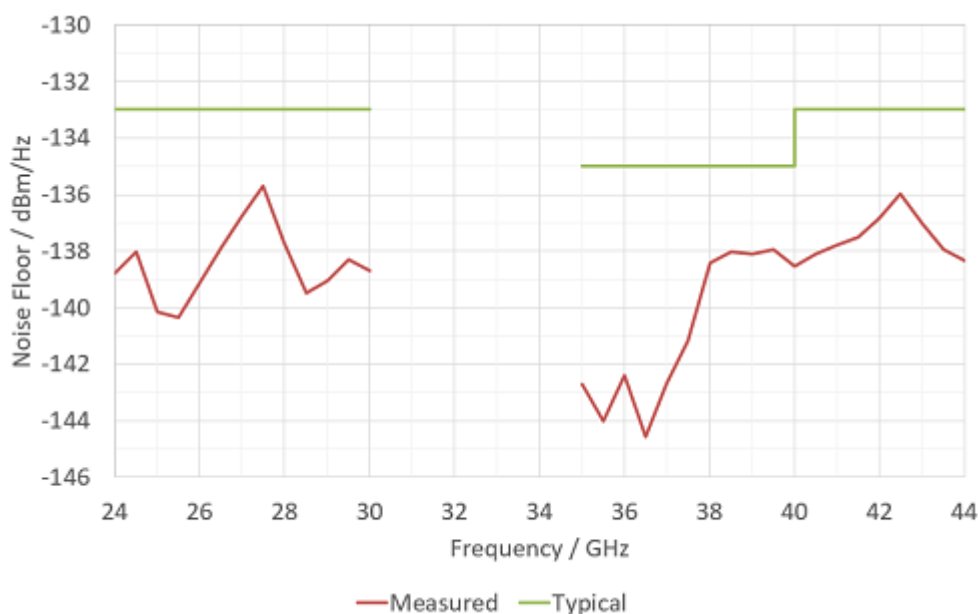
- All values are typical according to *Terms and conditions* on page 83, unless noted.
- Non-default parameters in SmarTest 8:
 - optimization = noiseFloor
- Stimulus CW frequency has an offset of 40 MHz^[20] against specified frequency.

RMM-44 without switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 30$	-133
$35 \leq f \leq 40$	-135
$40 < f \leq 44$	-133

Measured with carrier power = 0 dBm



RMM-44 without switch

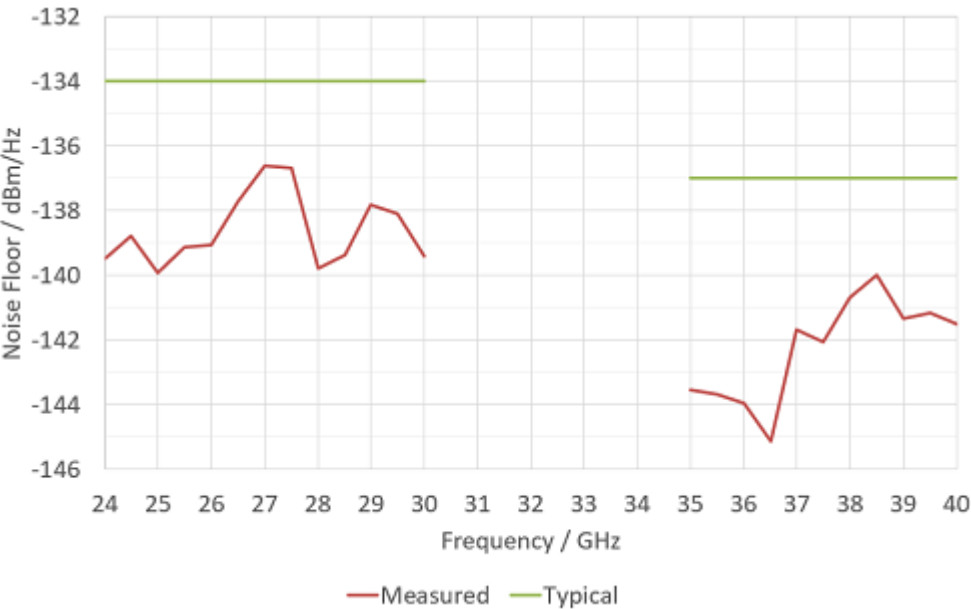
^[20] offset was adjusted for some frequencies to avoid spurs

RMM-44 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 30$	-134
$35 \leq f \leq 40$	-137

Measured with carrier power = -5 dBm



RMM-44 with 1:8 switch

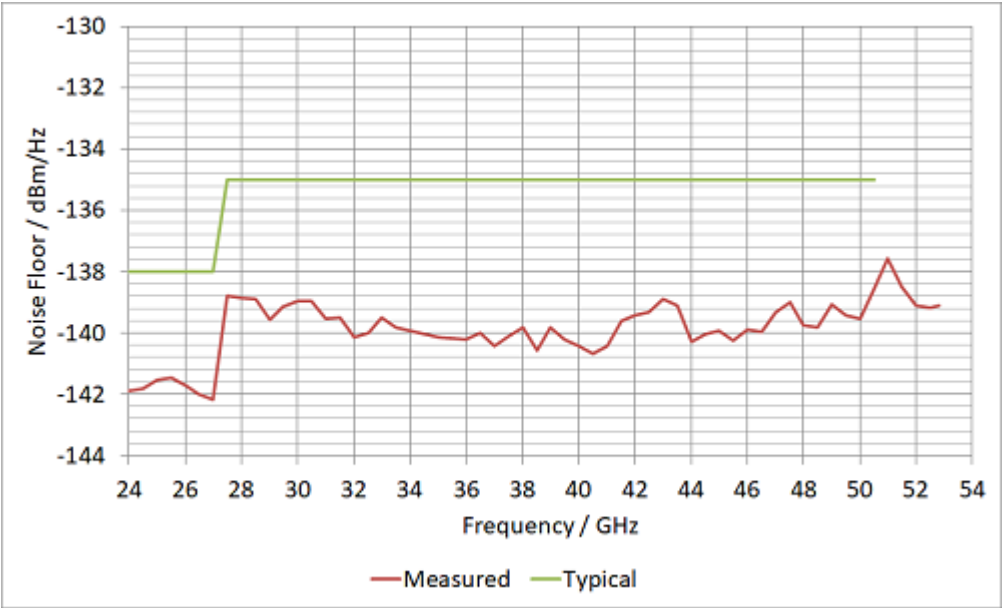
RMM-53 without switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 27$	-138
$27 < f \leq 50$	-135
$52 < f \leq 52.8$	-135 ^[21]

Measured with carrier power = -2 dBm

^[21] Measured



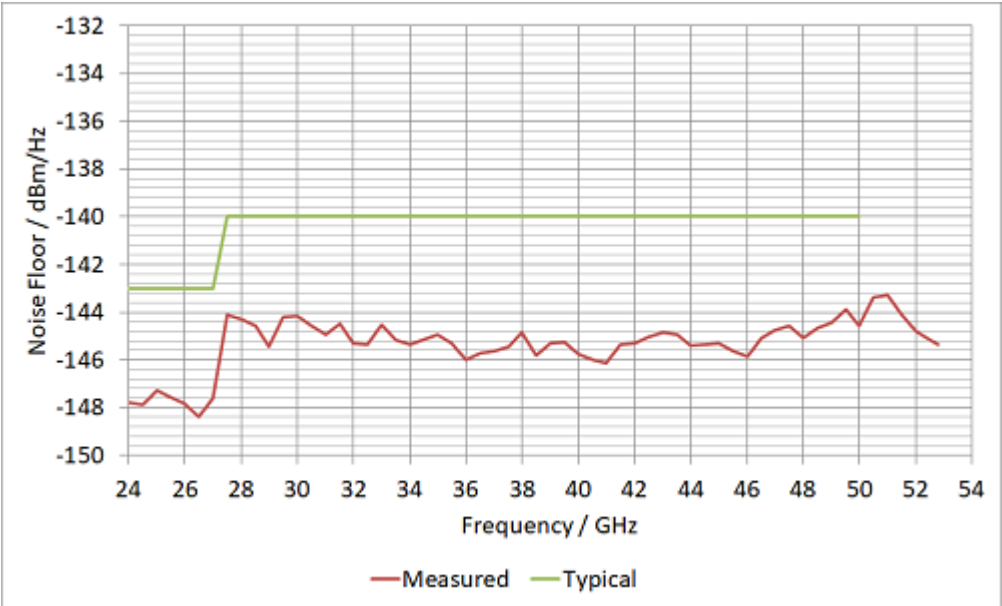
RMM-53 without switch

RMM-53 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 27$	-143
$27 < f \leq 50$	-140
$50 < f \leq 52.8$	-140 ^[21]

Measured with carrier power = -7 dBm



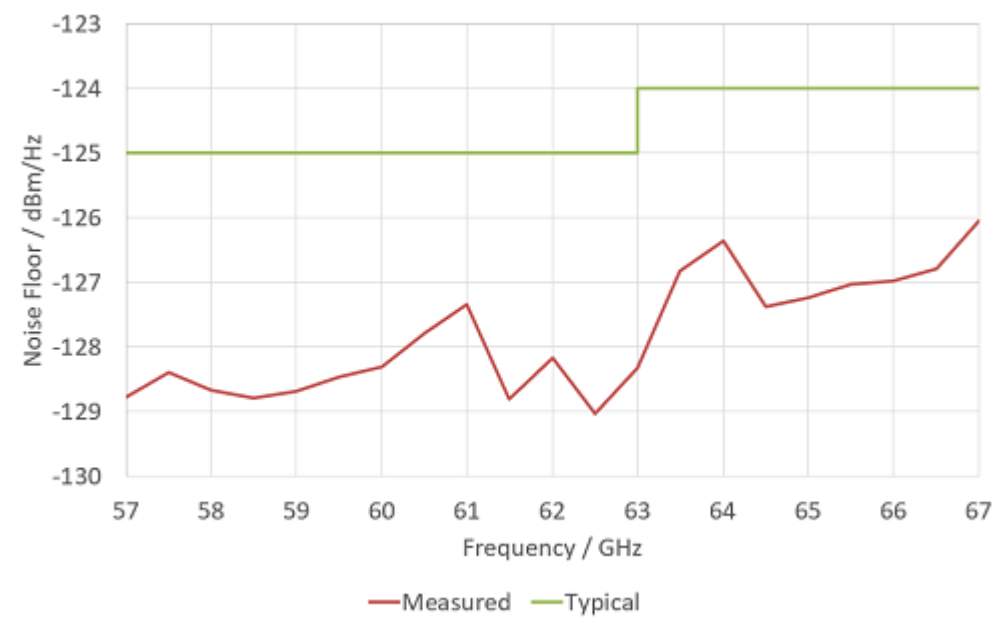
RMM-53 with 1:8 switch

RMM-72 without switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 63$	-125
$63 < f \leq 67$	-124

Measured with carrier power = -2 dBm



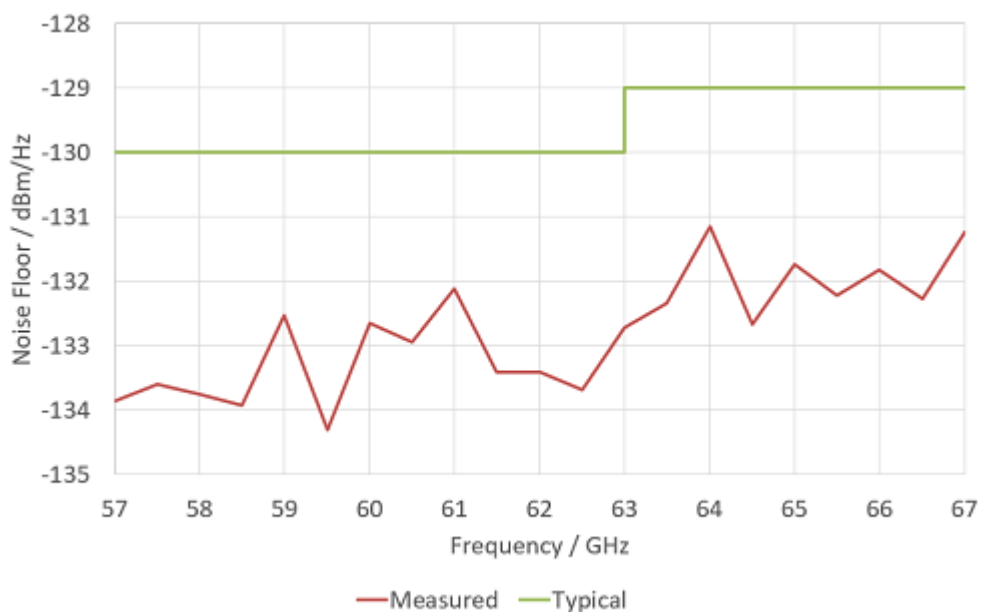
RMM-72 without switch

RMM-72 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 63$	-130
$63 < f \leq 67$	-129

Measured with carrier power = -7 dBm



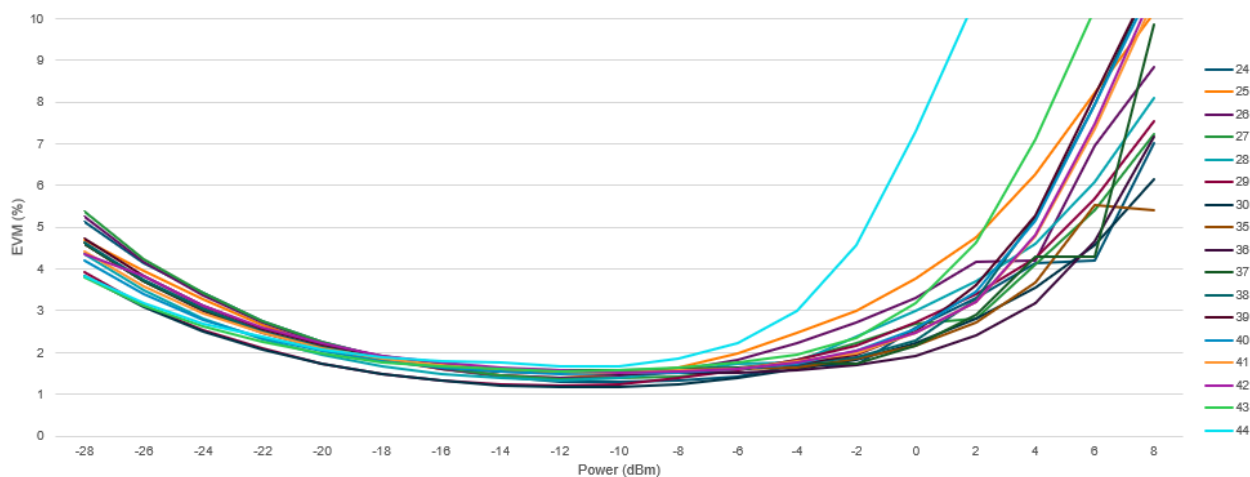
RMM-44 with 1:8 switch

WSMM stimulus EVM for RMM-44

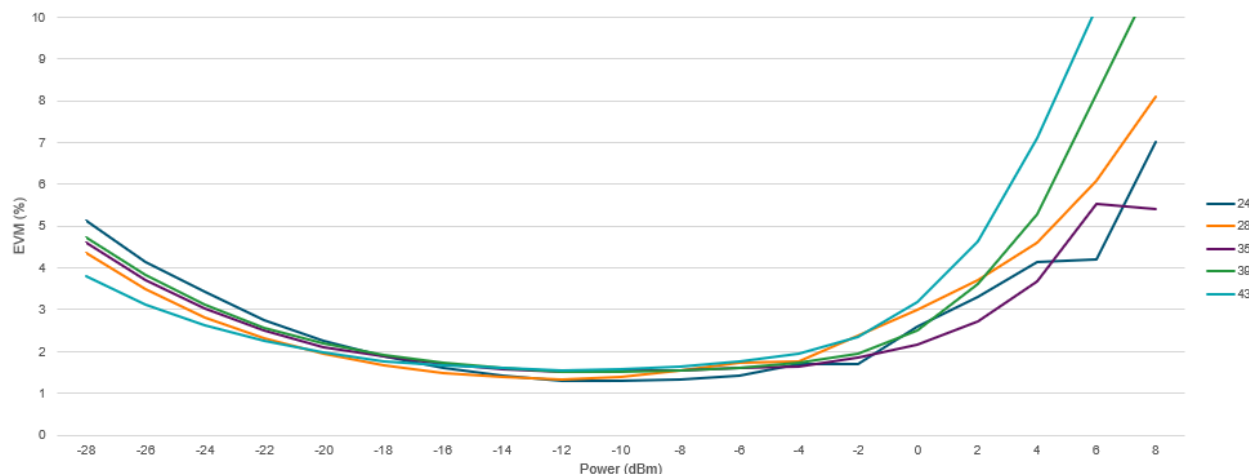
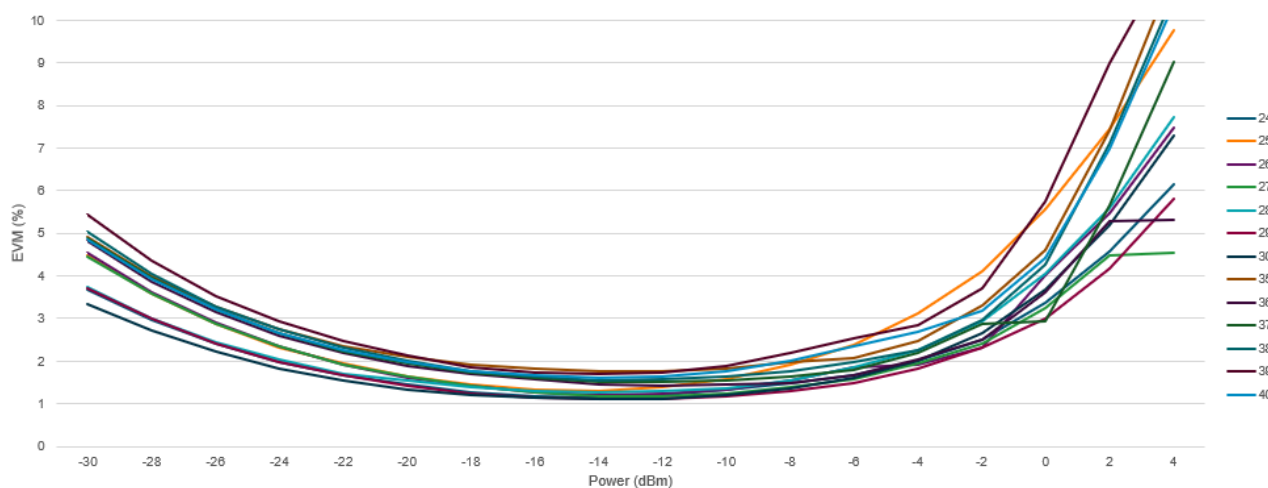
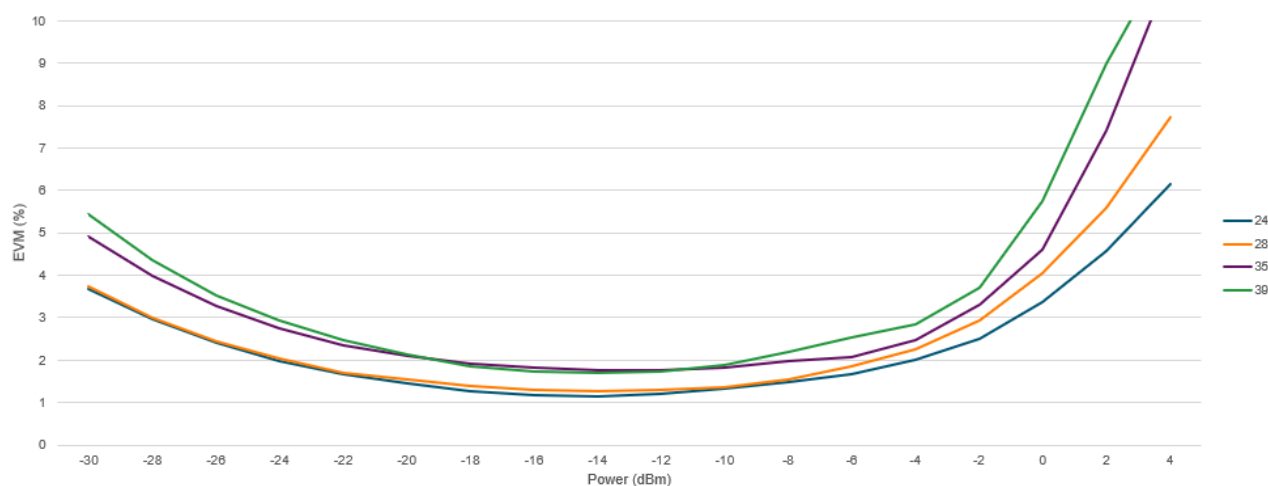
Stimulus EVM for RMM-44 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 83.
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e6
- Base Frequency left as default
- Waveform = NR5G_MCS27_UL_100MHz_64QAM

Stimulus EVM for RMM-44 without switch 24 GHz to 44 GHz



Stim EVM x1 RMM-44

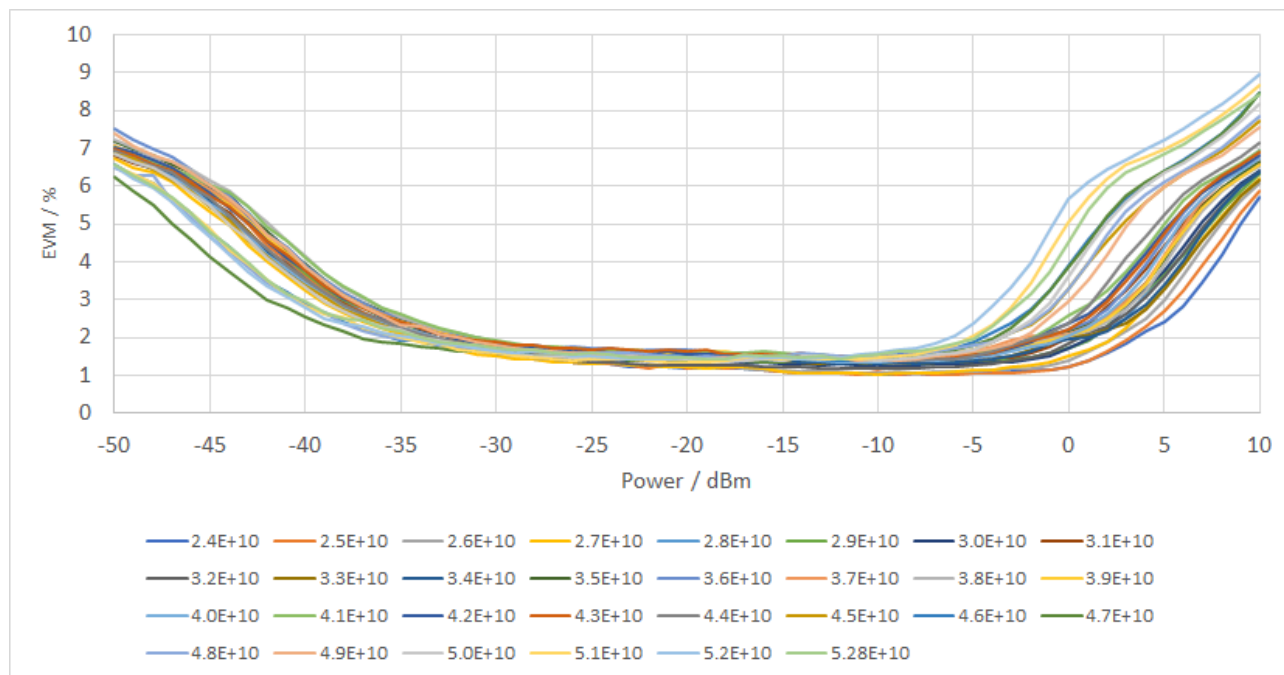
Stimulus EVM RMM-44 without switch 24 GHz to 44 GHz with few frequencies**Stimulus EVM (x1) RMM-44 direct 24 GHz to 44 GHz with few frequencies****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies**

WSMM stimulus EVM for RMM-53

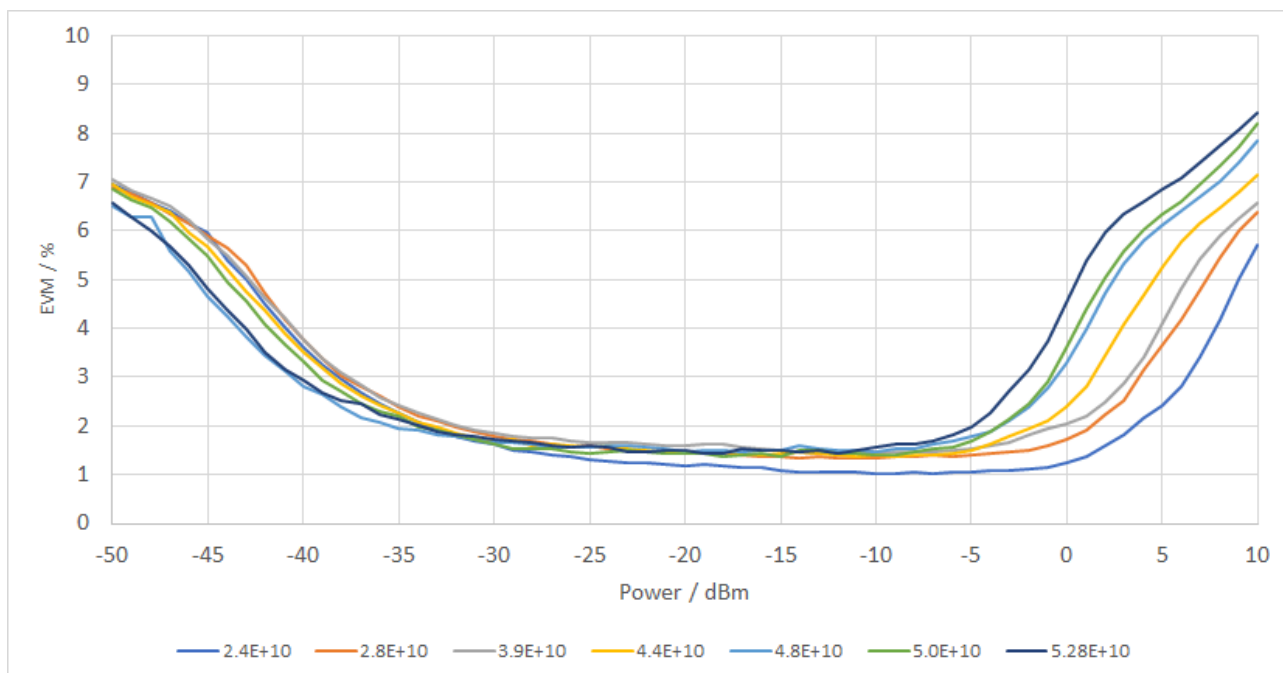
Stimulus EVM for RMM-53 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 83.
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 10e6
 - Base Frequency = 3.9e9
- Waveform = UL_MCS27_256QAM_100MHz_FR2_122p88Msps

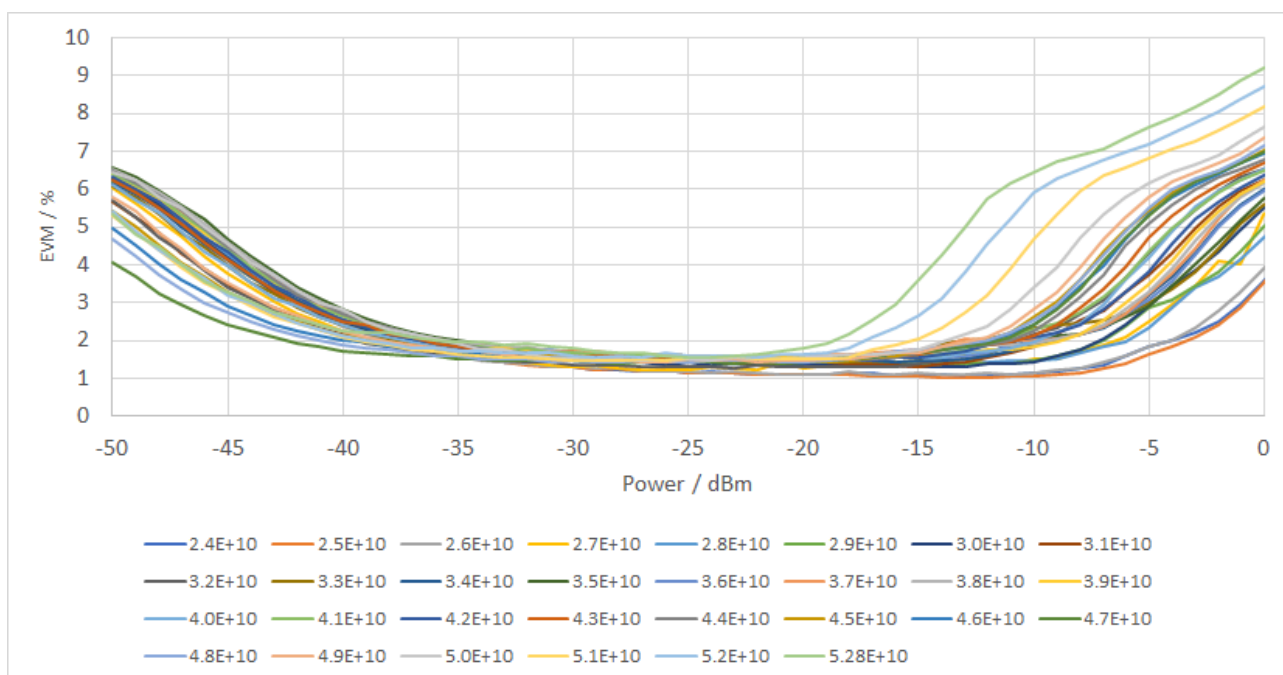
Stimulus EVM RMM-53 without switch 24 GHz to 52.8 GHz



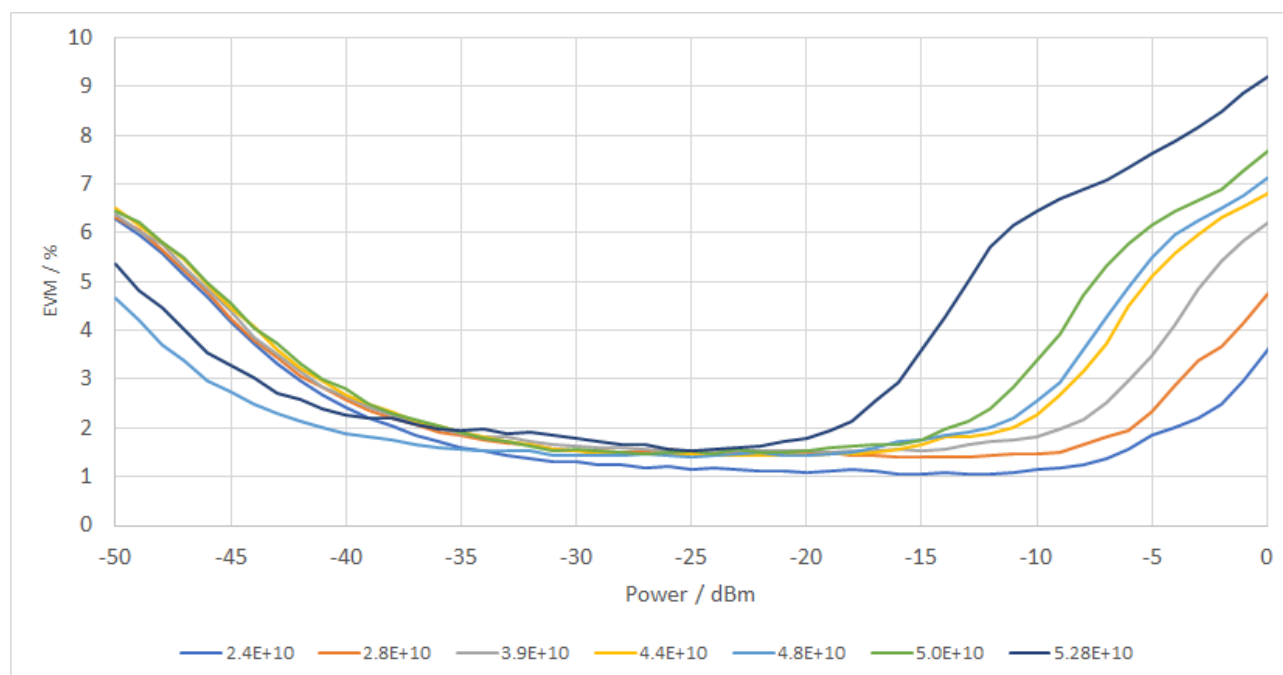
Stim EVM RMM53 without switch

Stimulus EVM RMM-53 without switch 24 GHz to 52.8 GHz with few frequencies

Stim EVM RMM53 without switch few freq

Stimulus EVM RMM-53 with 1:8 switch 24 GHz to 52.8 GHz

Stim EVM RMM53 with switch

Stimulus EVM RMM-53 with 1:8 switch 24 GHz to 52.8 GHz with few frequencies

Stim EVM Rmm53 with switch few freq

WSMM stimulus maximum output power

This is a measure of guaranteed maximum output power, measured as saturation power.

It is measured with the following conditions:

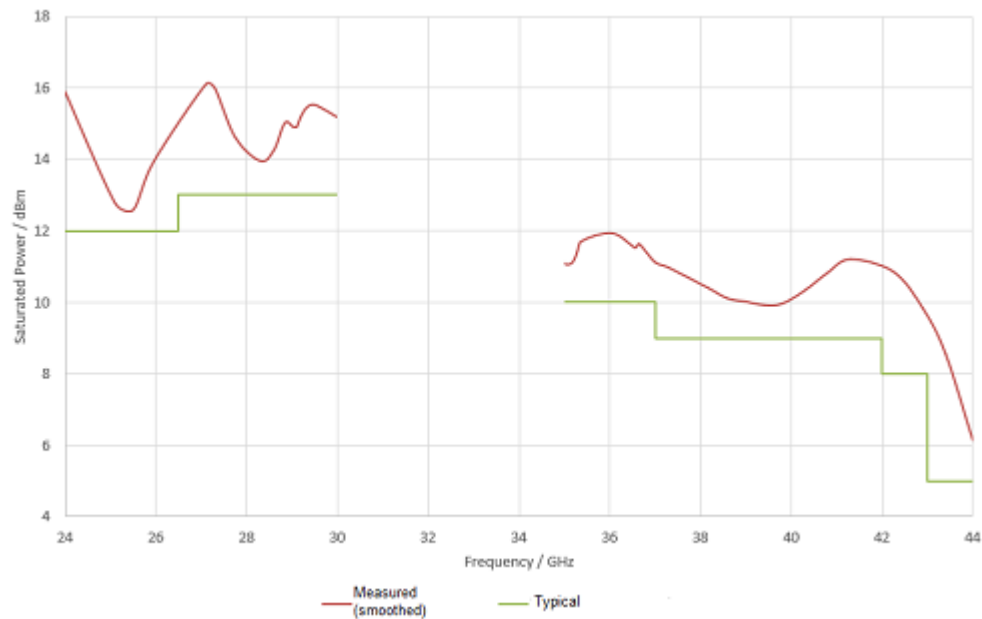
- All values are meas values, according to *Terms and conditions* on page 83.
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e-100 Hz

RMM-44 without Switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$24 \leq f < 26.5$	+12
$26.5 \leq f \leq 30$	+13
$35 \leq f \leq 37$	+10
$37 < f \leq 42$	+9
$42 < f \leq 43$	+8
$43 < f \leq 44$	+5

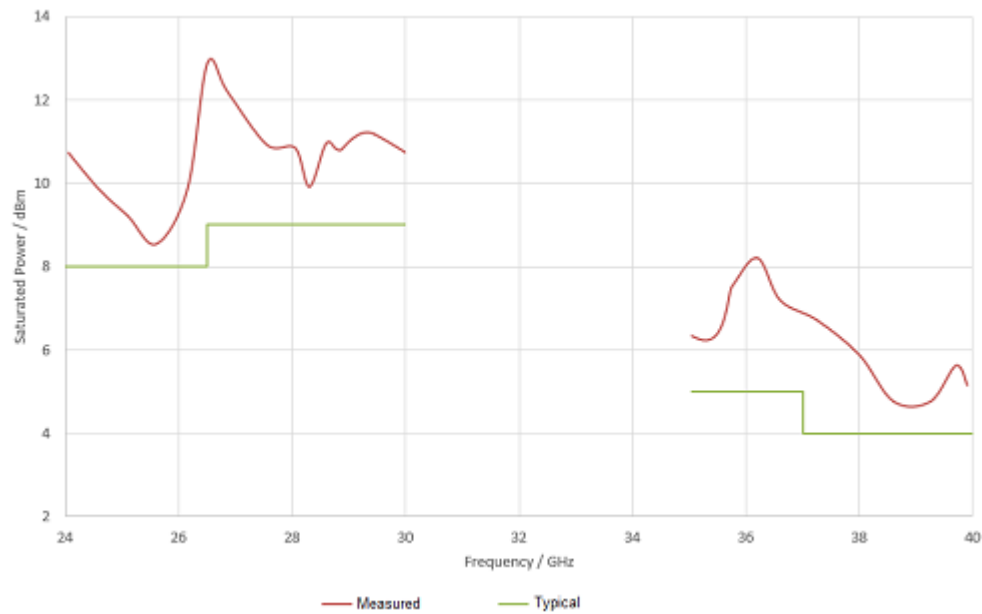


RMM-44 without switch

RMM-44 with 1:8 switch

Typical
power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$24 \leq f < 26.5$	+8
$26.5 \leq f \leq 30$	+9
$35 \leq f \leq 37$	+5
$37 < f \leq 40$	+4



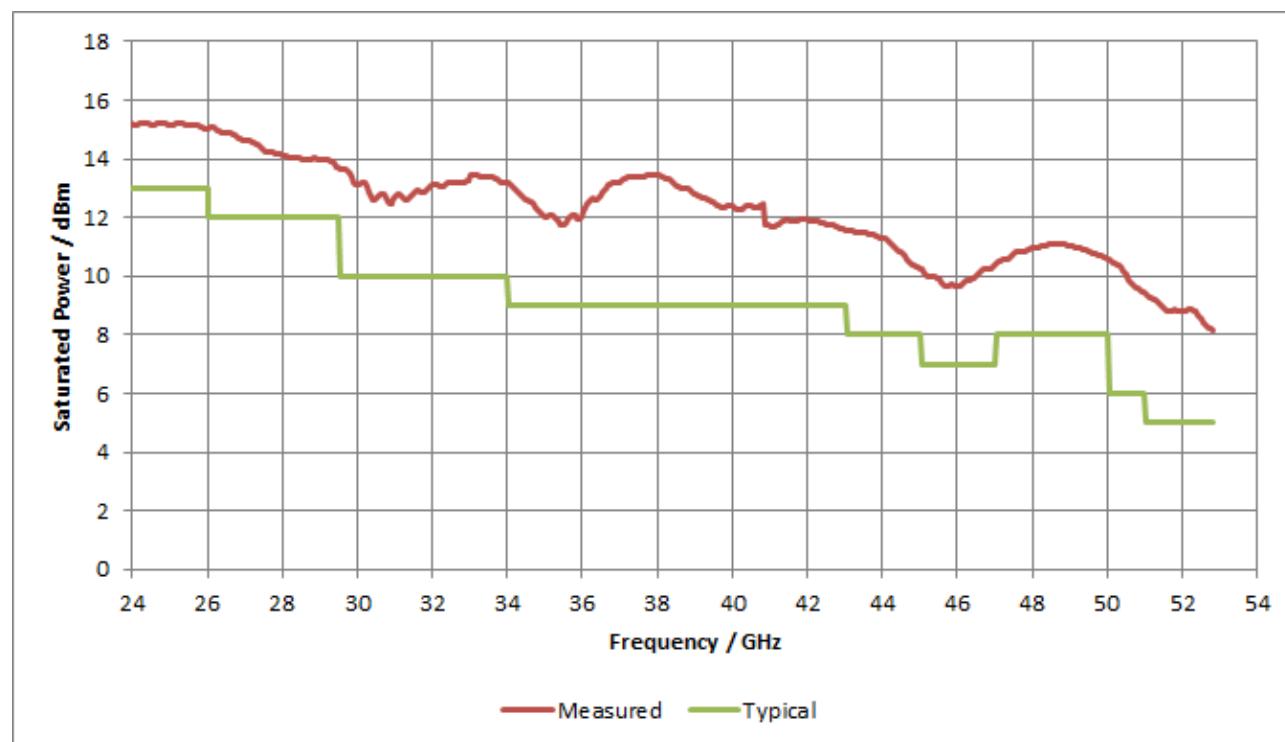
RMM-44 with 1:8 switch

RMM-53 without switch

Typical

Power setting = +20 dBm

Frequency (GHz)	Saturated power (dBm)
$24 \leq f \leq 26$	+13
$26.0 < f \leq 29.5$	+12
$29.5 < f \leq 34$	+10
$34 < f \leq 43$	+9
$43 < f \leq 45$	+8
$45 < f \leq 47$	+7
$47 < f \leq 50$	+8
$50 < f \leq 51$	+6
$51 < f \leq 52.8$	+5



Stim power saturated RMM-53 without switch

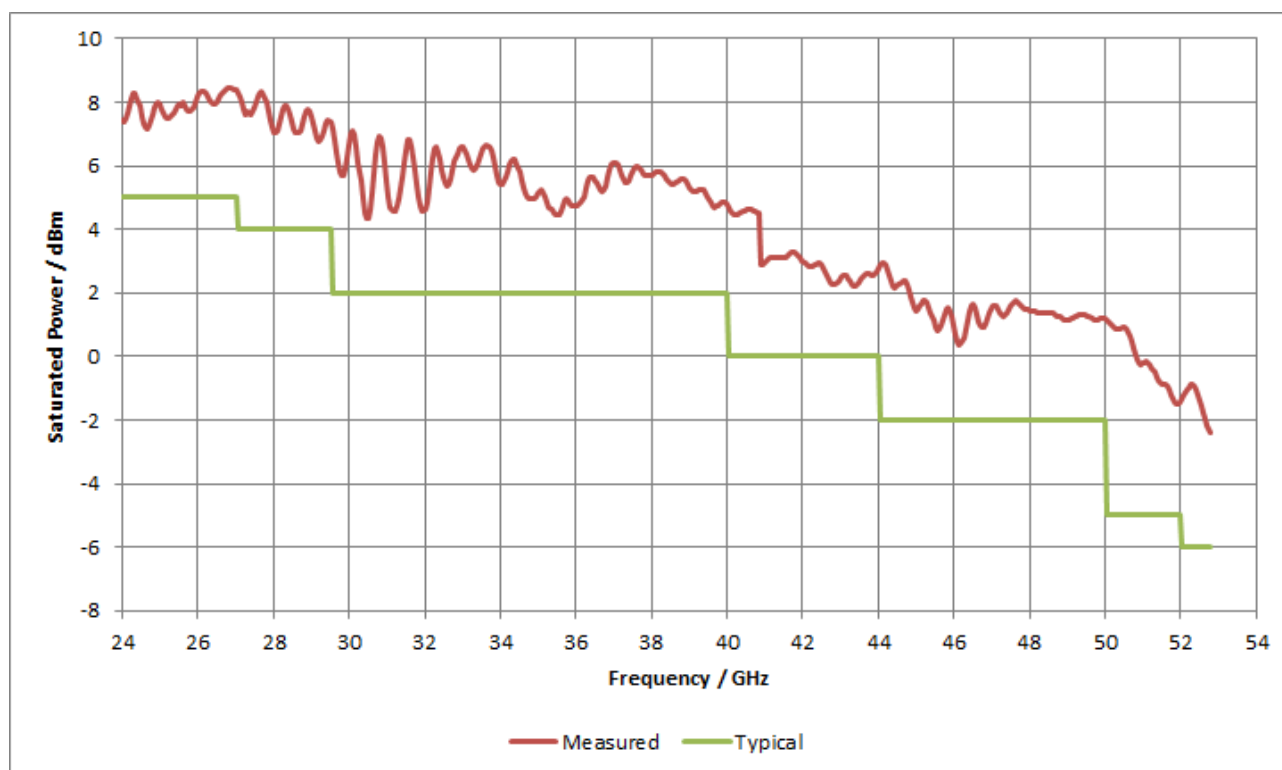
RMM-53 with 1:8 switch

Typical

Power setting = +20 dBm

Frequency (GHz)	Saturated power (dBm)
$24 \leq f \leq 27$	+5
$27 < f \leq 29.5$	+4
$29.5 < f \leq 40$	+2
$40 < f \leq 44$	0

Frequency (GHz)	Saturated power (dBm)
$44 < f \leq 50$	-2
$50 < f \leq 52$	-5
$52 < f \leq 52.8$	-6



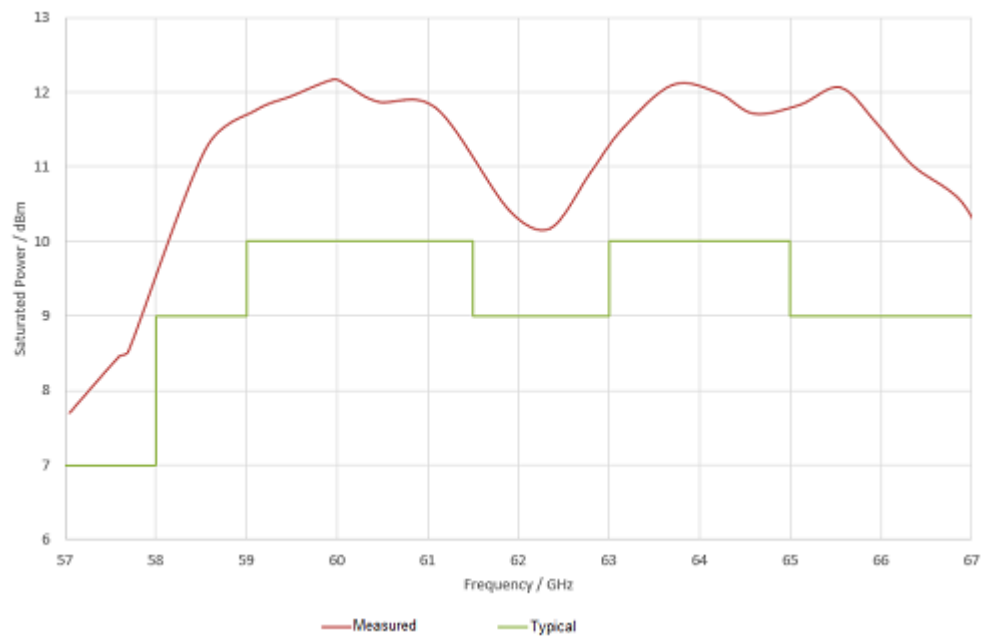
Stim power saturated RMM-53 with switch

RMM-72 without switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$57 \leq f < 58$	+7
$58 \leq f < 59$	+9
$59 \leq f \leq 61.5$	+10
$61.5 < f < 63$	+9
$63 \leq f \leq 65$	+10
$65 < f \leq 67$	+9



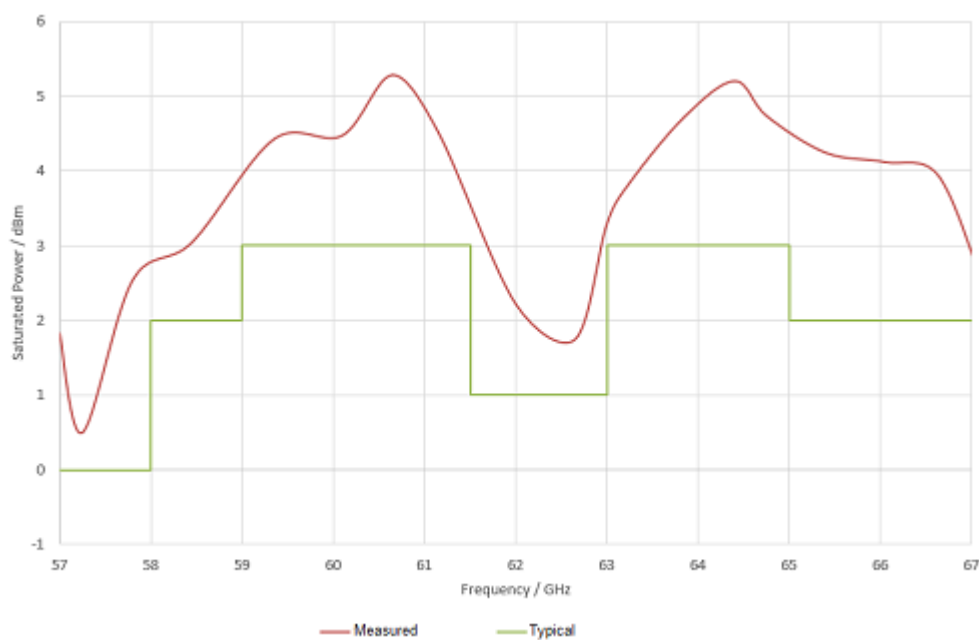
RMM-72 without switch

RMM-72 with 1:8 switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$57 \leq f < 58$	+0
$58 \leq f < 59$	+2
$59 \leq f \leq 61.5$	+3
$61.5 < f < 63$	+1
$63 \leq f \leq 65$	+3
$65 < f \leq 67$	+2



RMM-44 with 1:8 switch

WSMM stimulus phase noise

Stimulus phase noise conditions and notes:

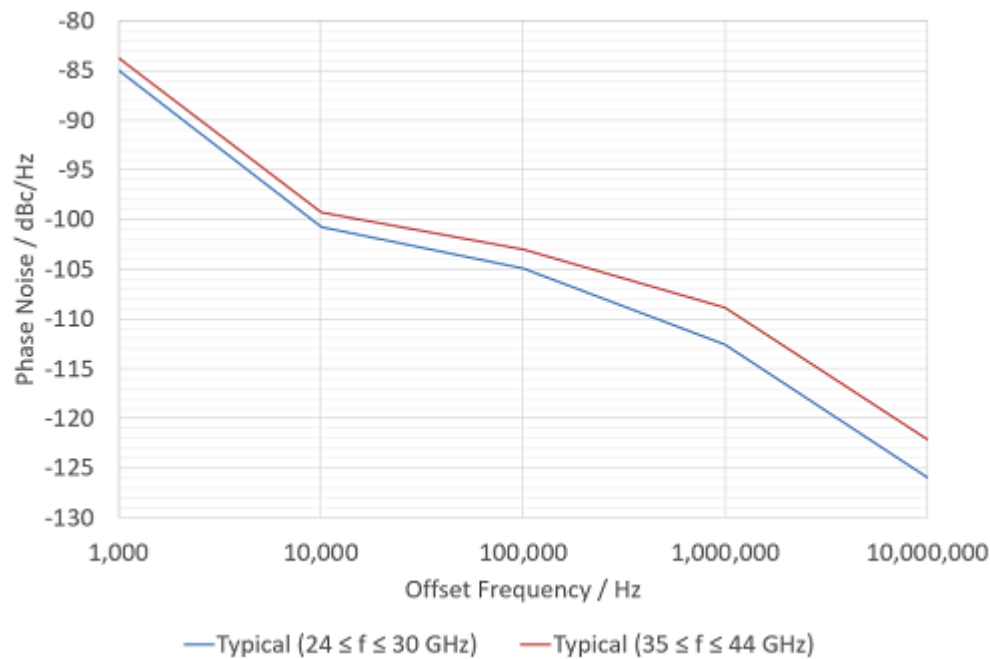
- All values are typical according to *Terms and conditions* on page 83, unless noted.
- Non-default parameters in SmarTest 8:
 - optimization = noiseFloor
 - optimizationNoiseBandwidth = 80 MHz
- Phase noise was measured with a -3 dBm carrier signal.
- Spurs are not taken into consideration.

Note All values are in dBc/Hz.

RMM-44 without switch

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$24 \leq f \leq 30$	-85.0	-100.8	-104.9	-112.6	-126.0
$35 \leq f \leq 44$	-83.7	-99.3	-103.0	-108.9	-122.2

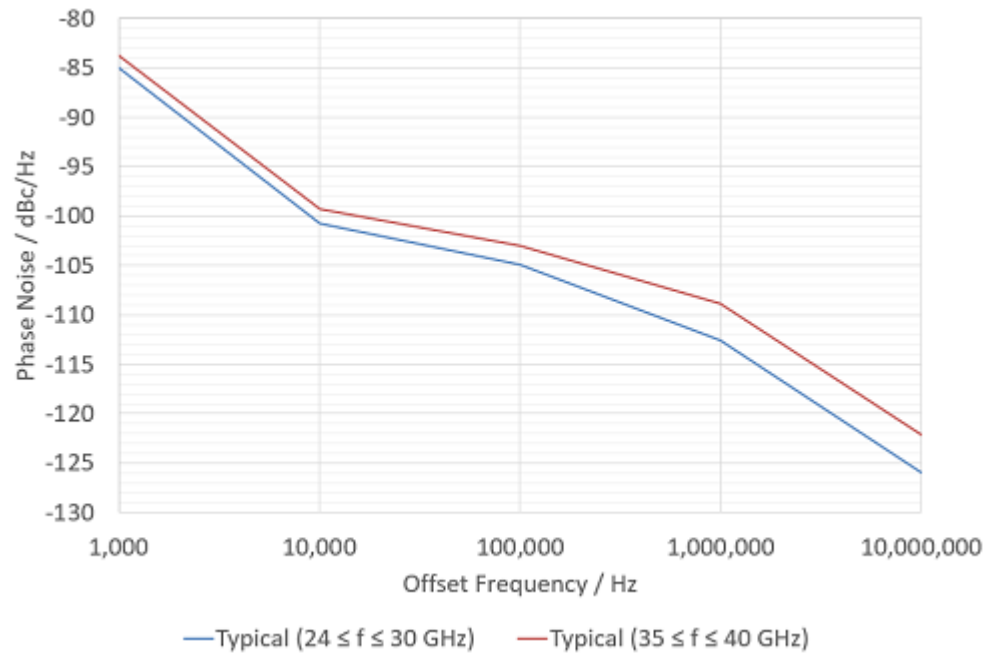


RMM-44 without switch

RMM-44 with 1:8 switch

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
24 ≤ f ≤ 30	-85.0	-100.8	-104.9	-112.6	-126.0
35 ≤ f ≤ 40	-83.7	-99.3	-103.0	-108.9	-122.2

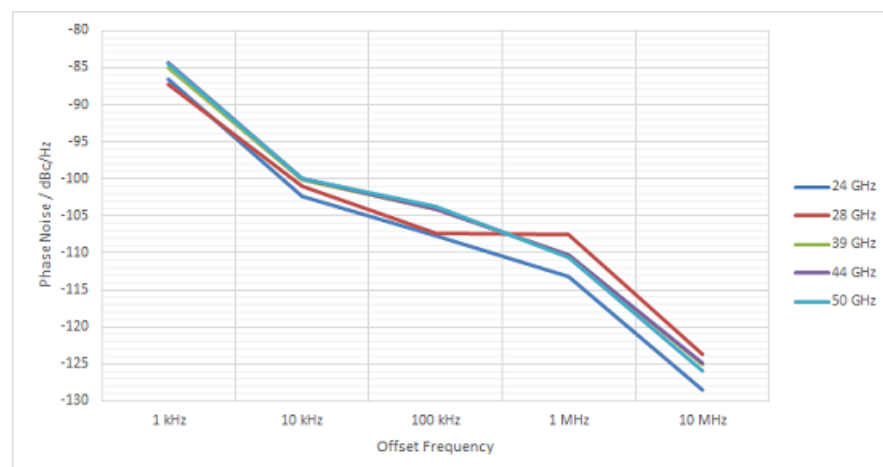


RMM-44 with 1:8 switch

RMM-53 without switch or with 1:8 switch

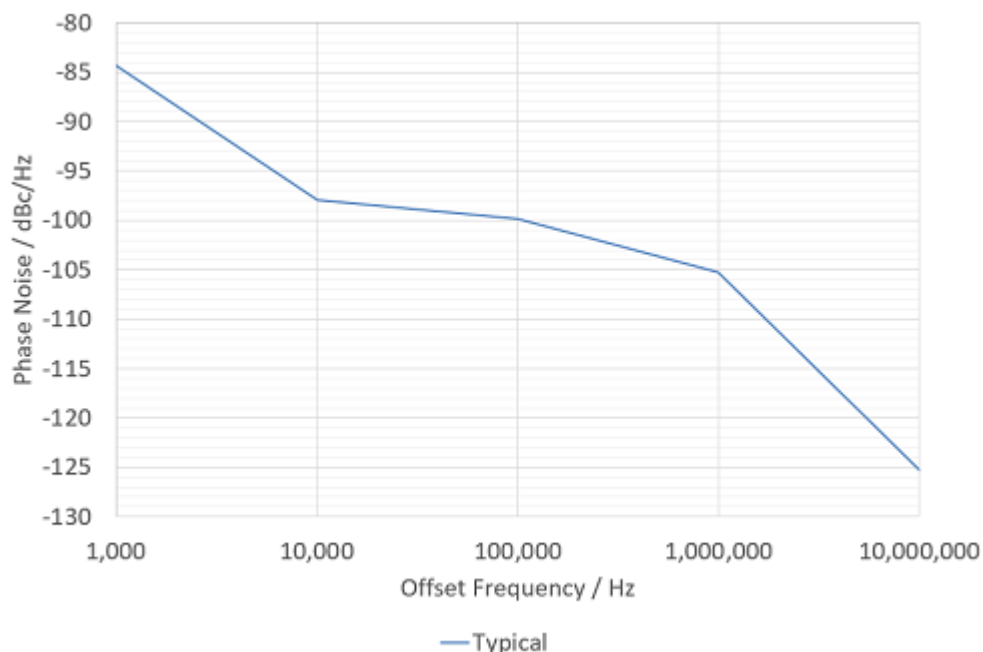
Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$24 \leq f \leq 27$	-85.0	-101.1	-105.5	-112.1	-127.3
$27 < f \leq 32$	-82.4	-99.5	-105.5	-106.4	-122.7
$32 < f \leq 44$	-82.4	-98.6	-102.3	-108.4	-123.1
$44 < f \leq 50$	-82.4	-98.6	-101.5	-108.4	-124

**RMM-53 with 1:8 switch****RMM-72 without switch or with 1:8 switch**

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$57 \leq f \leq 67$	-84.3	-97.9	-99.9	-105.2	-125.3



RMM-72 without switch

WSMM measure

WSMM measure accuracy

Measure accuracy conditions and notes:

- All values are typical according to *Terms and conditions* on page 83, unless noted.
- All power levels are CW signals, unless noted.
- Non-default parameters in SmarTest 8:
 - bandwidthOfInterest = 1 kHz
 - resultAveraging = 16

Note All values are in dB.

RMM-44 without Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)		
	$-85 \leq L_v < -60$	$-60 \leq L_v < -5$	$-5 \leq L_v \leq +4$
$24 \leq f < 26$	± 0.99	± 0.94	± 0.81
$26 \leq f \leq 30$	± 0.71	± 0.65	± 0.56
$35 \leq f < 40$	± 1.08	± 0.92	± 0.81
$40 \leq f \leq 44$	± 0.93	± 0.77	± 0.68

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$24 \leq f < 26$	-	± 0.07
$26 \leq f \leq 30$	-	± 0.07
$35 \leq f < 40$	-	± 0.09
$40 \leq f \leq 44$	-	± 0.12

RMM-44 with 1:8 Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)		
	$-85 \leq L_v < -60$	$-60 \leq L_v < -5$	$-5 \leq L_v \leq +4$
$24 \leq f < 26.5$	± 1.90	± 1.87	± 1.81
$26.5 \leq f \leq 30$	± 1.12	± 1.09	± 1.05
$35 \leq f \leq 40$	± 1.27	± 1.15	± 1.06

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

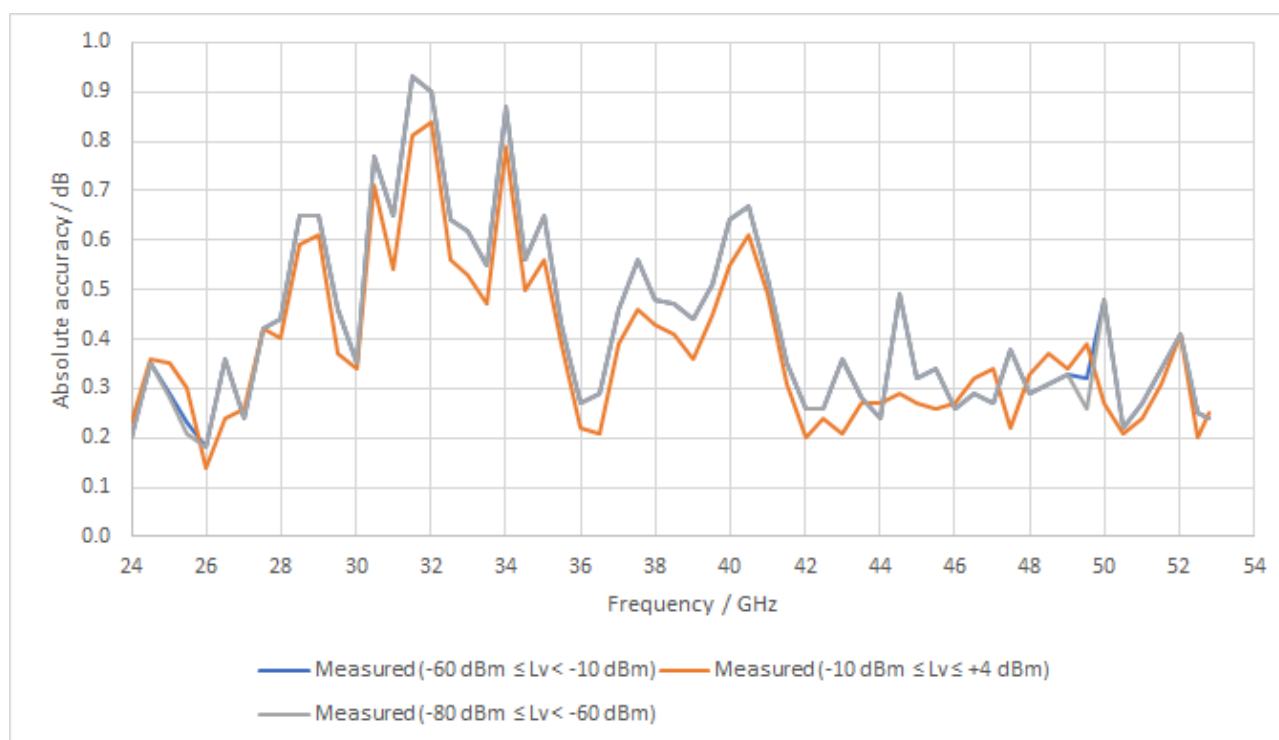
Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$24 \leq f < 26.5$	-	± 0.07
$26.5 \leq f \leq 30$	-	± 0.07
$35 \leq f \leq 40$	-	± 0.17

RMM-53 without switch

Typical

Frequency Range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v \leq +4$
$24 \leq f \leq 25$	± 0.73	± 0.71	± 0.75
$25 < f \leq 26$	± 0.53	± 0.65	± 0.66
$26 < f \leq 30$	± 0.86	± 0.98	± 0.92
$30 < f \leq 36$	± 1.12	± 1.22	± 1.11
$36 < f \leq 40$	± 0.67	± 0.76	± 0.66
$40 < f \leq 42$	± 0.77	± 0.88	± 0.81
$42 < f \leq 50$	± 0.86	± 0.74	± 0.61
$50 < f \leq 52.8$	± 0.38 ^[22]	± 0.41 ^[22]	± 0.41 ^[22]

^[22] Measured



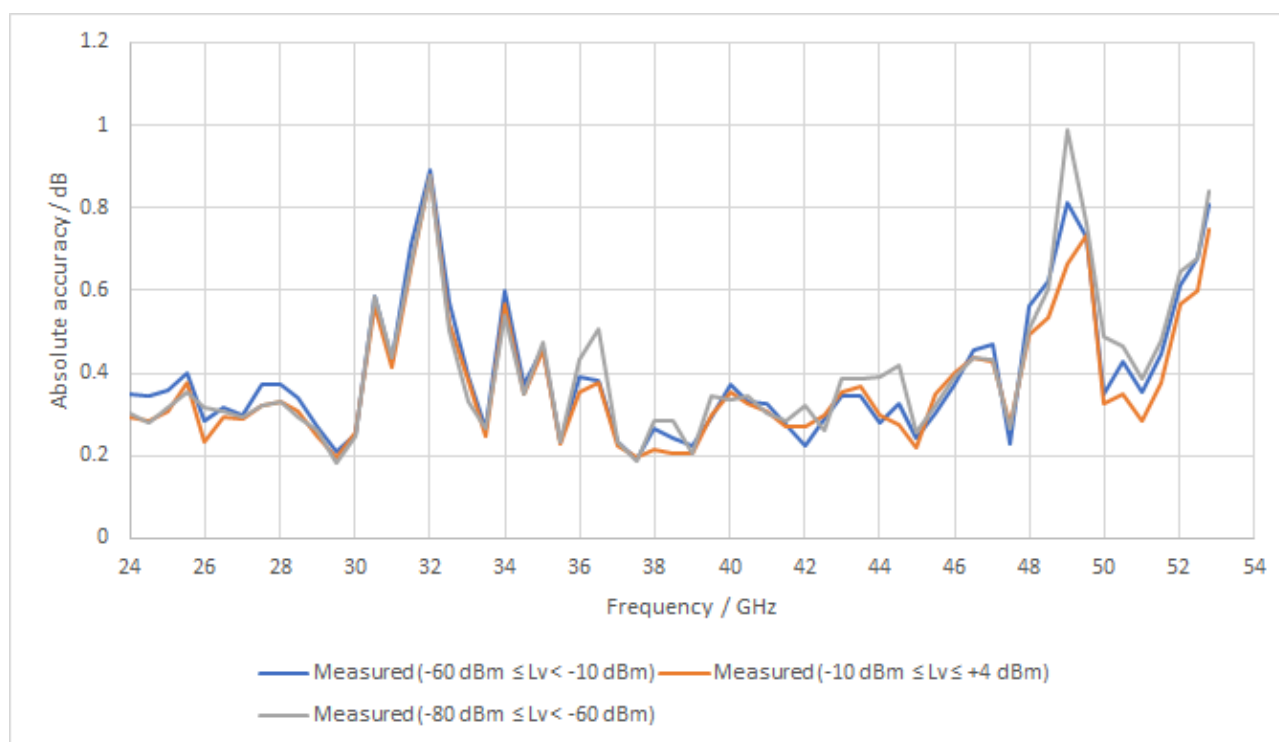
Meas accuracy RMM53 without switch

The given values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.

RMM-53 with 1:8 switch

Typical

Frequency Range	Power level range (dBm)		
	$-80 \leq Lv < -60$	$-60 \leq Lv < -10$	$-10 \leq Lv \leq +4$
$24 \leq f \leq 25$	± 0.75	± 0.80	± 0.73
$25 < f \leq 30$	± 0.67	± 0.71	± 0.68
$30 < f \leq 35$	± 1.42	± 1.44	± 1.43
$35 < f \leq 37$	± 0.97	± 0.87	± 0.86
$37 < f \leq 45$	± 0.82	± 0.71	± 0.70
$45 < f \leq 50$	± 1.37	± 1.13	± 1.03
$50 < f \leq 52.8$	± 0.85 ^[22]	± 0.81 ^[22]	± 0.75 ^[22]



Meas accuracy RMM53 with switch

The given values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.

RMM-72 without Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)	
	-85 ≤ Lv < -60	-60 ≤ Lv ≤ +5
57 ≤ f < 59.5	±1.58	±1.41
59.5 ≤ f ≤ 61.5	±1.42	±1.26
61.5 < f < 63	±2.50	±2.57
63 ≤ f ≤ 65	±1.61	±1.42 ^[23]
65 < f ≤ 67	±1.97	±1.75 ^[24]

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

^[23] for -60 ≤ Lv ≤ +4 dBm

^[24] for -60 ≤ Lv ≤ +3 dBm

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$57 \leq f < 59.5$	-	± 0.07
$59.5 \leq f \leq 61.5$	± 0.14	± 0.11
$61.5 < f < 63$	-	± 0.07
$63 \leq f \leq 65$	-	± 0.07
$65 < f \leq 67$	-	± 0.07

RMM-72 with 1:8 Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)	
	$-85 \leq L_v < -60$	$-60 \leq L_v \leq +1$
$57 \leq f \leq 60.5$	± 1.36	± 1.19
$60.5 < f \leq 61.5$	± 1.83	± 1.69
$61.5 < f < 62.5$	± 2.33	± 2.13
$62.5 \leq f < 65$	± 1.68	± 1.49 ^[25]
$65 \leq f \leq 67$	± 1.31	± 0.97 ^[25]

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$57 \leq f \leq 60.5$	± 0.18	± 0.12
$60.5 < f \leq 61.5$	± 0.02	± 0.07
$61.5 < f < 62.5$	± 0.05	± 0.07
$62.5 \leq f < 65$	± 0.05	± 0.07
$65 \leq f \leq 67$	± 0.03	± 0.07

WSMM measure power

Measure power conditions:

- All values are absolute maximum ratings.

RMM-44 without switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 44$	+14

^[25] for $-60 \leq L_v \leq 0$ dBm

RMM-44 with 1:8 switch

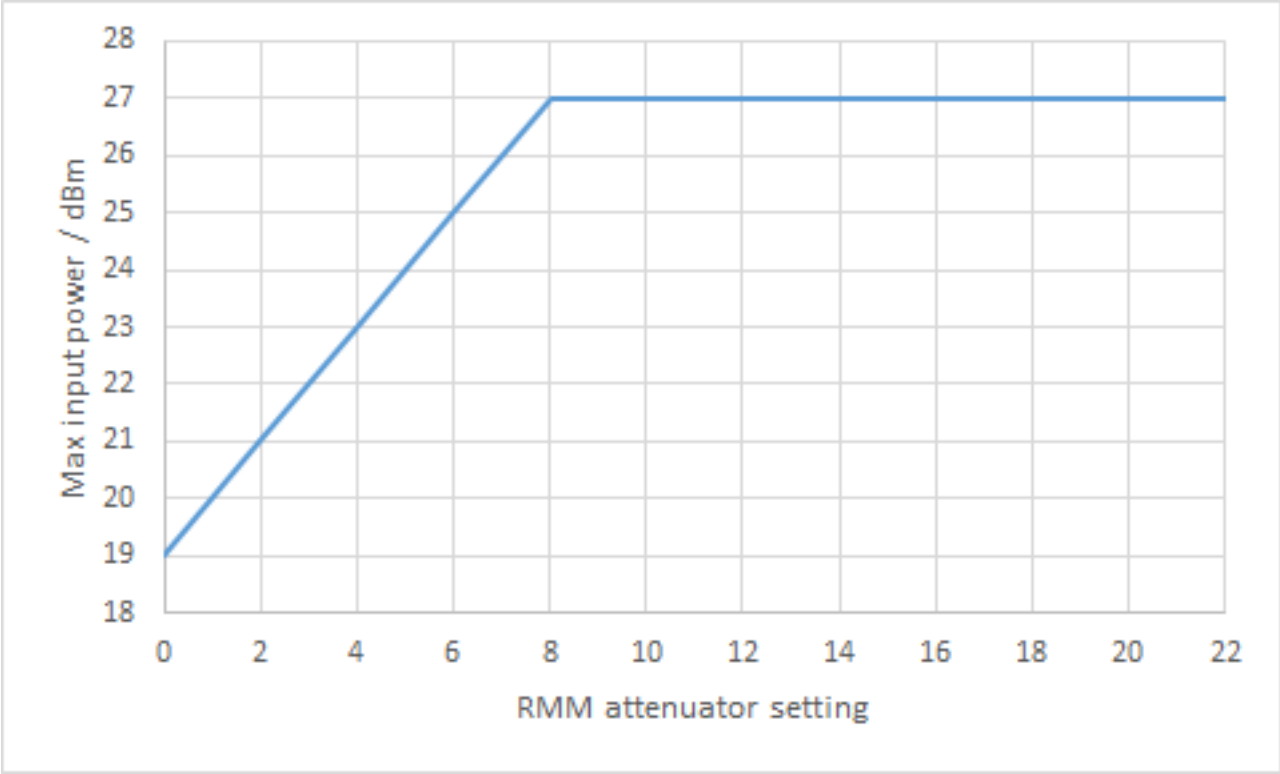
Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 40$	+20

RMM-53 without switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 52.8$	+27

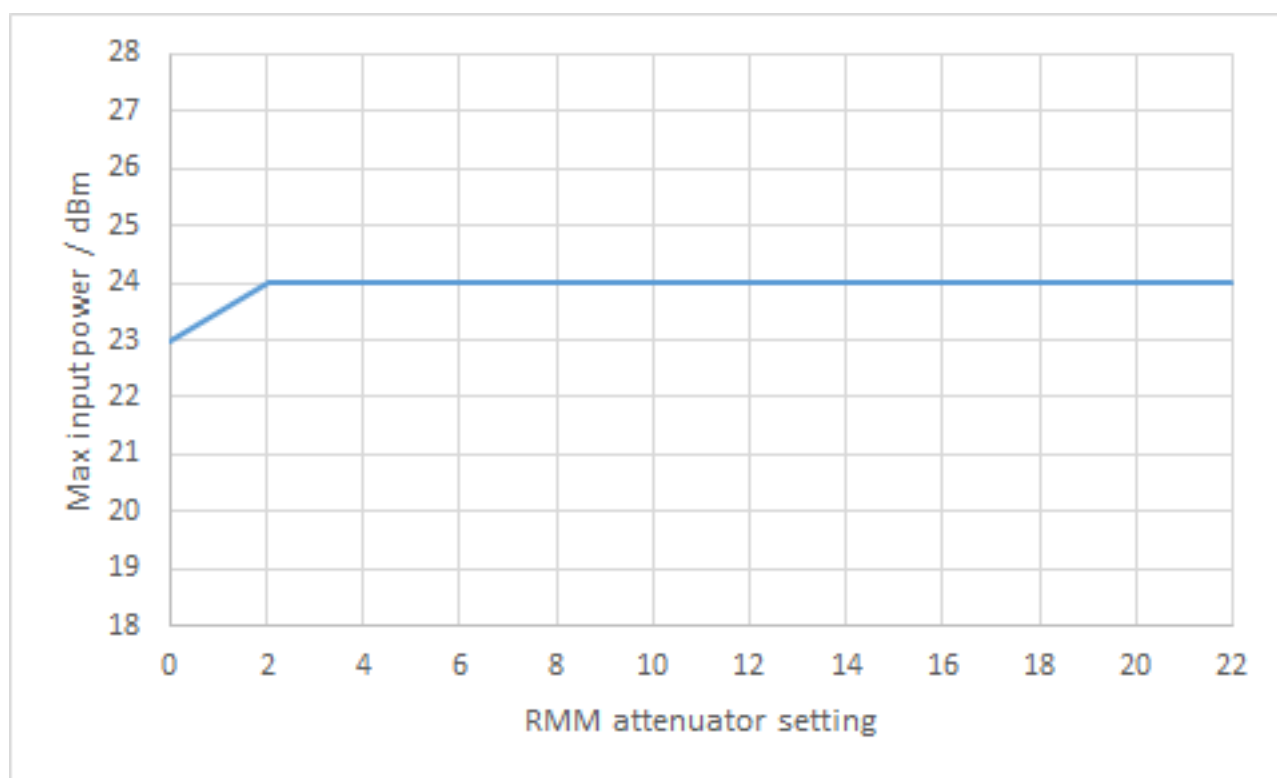


RMM53 measure power without switch

RMM-53 with 1:8 switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 52.8$	+24



RMM53 measure power with switch

RMM-72 without switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$57 \leq f < 61$	+8
$61 \leq f \leq 70$	+10

RMM-72 with 1:8 switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$57 \leq f \leq 70$	+15

WSMM measure noise floor

Measure noise floor conditions and notes:

- All values are typical.
- Non-default parameters in SmarTest 8:
 - expectedMaxPower = -130 dBm
 - samples = 1024
 - sampleRate = 250 MHz
 - bandwidthOfInterest = 1 MHz
 - resultAveraging = 8
 - ifFrequency = 15 MHz
- Center frequency is programmed to the specified frequency^[26].

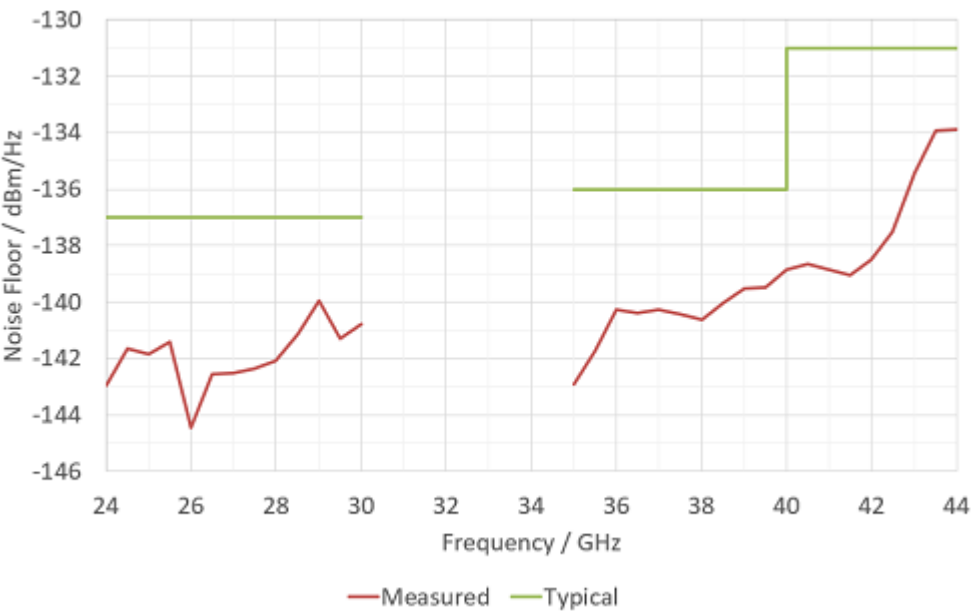
^[26] some center frequencies were shifted to avoid spurs

- Noise floor was measured with the RMM input terminated with 50 Ω.
- For optimal performance avoid generating or measuring strong signals on the neighbor channel.

RMM-44 without Switch

Typical
mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
24 ≤ f ≤ 30	-137
35 ≤ f ≤ 40	-136
40 < f ≤ 44	-131

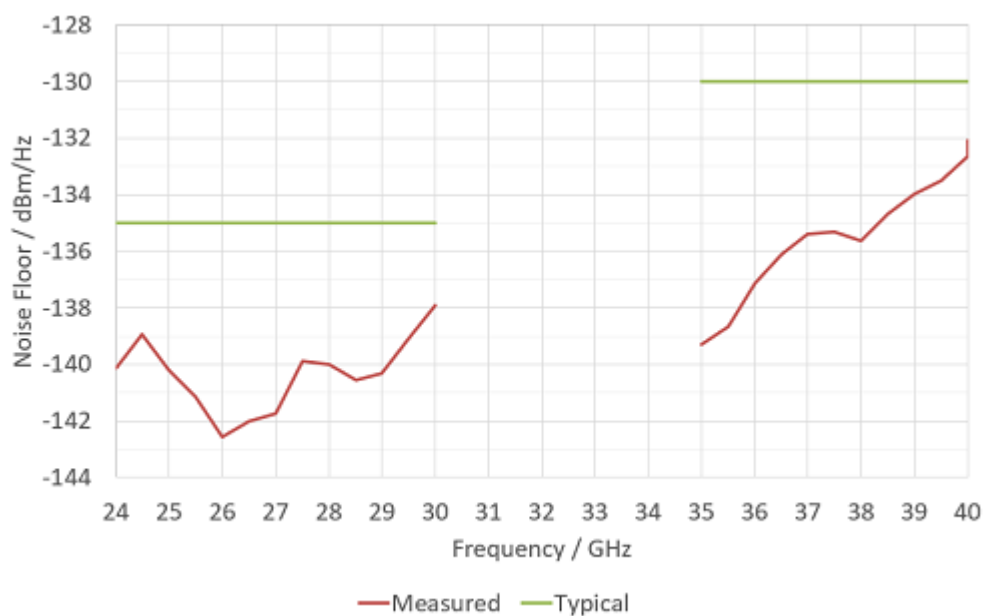


RMM-44 without switch

RMM-44 with 1:8 switch

Typical
mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
24 ≤ f ≤ 30	-135
35 ≤ f ≤ 40	-130

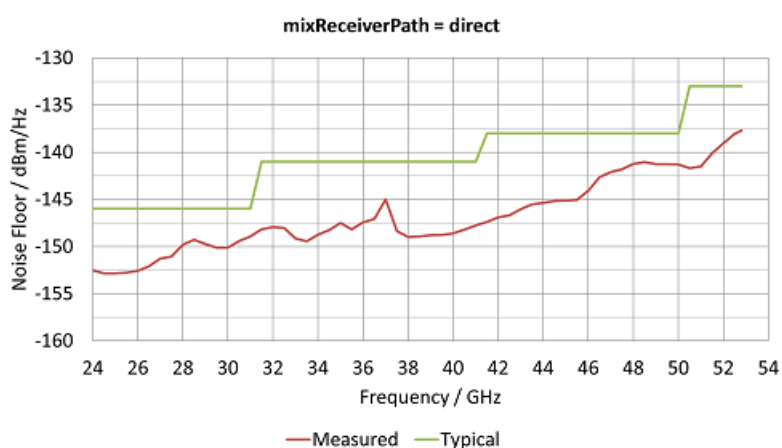


RMM-44 with 1:8 switch

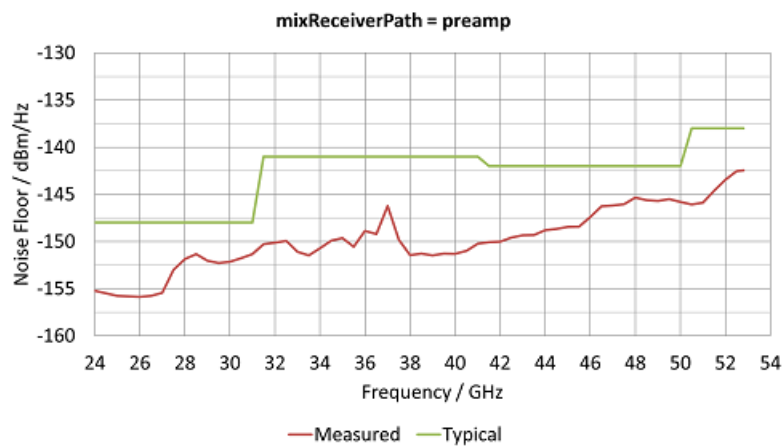
RMM-53 without switch

Typical

mixReceiverPath	Frequency range (GHz)	Noise floor (dBm/Hz)
direct	$24 \leq f \leq 31$	-146
	$31 < f \leq 41$	-141
	$41 < f \leq 50$	-138
	$50 < f \leq 52.8$	-133
preamp	$24 \leq f \leq 31$	-148
	$31 < f \leq 41$	-141
	$41 < f \leq 50$	-142
	$50 < f \leq 52.8$	-138



Meas Noise RMM53 without switch receiverPath direct

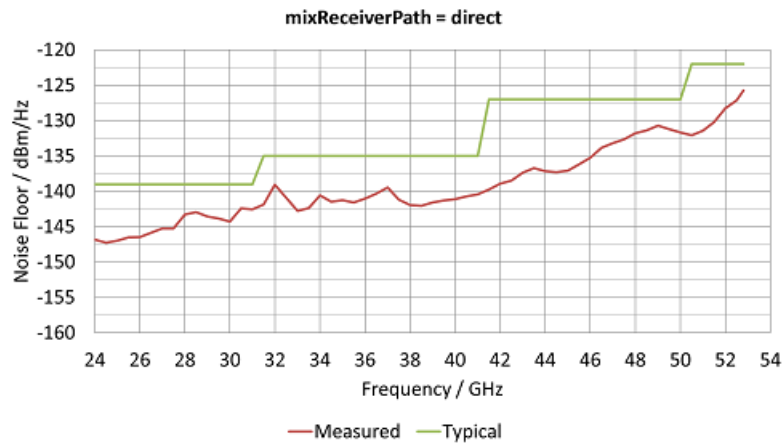


Meas Noise RMM53 without switch receiverPath preamp

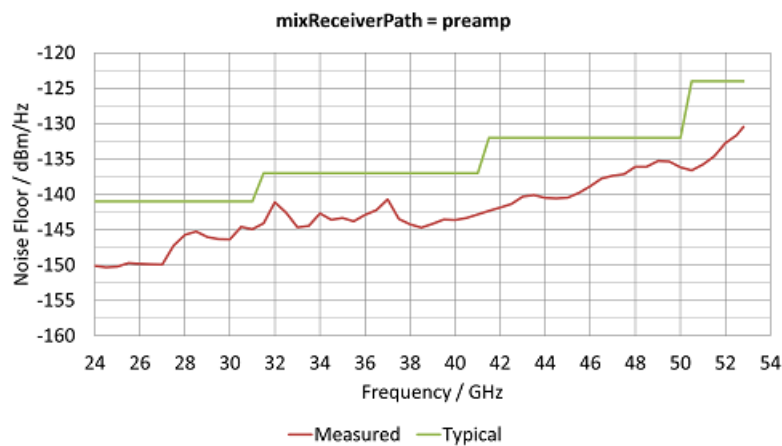
RMM-53 with 1:8 switch

Typical

mixReceiverPath	Frequency (GHz)	Noise floor (dBm/Hz)
direct	24 ≤ f ≤ 31	-139
	31 < f ≤ 41	-135
	41 < f ≤ 50	-127
	50 < f ≤ 52.8	-122
preamp	24 ≤ f ≤ 31	-141
	31 < f ≤ 41	-137
	41 < f ≤ 50	-132
	50 < f ≤ 52.8	-124



Meas noise RMM53 with switch receiverPath direct



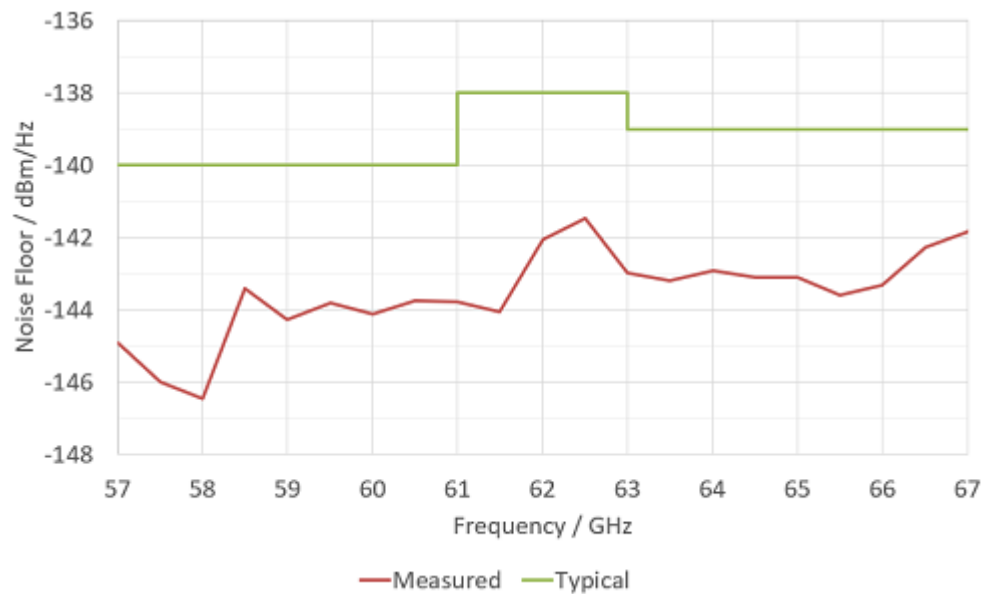
Meas noise RMM53 with switch receiverPath preamp

RMM-72 without switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 61$	-140
$61 < f < 63$	-138
$63 \leq f \leq 67$	-139



RMM-72 without switch

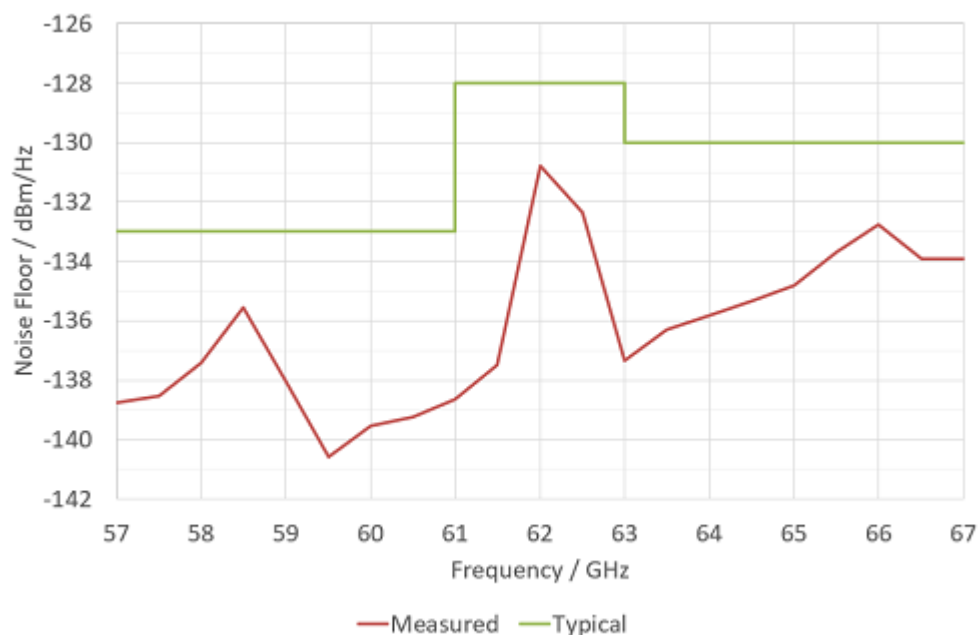
RMM-72 with 1:8 switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 61$	-133

Frequency range (GHz)	Noise floor (dBm/Hz)
$61 < f < 63$	-128
$63 \leq f \leq 67$	-130



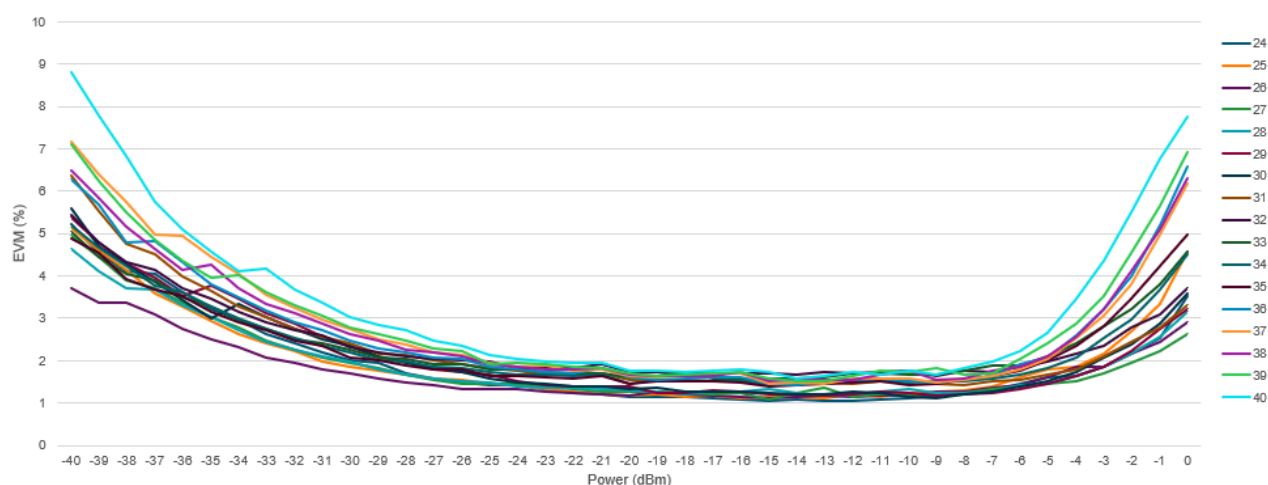
RMM-44 with 1:8 switch

WSMM measure EVM for RMM-44

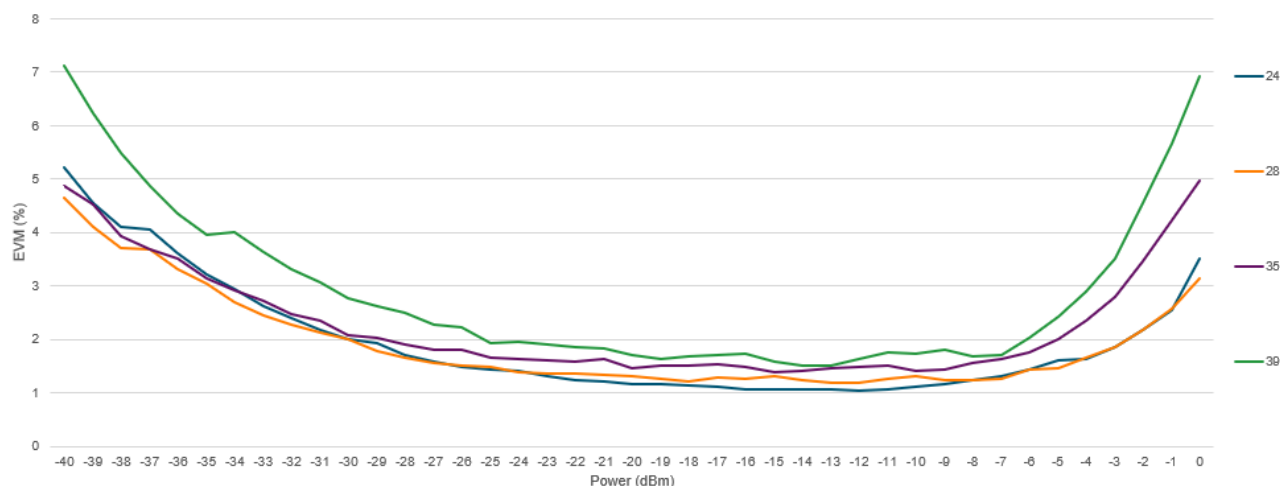
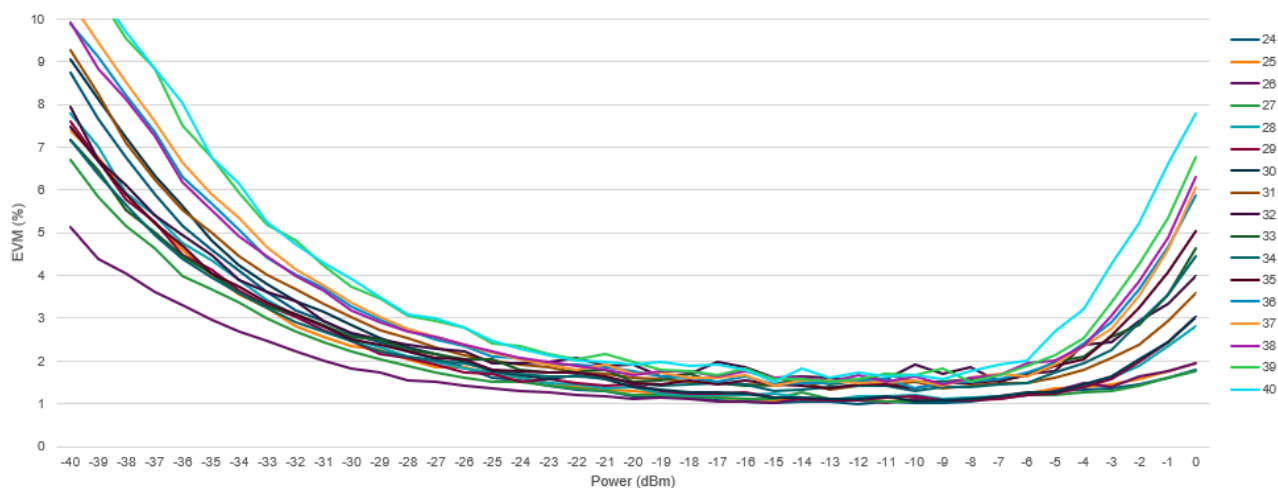
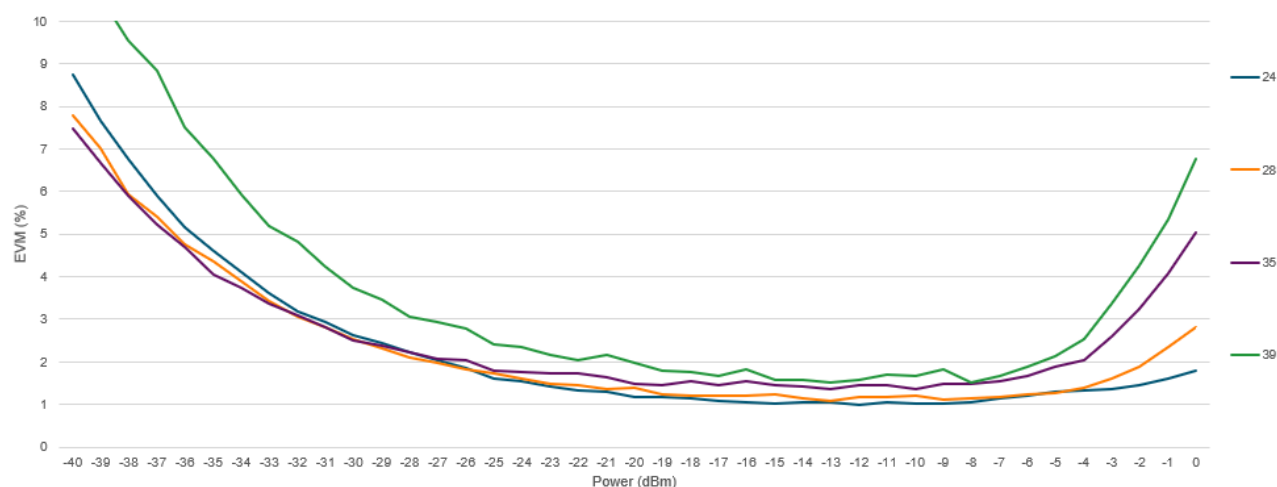
Measure EVM for RMM-44 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 83.
- Non-default parameters in SmarTest 8
 - RFIM18: Direct path
 - WSRF: default receiver path chosen by SmarTest
- Base Frequency left as default
- Waveform = NR5G_MCS27_UL_100MHz_64QAM

Measure EVM for RMM-44 without switch 24 GHz to 44 GHz



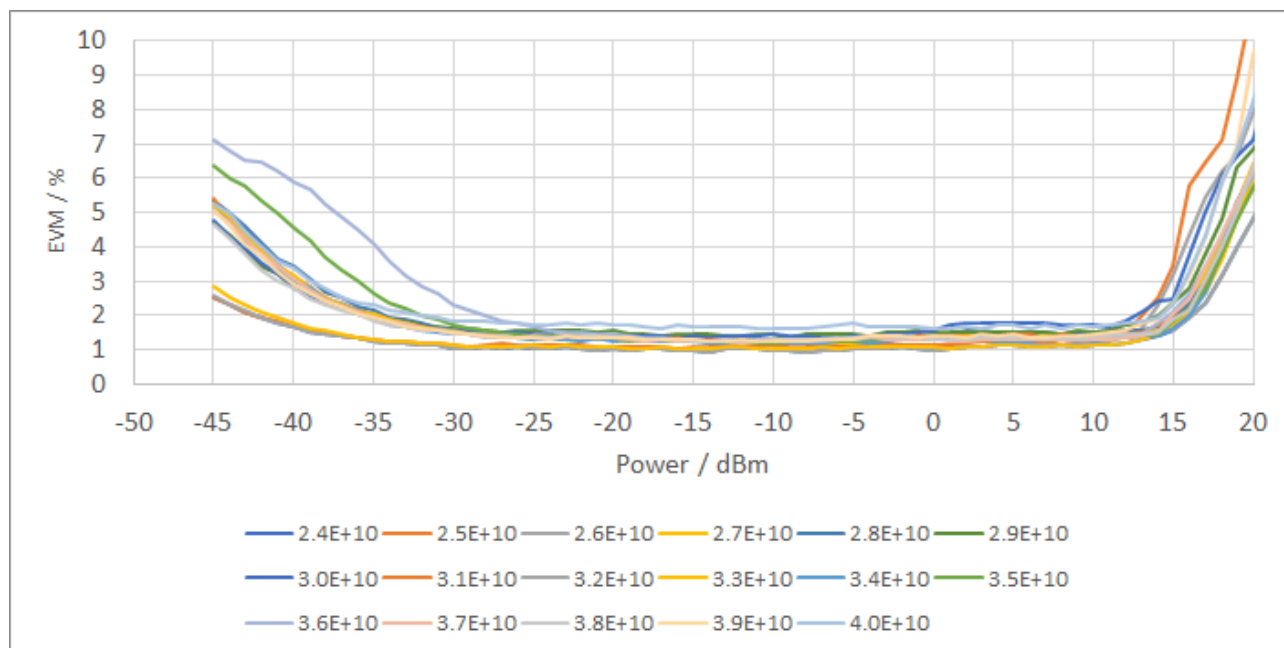
Meas EVM x1 RMM-44

Measure EVM RMM-44 without switch 24 GHz to 44 GHz with few frequencies**Measure EVM (x1) RMM-44 without switch 24 GHz to 44 GHz with few frequencies****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies**

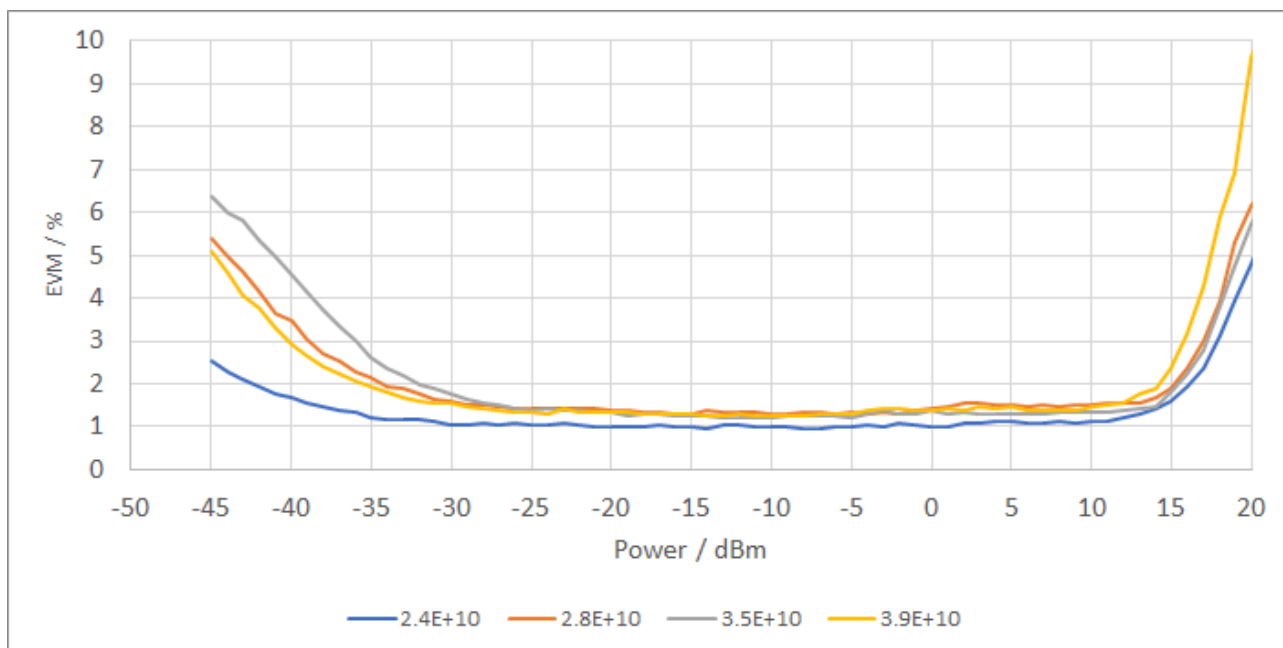
WSMM measure EVM for RMM-53

Measure EVM for RMM-53 conditions and notes:

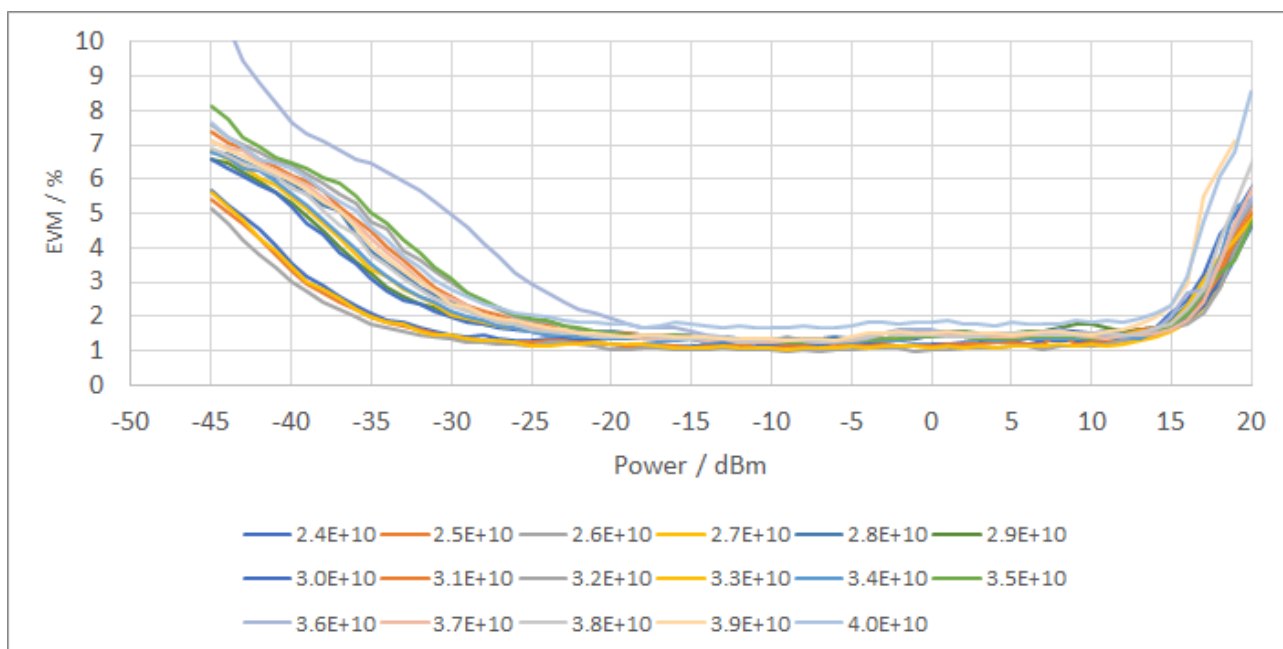
- All values are meas values, according to *Terms and conditions* on page 83.
- Non-default parameters in SmarTest 8
 - WSRF: default receiver path chosen by SmarTest
- Base Frequency = 5.1 GHz
- Waveform = UL_MCS27_256QAM_100MHz_FR2_122p88Msps_4slots_2subfr_short
- Frequency range limited by external equipment to 40 GHz

Measure EVM RMM-53 without switch 24 GHz to 40 GHz

Meas EVM RMM53 without switch

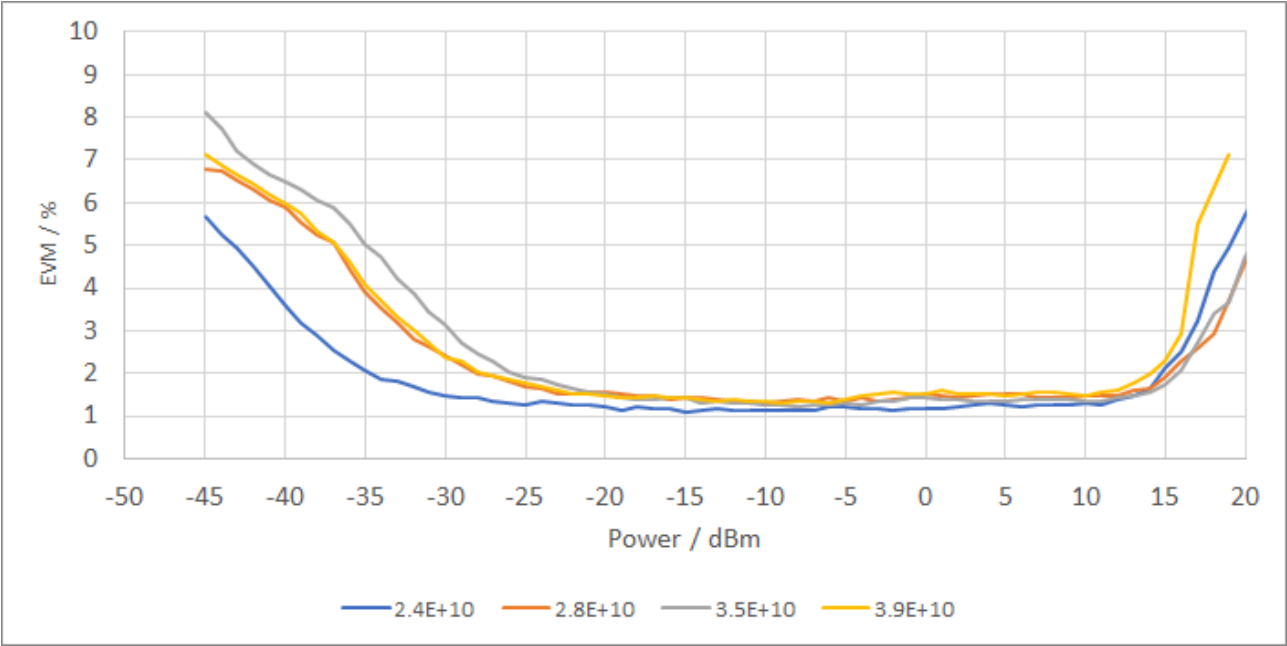
Measure EVM RMM-53 without switch with few frequencies

Meas EVM RMM53 without switch few freq

Measure EVM RMM-53 with 1:8 switch 24 GHz to 40 GHz

Meas EVM RMM53 with switch

Measure EVM RMM-53 with 1:8 switch with few frequencies



Meas EVM RMM53 with switch few freq

WSMM measure P1dB

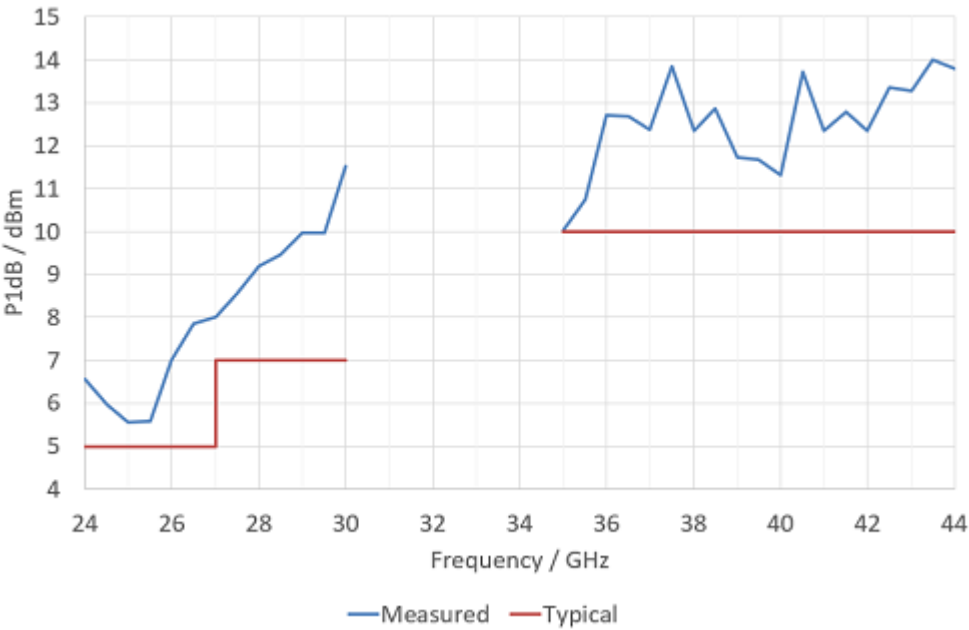
Measure P1dB conditions:

- Non-default parameters in SmarTest 8
 - mixReceiverPath = direct
 - bandwidthOfInterest = 1 kHz
 - resultAveraging = 16

RMM-44 without Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f < 27$	+5
$27 \leq f \leq 30$	+7
$35 \leq f \leq 44$	+10



RMM-44 without switch

RMM-44 with 1:8 Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f < 27$	+7.5
$27 \leq f \leq 30$	+11
$35 \leq f \leq 40$	+11



RMM-44 with switch

RMM-53 without switch

Measured

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f \leq 50$	+20

RMM-53 with 1:8 switch

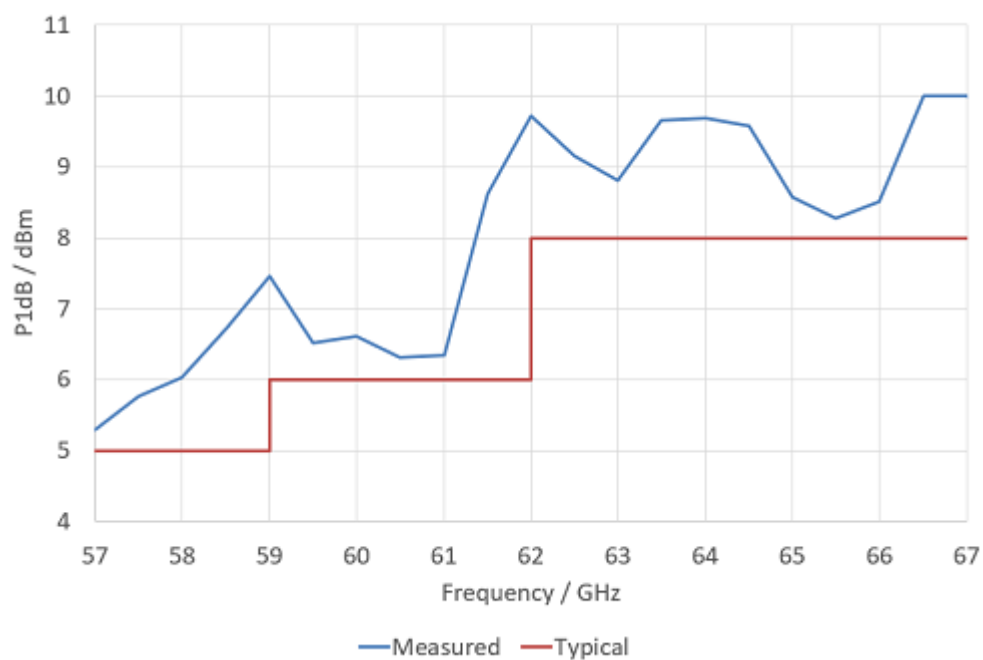
Measured

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f \leq 48$	+24 ^[27]
$48 < f \leq 50$	+21

RMM-72 without Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$57 \leq f < 59$	+5
$59 \leq f < 62$	+6
$62 \leq f \leq 67$	+8



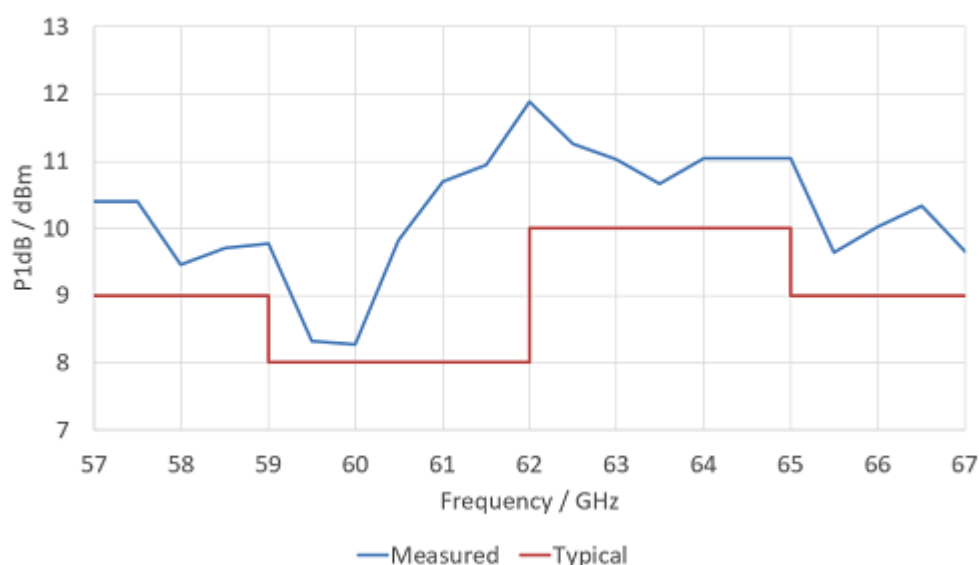
RMM-72 without switch

RMM-72 with 1:8 Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$57 \leq f \leq 59$	+9
$59 < f < 62$	+8
$62 \leq f \leq 65$	+10
$65 < f \leq 67$	+9

^[27] limited by abs. max. input power.



RMM-72 with switch

WSMM performance implications on other test head cards

Using the WSMM Family Board E8027FAA has performance implications on other test head cards.

Implications for specific cards:

- *DPS128* on page 119
- *PS1600* on page 120
- *PS9G* on page 120

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

DPS128

For general technical specification see

- See topic 148671
- See topic 141967
- See topic 141968

All DPS channels have a capacitor of 2.2 nF installed. The typical trace length of DPS locations is ~25 cm, resulting in ~0.2 Ohm per trace. MS-DPS1 and MS-DPS5 locations are longer and result in ~0.4 Ohm per trace.

Ganging of DPS channels is supported on WSMM only (for example ganging of channels on DUT board).

Ganging is not supported within DPS pogo cable(s) on family board (E8027-66402) within WSMM pogo blocks.

	dl	DUT Scale interface specification			WSMM specification		
Loadstep		1 μ F	47 μ F	100 μ F	1 μ F	47 μ F	100 μ F
t _{sett} (us)	0.1 A to 1 A	20 μ s	25 μ s	30 μ s	20 μ	25 μ	30 μ
	1 A to 0.1 A	20 μ s	25 μ s	30 μ s	20 μ	25 μ	30 μ
dV (mV)	0.1 A to 1 A	320 mV	120 mV	85 mV	370 mV	125 mV	90 mV
	1 A to 0.1 A	500 mV	125 mV	85 mV	550 mV	130 mV	90 mV

PS1600

This specification is based on typical trace length of 250 mm. Approximation for shorter lengths, assumed a somewhat linear relationship, for example shortest trace 0 mm (+80 mm cable which is ~20 mm trace) which is about 10% expected specification degradation: impact for 0 mm trace length is assumed to be ~10% of longest trace impact.

	Impacting or boundary condition	PS1600 DUT Scale interface specification	PS1600 WSMM specification (typical)
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	1...2 Ohm 0...251 mm trace length
Driver, AC Max transition times	swing 1.0 V / 1.8 V / 3.0 V	0.6 ns / 0.7 ns / 0.8 ns	0.67 ns / 0.78 ns / 0.9 ns
Driver, AC min pulse width	swing 1.0 V / 1.8 V / 3.0 V	0.7 ns / 0.8 ns / 0.9 ns	0.78 ns / 0.9 ns / 1.0 ns
Receiver, AC Min detectable pulse width		600 ps	620 ps
Receiver, AC Intrinsic transition time		350 ps	450 ps
OTA and EPA, system		< $\pm$ 200 ps (1600 Mbps) EPA = OTA / 2	< $\pm$ 220 ps (1600 Mbps) < $\pm$ 215 ps (1400 Mbps) EPA = OTA / 2
OTA and EPA, per-board		< $\pm$ 160 ps (1600 Mbps) EPA = OTA / 2	< $\pm$ 180 ps (1600 Mbps) < $\pm$ 175 ps (1600 Mbps) EPA = OTA / 2

PS9G

This specification is based on a typical trace length of 445 mm. Approximation for shorter lengths, assumed a somewhat linear relationship, for example, shortest trace 364 mm which is about 70% expected specification degradation: impact for 364 mm trace length is assumed to be ~70% of longest trace impact.

PS9G WSMM specificationification (typical), typical trace, 445 mm narrow

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 445 mm narrow
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	2...3 Ohm 364...557 mm trace length
Eye Height, single ended Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 89 % of Vswing 5.33 Gbps 70 % of Vswing 6.8 Gbps 70 % of Vswing	2.5 Gbps 70 % of Vswing 5.33 Gbps 40 % of Vswing 6.8 Gbps 35 % of Vswing
Eye Height, differential Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 85 % of Vswing 5.33 Gbps 70 % of Vswing 8 Gbps 70 % of Vswing	2.5 Gbps 65 % of Vswing 5.33 Gbps 40 % of Vswing 8 Gbps 36 % of Vswing
Eye Height, single ended Driver + Receiver	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 88 % of Vswing 5.33 Gbps 68 % of Vswing 6.8 Gbps 54 % of Vswing	2.5 Gbps 47 % of Vswing 5.33 Gbps 10 % of Vswing
Eye Height, differential Driver + Receiver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 80 % of Vswing 5.33 Gbps 68 % of Vswing 8 Gbps 35 % of Vswing	2.5 Gbps 39 % of Vswing 5.33 Gbps 3 % of Vswing
Eye Height, single ended Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 78 % of Vswing 5.33 Gbps 29 % of Vswing 6.8 Gbps 15 % of Vswing	2.5 Gbps 37 % of Vswing
Eye Height, differential Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 74 % of Vswing 5.33 Gbps 37 % of Vswing 8 Gbps 11 % of Vswing	2.5 Gbps 33 % of Vswing
OTA at 4.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 88 ps / < ± 99 ps

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 445 mm narrow
OTA at 2.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 2.5 Gbps per pin pair / per-board	Not specified	< ± 55 ps / < ± 66 ps

PS9G WSMM specification (typical), typical trace, 120 mm narrow

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 120 mm narrow
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	2..3 Ohm 364...557 mm trace length
Eye Height, single ended Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 89 % of Vswing 5.33 Gbps 70 % of Vswing 6.8 Gbps 70 % of Vswing	2.5 Gbps 84 % of Vswing 5.33 Gbps 65 % of Vswing 6.8 Gbps 62 % of Vswing
Eye Height, differential Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 85 % of Vswing 5.33 Gbps 70 % of Vswing 8 Gbps 70 % of Vswing	2.5 Gbps 80 % of Vswing 5.33 Gbps 65 % of Vswing 8 Gbps 61 % of Vswing
Eye Height, single ended Driver + Receiver	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 88 % of Vswing 5.33 Gbps 68 % of Vswing 6.8 Gbps 54 % of Vswing	2.5 Gbps 79 % of Vswing 5.33 Gbps 53 % of Vswing 8 Gbps 35 % of Vswing
Eye Height, differential Driver + Receiver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 80 % of Vswing 5.33 Gbps 68 % of Vswing 8 Gbps 35 % of Vswing	2.5 Gbps 70 % of Vswing 5.33 Gbps 46% of Vswing 8 Gbps 16 % of Vswing
Eye Height, single ended Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 78 % of Vswing 5.33 Gbps 29 % of Vswing 6.8 Gbps 15 % of Vswing	2.5 Gbps 68 % of Vswing 5.33 Gbps 15% of Vswing 8 Gbps 10 % of Vswing

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 120 mm narrow
Eye Height, differential Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 74 % of Vswing 5.33 Gbps 37 % of Vswing 8 Gbps 11 % of Vswing	2.5 Gbps 66 % of Vswing 5.33 Gbps 26% of Vswing
OTA at 4.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 88 ps / < ± 99 ps
OTA at 2.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 2.5 Gbps per pin pair / per-board	Not specified	< ± 55 ps / < ± 66 ps

PS9G WSMM specification (typical), typical trace, 300 mm wide

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 300 mm wide
OTA at 4.5 Gbps	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 41 ps / < ± 52 ps

PS9G WSMM specification (typical), typical trace, 250 mm wide

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 250 mm narrow
OTA at 4.5 Gbps	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 45 ps / < ± 56 ps

Assuming ~40% eye height the

- pogo block 301 (16 channels) can support 3 Gbps and the
- pogo blocks 302, 305, 306 (48 channels) can support 6 Gbps.